

A Reflexive Root-Polytope Model for Preorder h -Polynomials

Simplicial-polytopal realization, palindromicity, and unimodality

With Part II (added 28 September 2026): height-two preorders — matching supports, nonreal-root families, and crown obstructions

With Part III (added 29 September 2026): cactus rigidity for matching-support determinants

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Abstract

For a finite preorder τ on $[n]$, let $h_\tau(t)$ count the integer points of its preorder polytope by the number of nonzero coordinates. We give a proposed general proof that this polynomial is the h -polynomial of an n -dimensional simplicial polytope, establishing Conjecture 5.1(c), and hence (a) and (b), of Athanasiadis and Chapoton. The principal construction is the explicit n -dimensional lattice polytope

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : 1 \leq i \leq n\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}).$$

We prove $h^*(C_\tau, t) = h_\tau(t)$ by an integer-fiber decomposition of an augmented bipartite root polytope. Every boundary pulling triangulation of C_τ , coned from the origin, is unimodular; its boundary is combinatorially polytopal. We also prove reflexivity, an exact formula for the gauge of C_τ , and a Gorenstein translation formula for the larger root polytope. The higher-dimensional root-polytope identity and preorder duality are already results of Dai, Hou, Liu, Thawinrak, and Wang, and are not claimed as new here. A separate hypertree argument reconstructs that known identity. Exact verification code and complete small-case data accompany the manuscript.

Addition of 28 September 2026 (batch 39). Part II, from a later manuscript, proves that for a poset of height at most two with bipartite comparability graph G the support polynomial is $(1+t)^n p_G(t/(1+t)^2)$, where p_G counts perfectly matchable vertex sets; this gives γ -positivity (Conjecture 5.2) in height two. It records that real-rootedness (Conjecture 5.3) fails, crediting the first counterexample, an eight-element poset, to a posting by Shivam Patel, and proves an exact real-root count for an infinite family extending it. It shows that an induced crown on $2r \geq 6$ elements forces a minimal nonface of size r in every lattice triangulation of C_τ ; Conjecture 5.1(d) itself remains open.

Addition of 29 September 2026 (batch 42). Part III, from a further manuscript, proves that a bipartite graph carries a complex matrix on its edges whose square minors have squared modulus exactly the perfect-matchability indicator if and only if it is a cactus, and that this is strictly stronger than real stability of the signed matching-support polynomial; noncacti miss the exact model by the sharp universal defect $(\sqrt{17} - 3)/2$.

Scope and status. The intended new result is the general simplicial-polytopal realization, not the already established duality theorem. The proof uses explicitly identified results on support-enumerators and pulling triangulations. All steps specific to the construction are proved below; the external results are identified explicitly. The manuscript has not been independently refereed or checked by a proof assistant. The finite computations test the constructions; they do not replace the proof. Flag realizability, γ -positivity, and real-rootedness are not proved. [Added 28 September 2026, batch 39: this sentence describes Part I. Part II proves γ -positivity for posets of height at most two, records counterexamples to real-rootedness (the first credited to prior work), and obstructs flag lattice triangulations of C_τ ; see Section 11.3.]

Contents

Part I. A reflexive root-polytope model	5
1 The problem and the result	5
2 The augmented bipartite root polytope	6
3 A complete inequality description	7
4 The dimension-reducing lattice fibers	8
5 Reflexivity and simplicial-polytopal realization	10
6 A second geometric check in dimension $2n$	11
7 Consequences and quantitative formulas	12
8 Examples and the role of transitivity	14
9 Exact computational verification	15
10 What remains unresolved and how to audit the claim	16
Part II. Height-two preorders and crown obstructions	18
11 Provenance, conventions and what Part II answers	18
12 The research target and the distinction from prior work	22
13 Definitions and main statements	23
14 A matching-polymatroid counting lemma	25
15 The height-two formula and its consequences	26

16	A complete nonreal-root family at treewidth two	27
17	The 17-element counterexample in exact arithmetic	30
18	The canonical root polytope by orthant decomposition	33
19	Cactus graphs: matrices, stability, and interlacing	35
20	Crowns: explicit formulas with simple negative roots	37
21	Induced crowns obstruct every lattice triangulation	38
22	Exact verification and reproducibility	41
23	Proof dependencies and a formalization plan	43
24	Further research questions and topics	43
25	Conclusion	45
	 Part III. Cactus rigidity for matching-support determinants	 46
26	Provenance, conventions and what Part III answers	46
27	The question, its provenance, and the answer	50
28	Definitions and distinctions	51
29	The classification theorem	52
30	Cactus certificates and their complete phase freedom	53
31	Why every noncactus contains the right obstruction	54
32	A path-compression identity that preserves all supports	55
33	Three finite phase obstructions	56
34	The Hermitian representation has no hidden extra freedom	57
35	The same boundary over the smallest fields	57
36	A sharp universal approximation gap	58
37	Real stability is strictly more general	61
38	No fixed minor order can test exact representability	62

39	Consequences for the repository's preorder program	63
40	Exact verification and reproducibility	64
41	Proof audit, priority, and formalization boundary	65
42	Further research questions and topics	67
43	Conclusion	68
44	A sharper calculation for the shared-edge core	68
45	A compact dependency ledger	70
	Appendices to Part I	71
A	A hypertree proof of the known root identity	71
B	Proof dependencies and normalization audit	72
C	Reproducing the accompanying computations	72

Reading route. Sections 1–5 contain the main argument. Section 6 gives a second geometric check. Sections 8–10 provide examples, exact experiments, and limitations. Appendix A gives an alternative proof of the known higher-dimensional identity; Appendix B records the proof dependencies and the main normalization checks. Part II (Sections 11–25), added on 28 September 2026, treats height-two preorders; it precedes the appendices of Part I. Part III (Sections 26–45), added on 29 September 2026, answers Part II's question on exact cactus-style matrices; it follows Part II.

1 The problem and the result

A *preorder* is a reflexive, transitive binary relation; antisymmetry is not required. For a preorder τ on $E = [n]$, write $\mathcal{J}(\tau)$ for its order ideals, including $\emptyset$ and E . For a vector x and a subset $A \subseteq E$, write $x(A) = \sum_{i \in A} x_i$. The preorder polytope and its support polynomial are

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}_{\geq 0}^n : x(I) \leq |I| \text{ for every } I \in \mathcal{J}(\tau)\}, \quad (1.1)$$

$$P_\tau = \mathcal{Q}_\tau \cap \mathbb{Z}^n, \quad h_\tau(t) = \sum_{x \in P_\tau} t^{|\text{supp}(x)|} = \sum_{k=0}^n h_k(\tau) t^k. \quad (1.1)$$

Here $\text{supp}(x) = \{i : x_i \neq 0\}$. These are the objects of Athanasiadis and Chapoton [1]. Their Conjecture 5.1(c) asks whether $(h_0(\tau), \dots, h_n(\tau))$ is the h -vector of an n -dimensional simplicial polytope. Parts (a) and (b) ask, respectively, for palindromicity and palindromic unimodality. The stronger part (d) requires a flag simplicial polytope.

1.1 What was already known when this problem was selected

The arXiv version of [1] is dated 26 May 2026. A subsequent paper by Dai, Hou, Liu, Thawinrak, and Wang [2], in its 27 August 2026 revision, proves that the support-enumerator of a bipartite graph is the Ehrhart numerator of its augmented root polytope. Its Theorem 1.3 explicitly settles Conjecture 5.4 of [1], the identity $h_\tau = h_{\tau^*}$ under reversal of the preorder. These facts must therefore be treated as prior results, not as open problems. Problem 5.3 of [2] still discusses the coefficient-shape conjectures for preorders. Menon's August 2026 paper [6] gives further special cases of γ -positivity for arbors.

The target of the present manuscript is the *general* assertion in Conjecture 5.1(c). The advance proposed here is a dimension-reducing lattice fiber decomposition and the resulting explicit root-polytope realization. The literature check establishes these version-specific statements; it does not certify absolute priority against every unpublished or unindexed argument.

Theorem 1.1 (Main theorem). *Let τ be any preorder on $[n]$, where $n \geq 1$, and put*

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : i \in [n]\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}) \subset \mathbb{R}^n. \quad (1.2)$$

Then C_τ is reflexive and

$$h^*(C_\tau, t) = h_\tau(t). \quad (1.3)$$

Moreover, any pulling triangulation of ∂C_τ is combinatorially the boundary of an n -dimensional simplicial polytope S_τ satisfying

$$h(S_\tau, t) = h_\tau(t).$$

Consequently $h_k(\tau) = h_{n-k}(\tau)$ for all k , and

$$1 = h_0(\tau) \leq h_1(\tau) \leq \dots \leq h_{\lfloor n/2 \rfloor}(\tau).$$

In fact the full g -theorem conditions hold for $h(\tau)$.

A pulling triangulation is obtained by fixing an order on the vertices and recursively triangulating faces by cones from their first vertex. The actual root polytope C_τ need not be simplicial. The theorem asserts that its triangulated boundary has a simplicial convex realization with the desired h -vector. These two assertions should not be conflated.

The empty preorder is harmless: its support polynomial is 1, and both root models degenerate to a point. We treat $n \geq 1$ below to avoid unnecessary zero-dimensional boundary conventions.

1.2 A few useful sanity checks

Since E is an ideal, every $x \in P_\tau$ has $x(E) \leq n$. Hence P_τ is finite, $h_0 = 1$, and $h_n = 1$: the only point with all coordinates positive is $\mathbf{1}$. For a point supported just at i , the largest allowed value of its coordinate is $|\downarrow i|$, where $\downarrow i = \{j : j \leq_\tau i\}$. Therefore

$$h_1(\tau) = \sum_{i=1}^n |\downarrow i| = |\{(j, i) : j \leq_\tau i\}|. \quad (1.4)$$

Elements in the same equivalence class of the preorder remain separate coordinates throughout. Replacing a preorder by its quotient poset without retaining class sizes would change the problem.

2 The augmented bipartite root polytope

Let G_τ have left vertices $1_L, \dots, n_L$, right vertices $1_R, \dots, n_R$, and an edge $i_L j_R$ precisely when $j \leq_\tau i$. Adjoin 0_L and 0_R , with 0_L adjacent to every right vertex and 0_R adjacent to every left vertex. Denote the resulting graph by $\widehat{G}_\tau$. In particular $0_L 0_R$ is an edge. Its root polytope is

$$B_\tau = \text{conv}\{\mathbf{e}_i + \mathbf{f}_j : i_L j_R \in E(\widehat{G}_\tau)\} \subset \mathbb{R}^{n+1} \oplus \mathbb{R}^{n+1}. \quad (2.1)$$

The two coordinate blocks will be denoted (u, v) . We use the plus-sign convention for bipartite roots. Negating the second block converts this into the difference convention and preserves the lattice.

The affine span of B_τ is

$$H_1 = \{(u, v) : \sum_{i=0}^n u_i = \sum_{j=0}^n v_j = 1\}. \quad (2.2)$$

Indeed, the edges $i_L 0_R$ and $0_L j_R$ form a spanning tree, whose $2n + 1$ root vertices are affinely independent. Thus $\dim B_\tau = 2n$. At dilation m , the lattice is the full integer lattice in $H_m = \{\sum u_i = \sum v_j = m\}$, not a smaller lattice generated by selected coordinates.

2.1 The known support-enumerator theorem

For a bipartite graph with left neighborhoods $D_1, \dots, D_n$ in an n -element right shore, its draconian vectors are the nonnegative integer vectors x satisfying

$$x(A) \leq \left| \bigcup_{i \in A} D_i \right| \quad (A \subseteq [n]). \quad (2.3)$$

Theorem 1.2 of [2] identifies their support-enumerator with the h^* -polynomial of the graph augmented by the two universal vertices just used in (2.1).

Lemma 2.1. *For $D_i = \downarrow i$, the draconian vectors in (2.3) are exactly P_τ .*

Proof. If $x \in P_\tau$ and $A \subseteq E$, transitivity makes $\downarrow A$ an ideal. Nonnegativity gives

$$x(A) \leq x(\downarrow A) \leq |\downarrow A| = \left| \bigcup_{i \in A} D_i \right|.$$

Conversely, if (2.3) holds and I is an ideal, reflexivity and downward closure give $\bigcup_{i \in I} D_i = I$. Taking $A = I$ gives $x(I) \leq |I|$. $\square$

Proposition 2.2 (Known higher-dimensional identity). *For every preorder on $[n]$,*

$$\text{Ehr}_{B_\tau}(t) := \sum_{m \geq 0} |mB_\tau \cap \mathbb{Z}^{2n+2}| t^m = \frac{h_\tau(t)}{(1-t)^{2n+1}}. \quad (2.4)$$

Proof. Apply the support-enumerator theorem of [2] and Lemma 2.1. $\square$

Appendix A supplies another proof of this proposition using the Kálmán–Postnikov theorem [3]. Its elementary matching construction also explains why a support statistic, rather than a sum-of-coordinates statistic, occurs. Neither proof makes the already known proposition a novelty claim of this manuscript.

3 A complete inequality description

The special feature of a preorder is that all relevant neighborhood sets can be reduced to ideals. We need the reduction over the reals, not only for integer degree vectors.

Lemma 3.1 (Weighted Hall criterion). *Let G be a bipartite graph with shores L, R . Nonnegative margins u and v with common total m are the row and column sums of a nonnegative matrix supported on the edges of G if and only if*

$$u(A) \leq v(N(A)) \quad (A \subseteq L). \quad (3.1)$$

If the margins are integral, the matrix can be chosen integral.

Proof. Necessity follows by summing the entries in the rows of A . For sufficiency, make a network with source-to-left capacities u_i , right-to-sink capacities v_j , and capacity $m + 1$ on each allowed left-to-right edge. A cut of capacity at most m cannot cut an allowed left-to-right edge. If its source side contains the left set A , it must therefore contain $N(A)$ on the right; its capacity is at least $m - u(A) + v(N(A)) \geq m$. A flow of value m follows from the max-flow/min-cut argument. For completeness, that argument chooses a maximum feasible flow, which exists by compactness; if its residual graph contains a source-to-sink path, a positive augmentation contradicts maximality. Otherwise the vertices reachable from the source give a cut with capacity equal to the flow value. Integral capacities allow integral augmentations from the zero flow, so an integral maximum flow exists in the integral case. $\square$

Proposition 3.2 (Ideal inequalities for B_τ). *For every real $m \geq 0$, a pair $(u, v) \in H_m$ belongs to mB_τ exactly when all its coordinates are nonnegative and*

$$u(I) - v(I) \leq v_0 \quad (I \in \mathcal{J}(\tau)). \quad (3.2)$$

Here $u(I), v(I)$ involve only the coordinates indexed by $[n]$.

Proof. A point of mB_τ is precisely the pair of margins of a nonnegative matrix of total m supported on $\widehat{G}_\tau$. For a left subset containing 0_L , its neighborhood is the entire right shore, so (3.1) follows from nonnegativity and the total-sum condition. For $A \subseteq [n]$,

$$N(A) = \{0_R\} \cup (\downarrow A)_R.$$

Thus the remaining Hall conditions are $u(A) \leq v_0 + v(\downarrow A)$. If (3.2) holds, then

$$u(A) \leq u(\downarrow A) \leq v_0 + v(\downarrow A),$$

so all Hall conditions hold. Conversely, taking $A = I$ for an ideal gives $\downarrow I = I$ and recovers (3.2). The case $m = 0$ is immediate. $\square$

The empty-ideal inequality is just $v_0 \geq 0$, and the full-ideal inequality is equivalent to $u_0 \geq 0$. They may be redundant, but retaining them makes later formulas uniform. We do not assert that every inequality in this presentation defines a distinct facet.

4 The dimension-reducing lattice fibers

Define the projection

$$\pi(u, v) = (u_i - v_i)_{i=1}^n. \quad (4.1)$$

On the root vertices in (2.1), it has values 0, $\pm \mathbf{e}_i$, and $\mathbf{e}_i - \mathbf{e}_j$ for $j \leq_\tau i$. Therefore

$$\pi(B_\tau) = C_\tau. \quad (4.2)$$

Its kernel on the direction space of H_1 is spanned by $\mathbf{e}_i + \mathbf{f}_i - \mathbf{e}_0 - \mathbf{f}_0$, $1 \leq i \leq n$. The dimension loss is exactly n . A linear projection alone does *not* usually preserve an Ehrhart numerator. The following explicit fibers are what make this one work.

For $y \in \mathbb{R}^n$, put

$$\begin{aligned} y_i^+ &= \max(y_i, 0), & y_i^- &= \max(-y_i, 0), \\ p(y) &= \sum_i y_i^+, & q(y) &= \sum_i y_i^-, \end{aligned} \quad (4.3)$$

$$a_\tau(y) = \max_{I \in \mathcal{J}(\tau)} y(I), \quad g_\tau(y) = q(y) + a_\tau(y). \quad (4.4)$$

The empty and full ideals imply $a_\tau(y) \geq 0$ and $a_\tau(y) \geq p(y) - q(y)$.

Theorem 4.1 (Exact fibers and an integral gauge). *For $y \in \mathbb{R}^n$, define*

$$\begin{aligned} u^{\min}(y) &= (a_\tau(y) + q(y) - p(y), y_1^+, \dots, y_n^+), \\ v^{\min}(y) &= (a_\tau(y), y_1^-, \dots, y_n^-). \end{aligned} \quad (4.5)$$

Every point of mB_τ projecting to y has a unique expression

$$(u, v) = (u^{\min}(y), v^{\min}(y)) + \sum_{i=0}^n w_i(\mathbf{e}_i + \mathbf{f}_i), \quad (4.6)$$

where $w_i \geq 0$ and

$$\sum_{i=0}^n w_i = m - g_\tau(y). \quad (4.7)$$

Consequently,

$$y \in mC_\tau \iff g_\tau(y) \leq m. \quad (4.8)$$

For integral y, m , all terms in (4.5) are integral, and the lattice points in the fiber correspond exactly to nonnegative integer vectors w satisfying (4.7).

Proof. The two vectors in (4.5) are nonnegative. Their sums are both $a_\tau(y) + q(y) = g_\tau(y)$, and their difference on $[n]$ is y . Moreover,

$$u^{\min}(I) - v^{\min}(I) = y(I) \leq a_\tau(y) = v_0^{\min}.$$

Proposition 3.2 puts this pair in $g_\tau(y)B_\tau$. Adding a nonnegative multiple of a diagonal root leaves every difference $u(I) - v(I)$ unchanged. For the root indexed by 0, the right side v_0 also

increases, so all inequalities remain valid. This proves that every expression in (4.6) with (4.7) is an admissible point.

Conversely, suppose $(u, v) \in mB_\tau$ projects to y . For $1 \leq i \leq n$, nonnegativity and $u_i - v_i = y_i$ imply uniquely

$$u_i = y_i^+ + w_i, \quad v_i = y_i^- + w_i, \quad w_i = \min(u_i, v_i) \geq 0.$$

The ideal inequalities give $v_0 \geq \max_I y(I) = a_\tau(y)$. Set $w_0 = v_0 - a_\tau(y) \geq 0$. The equality of the two total sums then forces

$$u_0 = v_0 + q(y) - p(y) = u_0^{\min}(y) + w_0.$$

All coordinates in (4.6) are now determined. Summing either block yields (4.7). Projection proves (4.8). If y is integral, the maximum defining $a_\tau(y)$ is an integer, which establishes the final assertion without any lattice-saturation assumption. $\square$

Corollary 4.2 (Equality of Ehrhart numerators). *For every finite preorder,*

$$\text{Ehr}_{B_\tau}(t) = \frac{\text{Ehr}_{C_\tau}(t)}{(1-t)^n}, \quad h^*(B_\tau, t) = h^*(C_\tau, t) = h_\tau(t). \quad (4.9)$$

Proof. For integral y , the least admissible integral dilation is $g_\tau(y)$. Theorem 4.1 gives

$$\text{Ehr}_{B_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{(1-t)^{n+1}}, \quad \text{Ehr}_{C_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{1-t}.$$

These are legitimate formal power series: each coefficient receives only finitely many contributions because C_τ is bounded. Dividing the two identities proves the first equality in (4.9). The dimensions are $2n$ and n , respectively, so their Ehrhart numerators coincide. Finally apply Proposition 2.2. $\square$

There is a useful coefficient-level form. The fiber over an integral y at dilation m contains

$$\binom{m - g_\tau(y) + n}{n} \quad \text{if } g_\tau(y) \leq m, \quad 0 \quad \text{otherwise.} \quad (4.10)$$

Thus the factor $(1-t)^{-n}$ is not a heuristic cancellation of dimensions; it comes from a concrete, uniquely parametrized family of integer fibers.

4.1 An order-theoretic interpretation of the gauge

The gauge can also be written

$$g_\tau(y) = \max_{I \in \mathcal{J}(\tau)} \left(\sum_{i \in I} y_i^+ + \sum_{i \notin I} y_i^- \right). \quad (4.11)$$

Thus testing membership in a dilation of C_τ is a maximum-weight-ideal problem. One need not enumerate ideals in a practical implementation: put source capacity y_i^+ at i , sink capacity y_i^- at i , and sufficiently large capacities on arcs $i \rightarrow j$ for $j \leq_\tau i$. A finite minimum cut has a downward-closed source side I , and its capacity is $p(y) - y(I)$. Hence $a_\tau(y)$, and therefore $g_\tau(y)$, can be obtained by a network-flow calculation. The accompanying small-case verifier deliberately uses exhaustive ideal enumeration instead, keeping the check independent of a flow library.

5 Reflexivity and simplicial-polytopal realization

We now finish the proof of Theorem 1.1. The required geometry is particularly simple because all nonzero vertices of C_τ are type- A roots with one coordinate deleted.

5.1 The polar is integral

Since C_τ contains $\text{conv}\{\pm \mathbf{e}_i\}$, it is full-dimensional and contains 0 in its interior. Its polar is

$$C_\tau^\circ = \{z \in \mathbb{R}^n : -1 \leq z_i \leq 1, \quad z_i - z_j \leq 1 \text{ whenever } j \leq_\tau i\}. \quad (5.1)$$

Lemma 5.1 (Root determinants). *Put $\mathbf{e}_0 = 0 \in \mathbb{Z}^n$. Any n linearly independent vectors of the form $\mathbf{e}_i - \mathbf{e}_j$, with $i, j \in \{0, \dots, n\}$ and $i \neq j$, form a basis of $\mathbb{Z}^n$.*

Proof. Associate the vector $\mathbf{e}_i - \mathbf{e}_j$ to a directed edge from j to i on $\{0, \dots, n\}$. A cycle in the underlying undirected graph would give a linear dependence, with signs determined by the edge directions. Consequently n independent vectors give a forest with n edges on $n + 1$ vertices, hence a spanning tree. The matrix of these vectors is a directed incidence matrix with the row indexed by 0 deleted. Remove a leaf different from 0; its row has one nonzero entry, equal to 1 or -1 . Expanding the determinant along that row and repeating reduces the determinant to 1 or -1 . $\square$

Proposition 5.2. *The polar (5.1) is a lattice polytope. In fact all its vertices have coordinates in $\{-1, 0, 1\}$. Thus C_τ is reflexive.*

Proof. At a vertex of the full-dimensional polar, choose n linearly independent active constraints. With $z_0 = 0$, each selected equality is of the form $z_i - z_j = 1$. Its coefficient row is a root as in Lemma 5.1. The resulting square matrix has determinant ± 1 , so its solution is integral. The bounds $-1 \leq z_i \leq 1$ restrict these integers to $-1, 0, 1$. Both the original polytope and its polar are therefore integral, with the origin in the interior, which is precisely reflexivity. $\square$

This proof also supplies a finite exact way to test lattice-point membership: one may evaluate against all ternary vectors satisfying (5.1), since that set contains every vertex of the polar. The verifier uses this method as a check independent of the gauge formula.

5.2 Unimodular cones over the boundary

Recall that a lattice simplex with vertices $0, r_1, \dots, r_n$ is unimodular when $r_1, \dots, r_n$ form a lattice basis. A triangulation is unimodular if all its full-dimensional simplices are unimodular.

Proposition 5.3. *Let Δ be any triangulation of ∂C_τ using only vertices of C_τ . Then the star triangulation $0 * \Delta$ of C_τ is unimodular.*

Proof. Every boundary simplex is contained in a supporting hyperplane not containing 0. The n vertices of a maximal boundary simplex are therefore linearly independent: an affine dependence is already excluded by being a simplex, and a linear dependence with nonzero coefficient sum would place 0 in that supporting hyperplane. All its vertices are roots $\mathbf{e}_i - \mathbf{e}_j$, allowing index 0. Lemma 5.1 shows that the cone simplex from 0 is unimodular. Since 0 is interior, these cones form a triangulation of all of C_τ . $\square$

5.3 Why the triangulated boundary is polytopal

We use the following standard pulling fact, stated as Lemma 2.1 in Athanasiadis [4]: a pulling (reverse lexicographic) triangulation of the boundary of a convex n -polytope is combinatorially the boundary of a simplicial convex n -polytope. The geometric mechanism is successive small outward perturbations of the ordered vertices; at each step one chooses the perturbation small enough not to change the already fixed incidences. The resulting convex boundary realizes the prescribed pulling subdivision. This is an actual polytopality assertion, not merely shellability or the assertion that the complex is a sphere.

Here is the Ehrhart connection, with its normalization made explicit. If K is a d -dimensional unimodular triangulation of a lattice polytope P , then

$$\text{Ehr}_P(t) = \sum_{\sigma \in K} \left(\frac{t}{1-t} \right)^{|\sigma|} = \frac{h(K, t)}{(1-t)^{d+1}}, \quad (5.2)$$

where the empty face contributes 1. Indeed, the relative interior of an r -dimensional unimodular simplex contributes $\binom{m-1}{r}$ lattice points at positive dilation m , with generating series $t^{r+1}/(1-t)^{r+1}$. Relative interiors partition the triangulation, giving the first equality. The second equality is the definition of its h -polynomial. For a cone, the face generating function gains a factor $1 + t/(1-t)$, while the denominator exponent increases by one; hence

$$h(0 * \Delta, t) = h(\Delta, t). \quad (5.3)$$

Completion of the proof of Theorem 1.1. Choose a pulling triangulation Δ of ∂C_τ . By the pulling fact, Δ is the boundary of an n -dimensional simplicial polytope S_τ . Proposition 5.3 and (5.2)–(5.3) give

$$h(S_\tau, t) = h(\Delta, t) = h^*(C_\tau, t) = h_\tau(t),$$

where the last equality is Corollary 4.2. Reflexivity was proved in Proposition 5.2. The Dehn–Sommerville relations for simplicial polytopes give palindromicity, and the g -theorem gives the stated inequalities and its stronger numerical conditions; see also the polytopal conclusions summarized in [4, Theorem 3.5]. $\square$

6 A second geometric check in dimension $2n$

The larger polytope B_τ supplies two independent checks on the preceding argument: an interior-point translation and an explicit special simplex. They are useful safeguards against a sign error or a mismatch of dimensions.

6.1 An exact interior translation

Let $\mathbf{c} = (1, \dots, 1; 1, \dots, 1) \in \mathbb{Z}^{2n+2}$. For every $m \geq n+1$,

$$(\text{relint}(mB_\tau)) \cap \mathbb{Z}^{2n+2} = \mathbf{c} + ((m-n-1)B_\tau \cap \mathbb{Z}^{2n+2}). \quad (6.1)$$

To see this, note that every coordinate inequality is nonconstant on B_τ , and each ideal inequality has strict slack at $\mathbf{e}_0 + \mathbf{f}_0$. Consequently relative interior is characterized by positive coordinates and strict versions of (3.2). At integer points these strict inequalities are

$$u(I) - v(I) \leq v_0 - 1.$$

Subtracting 1 from every coordinate reduces each block sum by $n + 1$. The terms $|I|$ cancel on the left, and the right side becomes the new v_0 . This is exactly the weak inequality description of $(m - n - 1)B_\tau$. Conversely, adding $\mathbf{c}$ makes every defining inequality strict. There are no interior integer points for $1 \leq m < n + 1$, since $n + 1$ positive coordinates in either block already require total at least $n + 1$.

In particular, B_τ has codegree $n + 1$, with unique interior integer point $\mathbf{c}$ in its first nonempty interior dilation. It is Gorenstein of index $n + 1$. Ehrhart reciprocity translates (6.1) into

$$t^{2n+1}h^*(B_\tau, t^{-1}) = t^{n+1}h^*(B_\tau, t),$$

which again gives $h_\tau(t) = t^n h_\tau(t^{-1})$. This symmetry proof does not use the g -theorem or polytopality of a sphere.

6.2 The diagonal special simplex

A simplex with r vertices inside a polytope is *special* if each facet contains exactly $r - 1$ of those vertices. Consider

$$\Sigma_\tau = \text{conv}\{d_i : 0 \leq i \leq n\}, \quad d_i = \mathbf{e}_i + \mathbf{f}_i. \quad (6.2)$$

Reflexivity of the preorder makes these root vertices available, and they are affinely independent. A coordinate hyperplane $u_i = 0$ or $v_i = 0$ contains exactly the diagonal vertices other than d_i . On an ideal hyperplane $u(I) - v(I) = v_0$, every d_i with $i > 0$ lies on the hyperplane and d_0 does not. Every facet is furnished by at least one nonredundant inequality of Proposition 3.2; thus Σ_τ is special.

Every triangulation of a bipartite root polytope using its vertices is unimodular. Indeed, affine dependence of edge roots is equivalent to a cycle in the supporting bipartite graph; a full-dimensional simplex therefore corresponds to a spanning tree. Leaf elimination in its incidence matrix shows that its edge vectors form a basis of the integer lattice $\{(u, v) : \sum u_i = \sum v_j\}$. Intersecting with the height-one hyperplane shows affine unimodularity. In particular every pulling ordering is compressed in the terminology of [4].

Athanasiadis's special-simplex theorem [4, Theorem 3.5], applied with dimension $2n$ and $n + 1$ special vertices, therefore realizes $h^*(B_\tau, t)$ as the h -polynomial of a simplicial polytope of dimension $2n - (n + 1) + 1 = n$. This recovers the main theorem by another route. The broader Gorenstein/unimodular-triangulation framework of Bruns and Römer [5] provides a further compatible interpretation, but is not needed for our main proof.

7 Consequences and quantitative formulas

7.1 Duality, with its prior attribution

Swapping the coordinate blocks identifies B_τ with B_{τ^*} . This is the already established duality mechanism of [2]. In the smaller model it becomes the especially transparent equality

$$C_{\tau^*} = -C_\tau. \quad (7.1)$$

Thus $h_{\tau^*} = h_\tau$. The gauge gives a third check. Ideals of τ^* are complements of ideals of τ , so

$$a_{\tau^*}(-y) = a_\tau(y) - y(E), \quad g_{\tau^*}(-y) = g_\tau(y).$$

This rederivation is included to connect the new lower-dimensional model with the known theorem, not to reclassify Conjecture 5.4 as open.

7.2 Ehrhart polynomials and volumes

The ordinary Ehrhart polynomial of C_τ is now explicit in terms of the preorder support counts:

$$|mC_\tau \cap \mathbb{Z}^n| = \sum_{k=0}^n h_k(\tau) \binom{m+n-k}{n}, \quad m \in \mathbb{Z}_{\geq 0}. \quad (7.2)$$

Similarly,

$$|mB_\tau \cap \mathbb{Z}^{2n+2}| = \sum_{k=0}^n h_k(\tau) \binom{m+2n-k}{2n}. \quad (7.3)$$

A binomial term with nonnegative upper argument smaller than its lower argument is zero. These formulas also extend as polynomial identities in m . Normalized volume, meaning dimension factorial times volume in the relevant lattice, satisfies

$$\text{Vol}_{\mathbb{Z}}(C_\tau) = \text{Vol}_{\mathbb{Z}}(B_\tau) = h_\tau(1) = |P_\tau|. \quad (7.4)$$

The two volumes in this equality live in different dimensions and are both *normalized* volumes.

Remark 5.6 of [1] considers $h_\tau(t)/(1-t)^{n+1}$. Equation (7.2) identifies its coefficients as the Ehrhart polynomial of C_τ . This gives that series a concrete geometric interpretation. It does not prove that its polynomial has all complex roots on a particular vertical line; no such root-location assertion follows just from reflexivity.

7.3 Vertices, elementary bounds, and the g -vector

Every vector displayed in (1.2) is a vertex. To expose $\mathbf{e}_i$, choose a linear functional taking value 1 at coordinate i and a fixed $\varepsilon \in (0, 1)$ at all other coordinates. The negative case is analogous. To expose $\mathbf{e}_i - \mathbf{e}_j$, give coordinates i, j the values 1, -1 and all others the value 0; this root is the unique displayed vector with value 2. Thus

$$f_0(C_\tau) = n + |\{(j, i) : j \leq_\tau i\}| = n + h_1(\tau). \quad (7.5)$$

Pulling adds no vertices, so the same number occurs in S_τ .

The inclusions

$$[0, 1]^n \subseteq Q_\tau \subseteq \{x \geq 0 : x(E) \leq n\}$$

give the useful coefficientwise bounds

$$\binom{n}{k} \leq h_k(\tau) \leq \binom{n}{k}^2 \quad (0 \leq k \leq n). \quad (7.6)$$

For the upper bound, choose the support in $\binom{n}{k}$ ways; the positive integer values on it with total at most n can be chosen in $\binom{n}{k}$ ways by stars and bars. The lower bound uses the corresponding 0–1 points.

For completeness, the full numerical strengthening supplied by polytopality can be stated as follows. Put $g_0 = 1$ and $g_i = h_i - h_{i-1}$ for $1 \leq i \leq \lfloor n/2 \rfloor$. The g -theorem says this sequence is an M -sequence. In particular $g_i \geq 0$ and $g_{i+1} \leq g_i^{\langle i \rangle}$ whenever both sides are relevant. Here, if

$$a = \binom{b_i}{i} + \binom{b_{i-1}}{i-1} + \cdots + \binom{b_j}{j}, \quad b_i > b_{i-1} > \cdots > b_j \geq j \geq 1,$$

is the Macaulay expansion, then $a^{\langle i \rangle} = \binom{b_{i+1}}{i+1} + \cdots + \binom{b_{j+1}}{j+1}$, with $0^{\langle i \rangle} = 0$. These inequalities are stronger than unimodality and follow here from an actual simplicial-polytope realization.

8 Examples and the role of transitivity

8.1 Antichains and a single equivalence class

For an antichain, $\mathcal{Q}_\tau = [0, 1]^n$ and

$$h_\tau(t) = (1 + t)^n, \quad C_\tau = \text{conv}\{\pm \mathbf{e}_1, \dots, \pm \mathbf{e}_n\}.$$

The gauge is $\sum_i |y_i|$. The smaller model is already the simplicial cross polytope, so no boundary refinement is needed.

For the indiscrete preorder, in which every two elements are equivalent, $\mathcal{Q}_\tau = \{x \geq 0 : x(E) \leq n\}$. The same stars-and-bars argument used above shows

$$h_\tau(t) = \sum_{k=0}^n \binom{n}{k}^2 t^k.$$

Here $B_\tau = \Delta_n \times \Delta_n$, and the lower-dimensional polytope contains all roots $\pm \mathbf{e}_i$ and $\mathbf{e}_i - \mathbf{e}_j$. Its gauge is $\max(p(y), q(y))$, since its only ideals are $\emptyset, E$. These two extreme examples check both the coefficient bounds and the change of Ehrhart denominator.

8.2 A non-self-dual four-element example

Let τ have one minimum below three incomparable maximal elements. Its polynomial, obtained directly from (1.1), is

$$h_\tau(t) = 1 + 7t + 12t^2 + 7t^3 + t^4.$$

For example, for supports of size two not containing the minimum, the three pairs of maximal elements each allow the values $(1, 1), (1, 2), (2, 1)$, giving nine points; the three supports containing the minimum each allow only $(1, 1)$, giving three more. The dual order is not isomorphic to the original, but its directed root polytope is the negative of the original one and its support polynomial agrees. This is a useful test that duality is not merely invariance under relabeling.

8.3 A preorder with a nontrivial equivalence class

Use labels a, b, c, d, e , with $a \sim c, a < e, c < e, b < e$, and $b < d$. No other comparisons hold beyond those forced by reflexivity and transitivity. The principal ideals are

$$\{a, c\}, \quad \{b\}, \quad \{a, c\}, \quad \{b, d\}, \quad \{a, b, c, e\}.$$

Direct enumeration gives

$$h_\tau(t) = 1 + 11t + 34t^2 + 34t^3 + 11t^4 + t^5. \tag{8.1}$$

This agrees with the running example of [1]. There are 92 points in P_τ , 22 vertices of B_τ , and 16 vertices of C_τ . The number 92 is also the normalized volume of either root model.

For $x = (0, 0, 1, 2, 2)$, the matching construction in Appendix A can assign the five demands to the five right vertices as follows: the demand at c uses c_R , the two demands at d use b_R, d_R , and the two demands at e use a_R, e_R . Attach every left vertex to 0_R , then attach each right vertex to its assigned left vertex. This gives the spanning-tree certificate included in the JSON data.

8.4 Selected coefficient vectors

Preorder	n	$(h_0, \dots, h_n)$	$ P_\tau $
Antichain	4	(1, 4, 6, 4, 1)	16
Chain	4	(1, 10, 20, 10, 1)	42
Single equivalence class	4	(1, 16, 36, 16, 1)	70
One minimum, three maxima	4	(1, 7, 12, 7, 1)	28
Diamond	4	(1, 9, 18, 9, 1)	38
Example (8.1)	5	(1, 11, 34, 34, 11, 1)	92
Six-element crown	6	(1, 12, 48, 75, 48, 12, 1)	197

The crown has minimal elements a, b, c and maximal elements A, B, C , with $a, b < A, b, c < B$, and $c, a < C$. Every row of the table is recomputed by the supplied verifier; no floating-point root finder is involved. [Added 28 September 2026, batch 39: the crown is the case $r = 3$ of Part II's crown family. Its support polynomial has the closed form (20.2) and only simple negative zeros (Proposition 20.1), yet no lattice triangulation of its C_τ , pulling or not, is flag (Theorem 13.6).]

8.5 Why arbitrary reflexive relations are not enough

Transitivity cannot simply be dropped from the fiber theorem. Consider the neighborhoods

$$D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{2, 3\}.$$

They define a reflexive but nontransitive relation. Its draconian inequalities reduce to

$$x_i \geq 0, \quad x_1 \leq 1, \quad x_1 + x_2 \leq 2, \quad x_3 \leq 2, \quad x_1 + x_2 + x_3 \leq 3.$$

The support counts are (1, 5, 6, 1): one-coordinate supports contribute $1 + 2 + 2 = 5$, two-coordinate supports contribute $1 + 2 + 3 = 6$, and the full support contributes just (1, 1, 1). This is the nonpalindromic example also recorded in [1, Remark 5.5].

The general support-enumerator/root-polytope theorem remains valid for this graph. What fails is the reduction of all weighted Hall inequalities to matching ideal differences $u(I) - v(I)$. In particular, subtracting a common coordinate from the two margins can damage an inequality involving different row and column subsets. Thus the argument does not inadvertently imply palindromicity for all graph support-enumerators.

9 Exact computational verification

The program code/verify.py uses only Python's standard library and integer arithmetic. It stores a preorder by bitmasks

$$D_i = \{j : j \leq_\tau i\}.$$

It enumerates every reflexive relation on $[n]$ and retains precisely those for which $j \in D_i$ implies $D_j \subseteq D_i$. This is a direct test of transitivity. All weak compositions with total at most n are then checked against the ideal inequalities to obtain h_τ .

Ground-set size	Labeled preorders checked	Result
0	1	Pass
1	1	Pass
2	4	Pass
3	29	Pass
4	355	Pass
5	6,942	Pass
Total	7,332	Pass

For each of these preorders, the code checks duality, palindromicity, unimodality, the endpoint coefficients, the relation-count formula for h_1 , and the special-simplex slack identities. These are finite consistency checks, not the reason the theorem holds for arbitrary n .

Stronger, independent comparisons are performed for all 35 preorders on at most three elements. The program enumerates spanning trees directly, deduplicates their left degree-minus-one vectors, and compares them with $\{(n - |x|, x) : x \in P_\tau\}$. It computes internal inactivity by testing all allowed transfers to smaller coordinates, rather than replacing inactivity by support size in the code. Across these instances it enumerates 18,009 spanning trees and checks 399 explicit matching-tree certificates.

For the same cases it counts sums of edge-root vertices directly at every dilation $0 \leq m \leq 2n$. Those counts are compared both with Hall-margin enumeration and with (7.3). Finite differences recover all $2n + 1$ coefficients of the root-polytope Ehrhart numerator, including the predicted zero coefficients above degree n . Independently, integer points of C_τ are counted using the ternary polar description, compared with the gauge formula, and checked through $m = n$ against (7.2). Full-margin fiber identities and the interior translation are also checked exactly.

Five selected four-element preorders undergo full root Ehrhart checks through $m = 8$, quotient checks through $m = 4$, and direct edge-sum checks through $m = 3$. Further examples include the weighted five-element preorder and the six-element crown. The JSON file records all individual results, and the CSV file contains every preorder and support vector through size five.

The original conjecture paper reports tests through larger sizes for some coefficient properties [1]. The present computations are not offered as an improved size record. Their purpose is to compare independent realizations of the newly proposed construction and detect normalization, orientation, or lattice-index errors.

10 What remains unresolved and how to audit the claim

The proof gives an ordinary simplicial polytope, not a flag one. A flag complex is determined by its edges: every collection of pairwise adjacent vertices must form a face. Neither a pulling triangulation nor a general simplicial polytope automatically has that property. Therefore Conjecture 5.1(d) is not settled by the argument. [Added 28 September 2026, batch 39: Part II shows more. If τ contains an induced crown on $2r \geq 6$ elements, every lattice triangulation of C_τ or of ∂C_τ has a minimal nonface of size r (Theorem 13.6); so the polytope S_τ of Theorem 1.1 is then never flag. Conjecture 5.1(d), which asks for *some* flag polytope, remains open; see Section 11.3.]

Likewise, palindromic unimodality and the g -theorem do not establish γ -positivity or

real-rootedness. The manuscript proves neither the nonnegativity of all coefficients in

$$h_\tau(t) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_j t^j (1+t)^{n-2j}$$

nor the assertion that every zero of h_τ is real. Accordingly it does not claim to settle Conjectures 5.2 or 5.3 of [1]. [Added 28 September 2026, batch 39: the two preceding sentences describe Part I at its state of 20 September 2026. Conjecture 5.3 is false: the first counterexample known here is an eight-element height-two poset posted by Shivam Patel [11], which Part II checks and extends to the family of Theorem 13.3. Conjecture 5.2 holds for posets of height at most two, with γ_k the number of perfectly matchable vertex sets of size $2k$ in the comparability graph (Theorem 13.2), and is open in general. See Section 11.3.]

For independent mathematical review, the shortest audit is to verify four links: the identification of the draconian vectors with P_τ ; Proposition 3.2; the unique integer fibers in Theorem 4.1; and the determinant argument for cones over a polytopal pulling boundary. Once these are accepted, the cited support-enumerator theorem and the pulling lemma give the asserted simplicial-polytopal realization. Section 6 is a separate geometric cross-check, not a circular use of the desired symmetry.

The result should be understood as a proposed research proof with reproducible support, not as a claim of community acceptance. The distinction between the known August 2026 result and the additional construction developed here is part of the mathematical claim, not a bibliographic afterthought.

Part II. Height-two preorders: matching supports, nonreal roots, crown obstructions

Sections 1–10 and the appendices form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition. Part II was added on 28 September 2026 in batch 39 of ProveIt’s incoming-report intake, from one manuscript written against Part I as printed above. It addresses the three questions Part I left open in Section 10: Conjectures 5.3, 5.2 and 5.1(d) of Athanasiadis and Chapoton [1]. Its headline counterexample to Conjecture 5.3 is **not new**: it was posted earlier by Shivam Patel on MathDB [11], and the manuscript says so. Section 11 records the provenance, fixes the notation and states what is answered.

11 Provenance, conventions and what Part II answers

11.1 One manuscript, one addition

Part II is batch-39 manuscript 02, *Height-Two Preorder Polynomials: Matching supports, nonreal-root families, and crown obstructions*. With a spectral treatment of bipartite cactus graphs, a research manuscript developed with ChatGPT for Vladimir Reshetnikov and dated 28 September 2026 (26 pages as delivered). It arrived as the archive `ProveIt_Height_Two_Preorder_Research.zip` (inner directory `Preorder_Matching_Research/`) in ProveIt commit `74f7f5bdb` and was placed in `e2b1f016a`. It names no ProveIt commit; it records the Git blob `4c350f5652096b7db91d4a10075eb7905910f647` of Part I’s `article.tex` as the file it consulted. That blob is Part I exactly as printed above (it is unchanged at the placement commit), so every statement the manuscript makes about “the ProveIt report” refers to Part I. The manuscript’s code and data are shipped beside Part I’s files with the prefix `02-height-two-` (Part I’s unprefix files count as 01), and its status and source notes as `02-height-two-STATUS.md` and `02-height-two-sources.md`. Its manuscript, delivery README and PDF are not shipped and survive in the arrival commit; its checksum list was verified at placement and retired.

The manuscript is printed here as Sections 12 to 25: its Section k is Section $k + 11$ here, and its theorem and equation numbers shift accordingly (its Theorem 2.2 is Theorem 13.2); its two figures and ten research questions keep their numbers. Every result, proof, example, remark, limitation, question and non-claim of the manuscript is printed. For the reader’s convenience, its abstract is reproduced here as delivered.

Abstract of manuscript 02 (as delivered). For a finite poset P , let $h_P(t)$ enumerate the integer points of its preorder polytope by support size. If P has height at most two and G is its bipartite comparability graph, we prove

$$h_P(t) = (1+t)^{|P|} p_G\left(\frac{t}{(1+t)^2}\right), \quad p_G(u) = \sum_{\substack{W \subseteq V(G) \\ G[W] \text{ has a perfect matching}}} u^{|W|/2}.$$

Thus the γ -coefficients count distinct matchable vertex sets, not matchings, proving the height-two case of Athanasiadis–Chapoton Conjecture 5.2. For the graph Θ_m formed by m internally disjoint three-edge paths with common endpoints, we obtain an explicit polynomial and a complete real-root count. For $m \geq 3$, the associated height-two support polynomial has exactly two real roots when m is even and four when m is odd. Consequently its proportion of nonreal roots tends to one, even though the graph has treewidth two. The eight-element case was previously posted by Shivam Patel; we verify it and extend it rather than claiming its discovery. A separate 17-element construction transfers the classical Stembridge–Ohsugi–Tsuchiya example.

An orthant argument proves the height-two instance of the repository’s root-polytope identity. For bipartite cactus graphs, a sixth-root-of-unity matrix gives a determinantal formula and a multivariate stability refinement, with univariate real-rootedness credited to prior work. Induced crowns force arbitrarily large minimal nonfaces in every lattice triangulation of the canonical root polytope and indispensable toric relations of arbitrarily large degree. Exact Python and C++ certificates accompany the proofs.

Status and scope (manuscript 02). The proposed contributions are the general height-two connection, the complete root-distribution theorem for Θ_m , and the geometric and spectral refinements. The first counterexample to Conjecture 5.3 is *not* claimed as new: a smaller posted example was found during the source audit. Two classical root-polytope results are imported explicitly. The manuscript has not been independently refereed or checked in Lean. General flag realizability and γ -positivity beyond height two remain unsettled here; absolute publication priority is not certified.

Priority of the counterexample. The first known counterexample to Conjecture 5.3 of [1] printed in this report is the eight-element height-two poset P_{Θ_3} of Section 16. It is due to Shivam Patel, whose posted solution on MathDB [11] specifies the graph Θ_3 and both polynomials. The manuscript found that posting during its own source audit, after its 17-element example had been developed, and revised its claims accordingly; it reproduces and independently checks Patel’s example and claims only the all- m extension. The exact calendar date of the posting was not established: the page showed a relative date (about one month before 28 September 2026) and labelled the claim unverified. A withdrawn SSRN listing (abstract 7385158) for an eight-element counterexample was also found; its full text was unavailable, and it is not used as evidence for any mathematical or priority claim. The 17-element counterexample of Section 17 is a transfer of the classical Stembridge–Ohsugi–Tsuchiya graph polynomial [10, 8], neither a first disproof nor a minimum-size example. Univariate real-rootedness for bipartite cacti is Ohsugi–Tsuchiya’s [9, Theorem 7.2], and the even-cycle (crown) formula is Davis–Kohl’s [7]. The source audit behind these statements does not prove the absence of other publications, preprints or postings containing any of Part II’s proposed extensions.

Shortest audit (manuscript 02). Read the counting lemma in Section 14, the Boolean summation in Section 15, and the explicit graph and exact certificate in Section 17. Together with Section 16, these establish the root-distribution results without assuming any unrefereed theorem in ProvelT (in particular, without Part I’s general theorem). Sections 18–21 develop its geometric and spectral context. The companion programs provide independent finite checks rather than substitutes for the infinite-family proofs.

11.2 Notation of Part II

Part II works with *posets* of height at most two, which are preorders; for $\tau = P$ its $\mathcal{J}(P)$, $\mathcal{Q}(P)$, h_P and C_P are exactly Part I's $\mathcal{J}(\tau)$, $\mathcal{Q}_\tau$, h_τ and C_τ (compare (13.1) and (13.2) with the two displays opening Section 1, and Theorem 13.6 with (1.2)). Its γ -coefficients are taken in the basis $t^k(1+t)^{n-2k}$ of Section 10, and $\text{Vol}_\mathbb{Z}$ is the normalized volume of (7.4). Several letters mean different things in the two parts; Table 1 fixes them. No symbol of the manuscript was renamed and no normalization was changed.

Table 1: Symbols of Part II. The last column prints the tempting false reading, usually a Part I symbol with the same letter.

Symbol	Meaning in Part II	Not to be confused with
$P, P_G, P_0, P_{\Theta_m}, P_r$	finite posets: P_G has $a < b$ exactly for the edges ab of G ; $P_0 = P_{G_0}$; P_r is the crown	Part I's lattice-point set $P_\tau = \mathcal{Q}_\tau \cap \mathbb{Z}^n$, which Part II writes $\mathcal{Q}(P) \cap \mathbb{Z}^V$. P_r is not " P_τ at $\tau = r$ ".
$V, n = V $	ground set of P	Part I's ground set $E = [n]$.
$G = (A \sqcup B, E)$	bipartite comparability graph of a height-two poset on its n elements; E its edge set	Part I's doubled graph G_τ on $2n$ vertices i_L, j_R (edge iff $j \leq_\tau i$, loops $i_L i_R$ included) and its augmentation $\widehat{G}_\tau$;
$p = A , q = B $	shore sizes (Section 13)	Part I's ground set E .
$p_G(u), \mathcal{M}_k(G)$	perfectly matchable <i>vertex sets</i> of G ; p_G is the γ -polynomial of h_{P_G}	Part I's functions $p(y), q(y)$ of (4.3); the polynomial $q_G(x)$ of (19.5). the matching polynomial $M_G(u) = \sum_k m_k(G)u^k$; also p_{G_τ} of Part I's doubled graph, which is h_τ itself (Remark 11.1).
$H = (X \sqcup Y, E), \widehat{H}, B_{\widehat{H}}, \mathcal{R}_H$	any bipartite graph, its augmentation by x_0, y_0 , its edge polytope, its demand vectors	Part I's B_τ , which is $B_{\widehat{H}}$ for $H = G_\tau$ (same augmentation, same lattice); Part I's hyperplanes H_1, H_m . Hypertrees here are <i>right</i> hypertrees; Appendix A uses left ones.
$\Theta_m, p_m, R_m(y), k = m - 2$	m parallel three-edge paths, its matchable-set polynomial	Part I's dilation m . The vertex d of Θ_m is not a diagonal vertex d_i of (6.2).
$G_0, p_0, F_{i,j}(u)$	the 17-element example, its γ -polynomial, the board recurrence	—
D_G, T_G, g_T, g_D	one-sided root polytope lattice-equivalent to C_{P_G} , its positive part, their gauges	Part I's gauge g_τ on C_τ ; g_D is g_τ after negating the A -coordinates. Part I's neighbourhoods D_i .
$\mathcal{I}(G), I$ in Section 18	independent vertex sets	order ideals $I \in \mathcal{J}(\tau)$ (also written I in Section 15); the toric ideal $I_{\mathcal{A}}$. Part I's triangulation K in (5.2).
$K, \zeta, H_G, \Phi_G, q_G, \lambda_j$	cactus matrix, sixth root of unity, Hermitian block matrix, stable polynomial, its diagonal specialization, eigenvalues of KK^*	
S	a subset of B or Y ; in (19.7) the support size of a uniformly random lattice point	Part I's simplicial polytope S_τ .
$F, U, W, v_i, w_i, \mathcal{A}$	the exposed circuit face of a crown and its Radon halves	W also denotes a perfectly matchable set elsewhere in Part II; $F_{i,j}$ is the board recurrence.
$\nu(G)$	maximum matching size	—

Remark 11.1 (Merge note: Part I's input and Part II's input agree). This remark is the merge's own and is not in manuscript 02. Part I's augmented polytope B_τ of (2.1) is the edge polytope $B_{\widehat{H}}$

of Section 14 for $H = G_\tau$, with the same lattice. Proposition 2.2 and the imported identity (14.4) therefore give, for every finite preorder,

$$h_\tau(t) = h^*(B_\tau; t) = p_{G_\tau}(t) :$$

the support polynomial is itself the perfectly-matchable-set polynomial of the doubled graph G_τ . This is a consequence of two cited theorems, not a new result, and it is a different statement from Theorem 13.2, in which p_G of the n -vertex comparability graph is the γ -polynomial of h_{P_G} . As a normalization check, both sides were compared by direct enumeration (ideal inequalities against induced perfect matchings of G_τ) for all 390 labeled preorders with $n \leq 4$; they agree. That check is a scratch computation of the write phase and is not shipped.

11.3 What Part II answers in Section 10

Section 10 of Part I leaves three statements of [1] unsettled. Their status after Part II, as of 28 September 2026:

- **Conjecture 5.3 (real-rootedness of h_τ) is false.** The first counterexample known to this report is Patel’s eight-element poset P_{Θ_3} [11], *not* a discovery of Part II. Part II proves that $h_{P_{\Theta_m}}$ has exactly $2m$ (m even) or $2m - 2$ (m odd) nonreal zeros for every $m \geq 3$ (Theorem 13.3, Corollary 16.2), and transfers the classical 17-element example (Theorem 13.4). Part I never asserted real-rootedness, so nothing in it is retracted.
- **Conjecture 5.2 (γ -positivity) holds for posets of height at most two** (Theorem 13.2: $\gamma_k = |\mathcal{M}_k(G)|$). It remains open for general preorders, including height-two posets whose elements are blown up into nontrivial equivalence classes (Research question 1).
- **Conjecture 5.1(d) (a flag simplicial polytope) remains open.** Part II shows that no lattice triangulation of C_τ or of its boundary is flag when τ contains an induced crown on $2r \geq 6$ elements (Theorem 13.6). A pulling triangulation of ∂C_τ uses only vertices of C_τ , so for such τ the simplicial polytope S_τ of Theorem 1.1 is never flag; the smallest case is Part I’s six-element crown (Section 8). This does not disprove Conjecture 5.1(d), which asks for *some* flag polytope with h -polynomial h_τ .

Part II also reproves, for height two, three statements of Part I by different routes: the identity $h^*(C_{P_G}) = h_{P_G}$ and reflexivity by an orthant decomposition (Theorem 18.2, compare Corollary 4.2 and Proposition 5.2), invariance under reversal (Corollary 15.2, compare Section 7), and $h_1 = n + |E|$ (compare (1.4)). Its hypertree identification in Section 14 is the argument of Lemma A.1 for a general bipartite graph and right hypertrees. These are printed as the manuscript gives them, as second routes.

11.4 Where the merge had to choose

- *Placement.* Part II follows Section 10 and precedes Part I’s appendices, whose letters and numbers are unchanged.
- *Self-citation.* The manuscript cites Part I as an external repository report (its reference “RepoRoot”, with the blob above). Here those citations are replaced by references to Part I’s sections, and the phrases “the ProveIt report” and “the repository’s construction” by “Part I”.
- *File names.* The manuscript’s delivered names (`code/verify.py`, `data/board_certificate.csv`, ...) are replaced by the shipped names in Section 22, because `code/verify.py` in this directory is Part I’s different verifier. Its build commands are replaced by commands that run on a copy.

- *Bibliography.* The manuscript’s entries for Athanasiadis–Chapoton and Kálmán–Postnikov are Part I’s references [1] and [3] (the manuscript adds the DOI 10.1112/plms.12015 for the latter); its other five entries are appended.
- *Labels and front matter.* Labels carry the prefix `htwo:.` Two unlabelled corollaries, the conclusion and the build subsection received labels so that this section can cite them. The manuscript’s title page, abstract, status box and audit box are printed in this section.

12 The research target and the distinction from prior work

Part I, the ProveIt report *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, constructs a canonical root polytope for every finite preorder and gives a proposed proof of simplicial-polytopal realization of its support polynomial (Theorem 1.1). Its final section, Section 10, explicitly leaves flag realizability, γ -positivity, and real-rootedness unresolved. These are different properties, and separating them is essential here.

The underlying questions are Conjectures 5.1(d), 5.2, and 5.3 of Athanasiadis and Chapoton [1]. The first asks for *some* flag simplicial polytope with the prescribed h -vector. The second asks for a nonnegative expansion in the basis $t^k(1+t)^{n-2k}$. The third asserts that all zeros of the support polynomial are real. We focus on posets of height at most two: they are preorders, but unlike arbitrary preorders they admit a direct bipartite-graph description.

The graph polynomial needed for this description is classical. The perfectly matchable set polynomial is studied by Davis and Kohl [7]; its root-polytope realization originates in Ohsugi and Tsuchiya [8]. The nonreal-rooted example used below comes from Stembridge’s naturally labeled, width-two counterexample to the Neggers conjecture [10], as translated into a graph polynomial by Ohsugi and Tsuchiya [8, Example 3.7]. It is important not to confuse that older *width-two* poset with the new *height-two* poset obtained by orienting its bipartite incomparability graph.

12.1 What the article proves

The height-two matching-support identity is the central theorem. It proves the relevant γ -positivity conjecture for an entire class and converts real-root questions into graph enumeration. The principal negative-family result concerns Θ_m : a complete real-root count shows that almost all zeros become nonreal while graph treewidth stays two.

The source audit found an earlier eight-element counterexample posted by Shivam Patel on MathDB [11]. It is exactly P_{Θ_3} . The source labels the posting unverified; our independent finite calculation agrees with it, and the general proof below includes it. Consequently a claim to have first disproved Conjecture 5.3 would be inappropriate. The site’s exact posting date was not available, and its broad historical metadata is not used as evidence. The 17-element construction is retained as an independent bridge to classical literature, not as the smallest or first counterexample.

A second theme concerns a limitation of the canonical root-polytope construction. Every induced crown of size $2r$, $r \geq 3$, forces a minimal nonface of size r in any lattice triangulation. This rules out obtaining a flag complex by choosing a more clever lattice triangulation of that particular polytope. It does *not* rule out another flag polytope with the same h -vector.

12.2 A provenance ledger

Statement	Provenance and status
$h^*(T_G; u) = p_G(u)$ for bipartite G	Established input; Ohsugi–Tsuchiya, reformulated by Davis–Kohl.
Hypertrees count normalized root-polytope volume	Established input from Kálmán–Postnikov; used at $u = 1$.
Height-two formula $h_P = (1 + t)^n p_G(t/(1+t)^2)$	Proved here by a support partition and a counting lemma.
Nonreal-rooted degree-eight graph polynomial	Classical Stembridge–Ohsugi–Tsuchiya example, independently recomputed here.
Eight-element counterexample to AC Conjecture 5.3	Previously posted by Patel; independently checked here, not claimed as new.
Complete root count for P_{Θ_m}	Proved here for every m , extending the eight-element case to a connected treewidth-two family.
17-element counterexample	Independent transfer of the classical graph example; no claim of first disproof or minimality.
Cactus p_G is real-rooted	Already follows from Ohsugi–Tsuchiya [9, Theorem 7.2]; not claimed anew.
Sixth-root matrix and multivariate refinement	Explicit proof supplied here; no absolute novelty claim for the matrix technique.
Induced-crown obstruction in the Provel root model of Part I	Proved here, including a nonregular-triangulation obstruction and a toric degree lower bound.

13 Definitions and main statements

For a finite poset P on a ground set V of size n , let $\mathcal{J}(P)$ be its order ideals, including $\emptyset$ and V . Write $x(S) = \sum_{v \in S} x_v$. Its preorder polytope and support polynomial are

$$\mathcal{Q}(P) = \{x \in \mathbb{R}_{\geq 0}^V : x(I) \leq |I| \text{ for all } I \in \mathcal{J}(P)\}, \quad (13.1)$$

$$h_P(t) = \sum_{x \in \mathcal{Q}(P) \cap \mathbb{Z}^V} t^{|\text{supp}(x)|}. \quad (13.2)$$

The full-set inequality makes the sum finite. Its constant and leading coefficients are both one: the all-zero and all-one vectors are the unique points with support sizes zero and n , respectively.

Here height at most two means that every chain has at most two elements; we consider ordinary posets, not arbitrary equivalence-class blowups of them. Such a poset can be written $P = P_G$ for a bipartite graph $G = (A \sqcup B, E)$: the only strict comparisons are $a < b$ for $ab \in E$. Isolated elements may be assigned to either shore. Put $p = |A|$, $q = |B|$, $n = p + q$, and denote by $N_G(S) \subseteq A$ the neighborhood of $S \subseteq B$.

Definition 13.1. A subset $W \subseteq V(G)$ is *perfectly matchable* if the induced graph $G[W]$ has a perfect matching. The empty set is perfectly matchable. Let

$$\mathcal{M}_k(G) = \{W \subseteq V(G) : |W| = 2k, G[W] \text{ has a perfect matching}\}, \quad p_G(u) = \sum_{k \geq 0} |\mathcal{M}_k(G)| u^k.$$

For comparison, $M_G(u) = \sum_k m_k(G) u^k$ counts individual k -edge matchings. These two polynomials are generally different.

Theorem 13.2 (Height-two matching-support formula). *For every finite bipartite graph G on n vertices,*

$$h_{P_G}(t) = \sum_{k \geq 0} |\mathcal{M}_k(G)| t^k (1+t)^{n-2k} = (1+t)^n p_G\left(\frac{t}{(1+t)^2}\right). \quad (13.3)$$

In particular, every height-at-most-two poset has a γ -positive support polynomial, with $\gamma_k(P_G) = |\mathcal{M}_k(G)|$.

The rational substitution in (13.3) denotes its polynomial expansion; it is not an instruction to evaluate an undefined expression at $t = -1$.

Theorem 13.3 (A complete root-distribution family). *Let Θ_m have lower elements $d, a_1, \dots, a_m$, upper elements $b_0, b_1, \dots, b_m$, and edges $a_i b_0, a_i b_i, d b_i$. For $m \geq 2$,*

$$p_{\Theta_m}(u) = (1+u)^{m-2} (1 + (2m+3)u + (m^2+m+3)u^2 + u^3) - u. \quad (13.4)$$

The support polynomial of P_{Θ_m} is real-rooted exactly when $m \leq 2$ (for $m \geq 1$). For $m \geq 3$, it has precisely two real roots if m is even, and four if m is odd; these real roots are simple and negative. Its graph is connected and has treewidth at most two.

Theorem 13.4 (An explicit negative answer to real-rootedness). *Let G_0 have shores $A = \{a_1, \dots, a_9\}$ and $B = \{b_1, \dots, b_8\}$, with neighborhoods*

$$\begin{array}{c|cccccccc} j & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline N(b_j) & [1, 1] & [1, 2] & [1, 3] & [1, 4] & [1, 5] & [2, 6] & [3, 8] & [4, 9] \end{array} \quad (13.5)$$

where $[l, r]$ means $\{a_l, a_{l+1}, \dots, a_r\}$. Then the height-two poset P_{G_0} is a counterexample to Conjecture 5.3 of [1]. Its γ -polynomial is

$$\begin{aligned} p_0(u) = & 1 + 32u + 336u^2 + 1420u^3 + 2534u^4 \\ & + 1946u^5 + 658u^6 + 86u^7 + 3u^8. \end{aligned} \quad (13.6)$$

Its support polynomial is γ -positive despite having nonreal zeros.

Theorem 13.5 (Cactus matrix representation). *For every bipartite cactus graph $G = (A \sqcup B, E)$ there is an explicitly constructible complex matrix K , supported exactly on E , such that*

$$p_G(u) = \det(I_{|A|} + uKK^*). \quad (13.7)$$

The nonzero entries of K are sixth roots of unity, and each square minor has squared absolute value either zero or one, according as its support has no perfect matching or has one. Consequently h_{P_G} is real-rooted. A multivariate stable determinant and a vertex-deletion interlacing theorem are given in Section 19.

Theorem 13.6 (Induced-crown obstruction). *Let P be a finite preorder containing an induced height-two crown on $2r$ distinct elements, $r \geq 3$. The canonical root polytope*

$$C_P = \text{conv}(\{\pm \mathbf{e}_v : v \in P\} \cup \{\mathbf{e}_b - \mathbf{e}_a : a \neq b, a \leq_P b\})$$

uses all nonzero roots for distinct comparable elements; for equivalent elements both directions are included. Every lattice triangulation of C_P or of its boundary has a minimal nonface of cardinality r . The toric ideal on $C_P \cap \mathbb{Z}^n$ has an indispensable binomial of degree r , and every initial ideal has a minimal generator of degree r .

For preorders, all distinct pairs $a \leq_P b$ are used, including both directions within an equivalence class.

14 A matching–polymatroid counting lemma

For a bipartite graph $H = (X \sqcup Y, E)$, define the finite set

$$\mathcal{R}_H = \{c \in \mathbb{Z}_{\geq 0}^Y : c(S) \leq |N_H(S)| \text{ for every } S \subseteq Y\}. \quad (14.1)$$

This set describes nonnegative integer demands that can be supplied by unit-capacity vertices in X , with adjacency specifying allowable supply.

Lemma 14.1 (Counting demands by matching supports). *For every finite bipartite graph H ,*

$$|\mathcal{R}_H| = p_H(1) = \#\{W \subseteq V(H) : H[W] \text{ has a perfect matching}\}. \quad (14.2)$$

We give the reduction to two established root-polytope statements in full. This avoids treating (14.2) as an experimental identity or as an unexplained application of a lattice-point package.

14.1 The two classical inputs

Adjoin x_0 and y_0 to the shores of H , joining x_0 to every old right vertex, y_0 to every old left vertex, and x_0 to y_0 . Write $\widehat{H}$ for this connected augmented bipartite graph. Its edge polytope is

$$B_{\widehat{H}} = \text{conv}\{\mathbf{e}_x + \mathbf{f}_y : xy \in E(\widehat{H})\}.$$

A *right hypertree* is a vector $(\deg_T(y) - 1)_{y \in Y \cup \{y_0\}}$ realized by a spanning tree T of $\widehat{H}$. Multiple spanning trees with the same vector count only once.

We use the following published facts, with the full affine integer lattice on the edge polytope:

$$\text{Vol}_{\mathbb{Z}}(B_{\widehat{H}}) = \#\{\text{right hypertrees of } \widehat{H}\}, \quad (14.3)$$

$$h^*(B_{\widehat{H}}; u) = p_H(u). \quad (14.4)$$

The first is the evaluation at $u = 1$ of the interior-polynomial/root-polytope theorem of Kálmán and Postnikov [3]. The second is Ohsugi–Tsuchiya’s matching-support interpretation, stated explicitly in Davis–Kohl [7, Theorem 3.10]. Recall that $h^*(B; 1) = \text{Vol}_{\mathbb{Z}}(B)$. No result about preorder real-rootedness is an input to either statement.

14.2 Explicit identification of the hypertrees

Let $s = |X|$, $r = |Y|$. A spanning tree of $\widehat{H}$ has $s + r + 1$ edges. Its right degree-minus-one vector has total s . Write it as $(c_0, (c_y)_{y \in Y})$. For nonempty $S \subseteq Y$, the tree edges incident with S form a forest on $S \cup N_{\widehat{H}}(S)$. Their number is $|S| + c(S)$, so

$$|S| + c(S) \leq |S| + |N_{\widehat{H}}(S)| - 1 = |S| + |N_H(S)|.$$

Thus $c \in \mathcal{R}_H$, and $c_0 = s - c(Y)$.

Conversely, take $c \in \mathcal{R}_H$ and replace each $y \in Y$ by c_y clones, each with neighborhood $N_H(y) \subseteq X$. A set of clones with labels in S has at most $c(S) \leq |N_H(S)|$ members. Hall’s criterion therefore matches all clones to distinct vertices of X . One can prove this form of Hall’s criterion directly by augmenting paths: failure to match a clone produces an alternating-reachable set of clones with fewer neighbors than members, contradicting the stated inequality.

Start a spanning tree with the star joining x_0 to every vertex of $Y \cup \{y_0\}$. Attach each old $x \in X$ as a leaf to the label of its matched clone, or to y_0 if no clone was assigned to it. The resulting graph is a tree and has right degree-minus-one vector

$$(s - c(Y), (c_y)_{y \in Y}).$$

Hence hypertree vectors correspond bijectively to $\mathcal{R}_H$. Combining this bijection with (14.3) and (14.4) proves Lemma 14.1. This argument also handles an empty shore: $\mathcal{R}_H$ then contains only the zero demand and $p_H = 1$.

Remark 14.2. The lemma equates two cardinalities; it does not provide a canonical bijection between demand vectors and perfectly matchable vertex sets. The tree construction is a bijection with *hypertree vectors*, not with their realizing trees. Counting matchings or spanning trees with multiplicity at this step would invalidate the main formula.

15 The height-two formula and its consequences

15.1 Partitioning lattice points by zeros and support

Write a lattice point of $\mathcal{Q}(P_G)$ as (a, b) on the two shores. Each singleton in A is an ideal, so $a_i \in \{0, 1\}$. Let

$$Z = \{i \in A : a_i = 0\}, \quad S = \{j \in B : b_j > 0\}.$$

An ideal is $U \sqcup T$ with $T \subseteq B$ and $N_G(T) \subseteq U \subseteq A$. Its inequality is

$$b(T) \leq |T| + |U \cap Z|.$$

For fixed T , the strongest choice is $U = N_G(T)$. Thus all ideal inequalities are equivalent to

$$b(T) \leq |T| + |N_G(T) \cap Z| \quad (T \subseteq B). \quad (15.1)$$

On S put $b_j = 1 + c_j$, with $c_j \geq 0$, and put $b_j = 0$ elsewhere. For $T \subseteq S$, equation (15.1) becomes

$$c(T) \leq |N_G(T) \cap Z|.$$

These inequalities also imply (15.1) for arbitrary $T \subseteq B$, by applying them to $T \cap S$. Therefore the possible c are exactly $\mathcal{R}_{G[Z \sqcup S]}$.

Every point in this part has support size $p - |Z| + |S|$. Lemma 14.1 gives

$$h_{P_G}(t) = \sum_{Z \subseteq A} \sum_{S \subseteq B} t^{p-|Z|+|S|} \sum_{W \in \mathcal{M}(G[Z \sqcup S])} 1, \quad (15.2)$$

where $\mathcal{M}(H)$ denotes all perfectly matchable vertex sets of H .

15.2 The Boolean summation

Fix a perfectly matchable set $W = X \sqcup Y$ of G , with $X \subseteq A$, $Y \subseteq B$, and $|X| = |Y| = k$. It is counted in (15.2) exactly when $X \subseteq Z$ and $Y \subseteq S$. Its total contribution is

$$\begin{aligned} \left(\sum_{Z \supseteq X} t^{p-|Z|} \right) \left(\sum_{S \supseteq Y} t^{|S|} \right) &= (1+t)^{p-k} t^k (1+t)^{q-k} \\ &= t^k (1+t)^{n-2k}. \end{aligned}$$

Summing over W proves Theorem 13.2. In particular, the γ -positivity conclusion is not deduced from an assumed flag complex; it follows from an explicit nonnegative counting interpretation.

15.3 Roots, matching number, and small coefficients

Let $\nu(G)$ be the maximum matching size. Then $\deg p_G = \nu(G)$, and all its coefficients from degree zero through $\nu(G)$ are positive: a maximum matching has submatchings of every smaller size.

Proposition 15.1 (Exact transfer of real-rootedness). *For a bipartite graph G , the polynomial h_{p_G} is real-rooted if and only if p_G is real-rooted. If p_G has only real roots, they are strictly negative. In all cases, the root -1 of h_{p_G} has multiplicity $n - 2\nu(G)$. In the real-rooted case, its other roots occur in reciprocal negative pairs.*

Proof. Write $p_G(u) = \prod_{j=1}^{\nu} (1 + \lambda_j u)$ over $\mathbb{C}$, with $\lambda_j \neq 0$. Formula (13.3) gives

$$h_{p_G}(t) = (1+t)^{n-2\nu} \prod_{j=1}^{\nu} ((1+t)^2 + \lambda_j t). \quad (15.3)$$

If p_G is real-rooted, its positive coefficients and constant term one exclude nonnegative roots. Hence every $\lambda_j > 0$, and each quadratic $t^2 + (2 + \lambda_j)t + 1$ has two distinct negative reciprocal roots. Its value at -1 is $-\lambda_j \neq 0$, proving the stated multiplicity.

Conversely, if u_0 is a nonreal zero of p_G , then $u_0(1+t)^2 - t = 0$ has two roots different from -1 . Neither is real, since $t/(1+t)^2$ is real for real $t \neq -1$. Both are zeros of h_{p_G} . Thus real-rootedness of h_{p_G} forces that of p_G . $\square$

Corollary 15.2. *The height-two polynomial is unchanged by reversing the poset, and it is multiplicative under disjoint unions. Its first coefficients satisfy*

$$\gamma_0 = 1, \quad \gamma_1 = |E|, \quad \gamma_2 = \binom{|E|}{2} - \sum_{v \in V} \binom{\deg(v)}{2} - c_4(G), \quad h_1 = n + |E|,$$

where $c_4(G)$ is the number of four-cycles, each counted once.

Proof. Reversing the order exchanges the shores of G , leaving p_G unchanged. Perfectly matchable subsets split independently across components. For γ_2 , subtract adjacent pairs of edges from all edge pairs to obtain $m_2(G)$. A four-vertex bipartite subgraph has two perfect matchings exactly when it is a four-cycle; otherwise it has at most one. Thus $\gamma_2 = m_2 - c_4$. Finally, the coefficient of t in (13.3) is $n + \gamma_1$. $\square$

For a forest, every perfectly matchable induced subgraph has a unique perfect matching: the symmetric difference of two perfect matchings would contain a cycle. Hence $p_G = M_G$. More generally, $|\mathcal{M}_k(G)| \leq m_k(G)$, and equality for all k holds for a bipartite graph exactly when it is a forest. An even cycle supplies two distinct matchings with the same covered set, proving the converse.

16 A complete nonreal-root family at treewidth two

This section strengthens a single counterexample to a uniform theorem with an exact count of real roots. The mechanism is distinct from taking disjoint unions: every graph in the family is connected, and its treewidth is bounded.

16.1 Parallel paths and the matching-support polynomial

Let Θ_m be the union of the m paths

$$d - b_i - a_i - b_0 \quad (1 \leq i \leq m),$$

whose only common vertices are the endpoints d, b_0 . Its bipartition is $A = \{d, a_1, \dots, a_m\}$ and $B = \{b_0, b_1, \dots, b_m\}$. For $m = 1$ this is a four-vertex path; for $m = 2$ it is a six-cycle. For $m = 3$ the corresponding height-two poset is the previously posted counterexample of Patel [11].

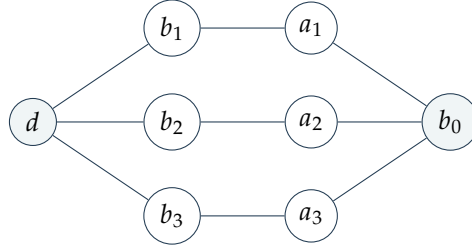


Figure 1: The graph Θ_3 . To obtain its height-two poset, orient every edge from the shore $\{d, a_1, a_2, a_3\}$ to the other shore. The horizontal layout shows paths, not increasing order in the poset.

Proposition 16.1. *For $m \geq 2$, the perfectly matchable set polynomial of Θ_m is (13.4). For $m = 1$, it is $1 + 3u + u^2$.*

Proof. Partition matchable vertex sets according to the presence of d, b_0 . If neither endpoint is present, each branch contributes either its two internal vertices or neither; the contribution is $(1 + u)^m$.

If d alone is present, it must be matched to a unique included vertex b_i whose partner a_i is absent. Every other included branch is a complete internal pair. There are m choices of i , giving $mu(1 + u)^{m-1}$. The case with b_0 alone has the same contribution.

Suppose both endpoints are present. If some branch contributes just b_i , then d must match it. There must be exactly one different branch contributing just a_j , to be matched to b_0 ; all other branches are empty or complete internal pairs. This gives $m(m - 1)u^2(1 + u)^{m-2}$. In the remaining case all selected branches are complete pairs. At least one such branch is needed to match the endpoints, and the selected vertex set is counted once, regardless of how many of its branches could be used for this purpose. This gives $u((1 + u)^m - 1)$.

Thus the sum is

$$\begin{aligned} & (1 + u)^m + 2mu(1 + u)^{m-1} + m(m - 1)u^2(1 + u)^{m-2} + u((1 + u)^m - 1) \\ &= (1 + u)^{m-2}(1 + (2m + 3)u + (m^2 + m + 3)u^2 + u^3) - u. \end{aligned}$$

The four-vertex path is counted directly. Notice that replacing matchable sets by matchings would overcount the case where both endpoints and several complete branches are present; that distinction is essential to the formula. $\square$

16.2 A positive interval with no roots

Write $p_m = p_{\Theta_m}$. For $m \geq 3$, put $k = m - 2 \geq 1$. A useful rearrangement, valid for $0 < x < 1$, is

$$\begin{aligned} p_m(-x) &= (1 - x)^k(1 - (m + 1)x)^2 \\ &\quad + x[1 - (1 - x)^k(1 + kx + x^2)]. \end{aligned} \tag{16.1}$$

The second line is strictly positive. Indeed, the binomial series gives

$$(1-x)^{-k} = 1 + kx + \frac{k(k+1)}{2}x^2 + \sum_{j \geq 3} \binom{k+j-1}{j} x^j > 1 + kx + x^2.$$

All terms are real and positive on this interval; the inequality holds also for $k = 1$ because the tail is then nonzero. Moreover, $p_m(0) = p_m(-1) = 1$. We have proved the particularly strong exclusion

$$p_m(u) > 0 \quad (-1 \leq u \leq 0, \quad m \geq 3). \quad (16.2)$$

The coefficients of p_m are nonnegative by their interpretation as counts, so there are no nonnegative real roots either.

Already this disproves real-rootedness: p_m is monic with constant term one. If all its roots were real, (16.2) would put them strictly below -1 , making the absolute value of their product larger than one. We next obtain the exact count, rather than just this contradiction.

16.3 Exact root counts by two sign changes

Set $u = -1 - y$ with $y > 0$ and define

$$R_m(y) = m(m-1) + 2m^2y + m(m+1)y^2 - y^3.$$

An exact change of variables gives

$$p_m(-1-y) = 1 + y + (-1)^{m-2}y^{m-2}R_m(y). \quad (16.3)$$

We use Descartes' elementary rule of signs: the number of positive roots, counted with multiplicity, is at most the number of changes in the sequence of nonzero coefficients. One way to prove the needed bound is to factor out positive linear factors successively; multiplying a real polynomial by $y - a$, $a > 0$, increases the number of sign changes by a positive odd integer. Factoring over $\mathbb{R}$ then yields the bound.

If m is even, $m \geq 4$, the polynomial on the right of (16.3) has positive coefficients except for its leading coefficient, which is negative. It has one sign change, is positive at $y = 0$, and tends to $-\infty$. It consequently has exactly one positive root, and that root is simple.

If m is odd, $m \geq 3$, its coefficients have signs

$$+, \quad \underbrace{- , \dots , -}_{\text{nonzero middle block}}, \quad +$$

when zero coefficients are omitted; for $m > 3$ there is an additional initial positive coefficient, which does not change the count of two sign changes. For $m = 3$, the merged coefficient of y is $1 - m(m-1) < 0$. At $y = 0$ the value is one, at $y = 1$ it is

$$2 - (4m^2 - 1) = 3 - 4m^2 < 0,$$

and it tends to $+\infty$ as $y \rightarrow \infty$. There is a root in $(0, 1)$ and another in $(1, \infty)$. The sign bound makes these the only positive roots, both simple.

By (16.2), these exhaust the real roots of p_m . It has degree $m+1$, so it has m nonreal roots when m is even and $m-1$ when m is odd, counted with multiplicity. The substitution in Proposition 15.1 sends each negative root to two distinct negative reciprocal roots of $h_{p_{\ominus m}}$ and each nonreal root to two nonreal roots. There is no root at -1 , because $n = 2m + 2 = 2 \deg p_m$.

This proves the root count in Theorem 13.3. The cases $m = 1, 2$ are a path and a cycle, respectively, and are real-rooted by the independent cactus proof in Section 19.

For completeness, a width-two tree decomposition is explicit. Use bags $\{d, b_0, a_i\}$ and $\{d, a_i, b_i\}$ for each i , connect the first bags in a path, and attach each second bag to its corresponding first bag. Every edge is covered; bags containing any fixed vertex form a connected subtree. All bags have size three. Thus the asserted treewidth bound requires no general graph-minor theorem.

Corollary 16.2 (Most roots nonreal in a bounded-treewidth family). *The connected height-two posets P_{Θ_m} satisfy*

$$\#\{\text{nonreal zeros of } h_{P_{\Theta_m}}\} = \begin{cases} 2m, & m \text{ even}, \\ 2m - 2, & m \text{ odd}, \end{cases} \quad (m \geq 3),$$

with multiplicities. Their proportion among all $2m + 2$ zeros tends to one. Thus bounded treewidth, even together with height two and connectedness, does not force real-rootedness or even a positive limiting proportion of real roots.

16.4 The eight-element specialization and its status

The specialization $m = 3$ gives

$$p_3(u) = 1 + 9u + 24u^2 + 16u^3 + u^4,$$

$$h_{P_{\Theta_3}}(t) = 1 + 17t + 106t^2 + 303t^3 + 427t^4 + 303t^5 + 106t^6 + 17t^7 + t^8.$$

These are the polynomials in Patel's posting [11], not newly claimed examples. Our direct enumeration from the ideal inequalities finds 1281 lattice points and reproduces every coefficient. Exact arithmetic also verifies the posted Laguerre obstruction

$$(p'_3(-1/4))^2 - p_3(-1/4)p''_3(-1/4) = -95/1024 < 0.$$

The additional result here is the all- m formula and complete root count, not a priority claim for this specialization.

m	Poset size	Real roots of p_m	Real roots of h	Nonreal roots of h
3	8	2	4	4
4	10	1	2	8
5	12	2	4	8
6	14	1	2	12
7	16	2	4	12
8	18	1	2	16

17 The 17-element counterexample in exact arithmetic

17.1 A complete graph specification

Use the graph G_0 of (13.5); there are no other edges. Orient each edge from a_i to b_j to define $P_0 = P_{G_0}$. Since all strict comparisons go from A to B , the relation is automatically transitive and antisymmetric, and has height two. There are $1 + 2 + 3 + 4 + 5 + 5 + 6 + 6 = 32$ strict comparisons.

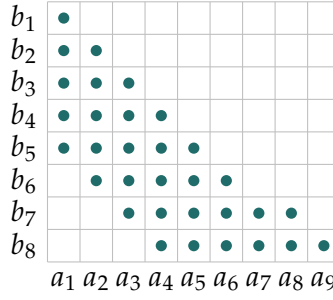


Figure 2: The complete 8×9 incidence board of the counterexample. A dot means $a_i < b_j$. Increasing pairs of dots give the canonical matching of any matchable support. The diagram is drawn from the exact interval list, not inferred from a numerical polynomial.

17.2 Why an increasing matching suffices

Lemma 17.1 (Uncrossing on a monotone interval board). *Suppose $N(b_j) = [\ell_j, r_j]$ with both (ℓ_j) and (r_j) nondecreasing. For sorted subsets $X = \{a_{i_1}, \dots, a_{i_k}\}$ and $Y = \{b_{j_1}, \dots, b_{j_k}\}$, the induced graph on $X \sqcup Y$ has a perfect matching if and only if every $a_{i_s} b_{j_s}$ is an edge.*

Proof. The backward implication is immediate. In a matching with an inversion, suppose $j < j'$ and $i < i'$ but the matching uses $a_i b_{j'}$ and $a_{i'} b_j$. Then

$$\ell_j \leq \ell_{j'} \leq i < i' \leq r_j \leq r_{j'}.$$

Therefore $a_i b_j$ and $a_{i'} b_{j'}$ are also edges. Swap the inverted pair. Choosing an adjacent inversion in the permutation of matched row indices reduces the inversion number; iteration gives the unique sorted matching. Thus existence is equivalent to the sorted-edge test. $\square$

Let $F_{i,j}(u)$ count increasing matched pairs in the first i vertices of A and first j vertices of B , by their number of pairs. With $\varepsilon_{i,j} = 1$ for an edge and zero otherwise,

$$\begin{aligned} F_{i,0}(u) &= F_{0,j}(u) = 1, \\ F_{i,j}(u) &= F_{i-1,j}(u) + F_{i,j-1}(u) + (u\varepsilon_{i,j} - 1)F_{i-1,j-1}(u). \end{aligned} \tag{17.1}$$

Indeed, those objects omitting the last row or the last column are counted by inclusion–exclusion. An object using both must pair them with each other and leave an arbitrary increasing matching on the smaller board. Lemma 17.1 implies $F_{9,8} = p_{G_0}$.

For a convenient hand-checkable summary, the last row of this polynomial recurrence is as follows. The complete 90-cell table, including boundary cells, is in `data/02-height-two-board_certificate.csv`.

j	$F_{9,j}(u)$
0	1
1	$1 + u$
2	$1 + 3u + u^2$
3	$1 + 6u + 6u^2 + u^3$
4	$1 + 10u + 20u^2 + 10u^3 + u^4$
5	$1 + 15u + 50u^2 + 50u^3 + 15u^4 + u^5$
6	$1 + 20u + 105u^2 + 175u^3 + 105u^4 + 21u^5 + u^6$
7	$1 + 26u + 205u^2 + 595u^3 + 665u^4 + 301u^5 + 49u^6 + 2u^7$
8	$1 + 32u + 336u^2 + 1420u^3 + 2534u^4 + 1946u^5 + 658u^6 + 86u^7 + 3u^8$

Thus the coefficient formula (13.6) is an exact finite recurrence calculation, not a fit to sampled values. There are $p_0(1) = 7016$ perfectly matchable sets.

17.3 An elementary obstruction to real-rootedness

Lemma 17.2 (Laguerre inequality). *If a nonzero real polynomial f has only real zeros, then*

$$(f'(x))^2 - f(x)f''(x) \geq 0 \quad (x \in \mathbb{R}). \quad (17.2)$$

Proof. Write $f(x) = c \prod_i (x - r_i)$ with $r_i \in \mathbb{R}$, repeated according to multiplicity. Away from the roots,

$$\frac{(f')^2 - ff''}{f^2} = -\left(\frac{f'}{f}\right)' = \sum_i \frac{1}{(x - r_i)^2} \geq 0.$$

Continuity extends the inequality to the roots. $\square$

Direct substitution into (13.6) gives

$$p_0(-2) = 65, \quad p'_0(-2) = -560, \quad p''_0(-2) = 4912. \quad (17.3)$$

Consequently

$$(p'_0(-2))^2 - p_0(-2)p''_0(-2) = 313600 - 319280 = -5680 < 0. \quad (17.4)$$

By Lemma 17.2, p_0 has nonreal roots. Proposition 15.1 proves Theorem 13.4. This proof requires no isolation of a complex root and no trust in a floating-point solver.

Expanding the height-two formula gives

$$\begin{aligned} h_{p_0}(t) = & 1 + 49t + 952t^2 + 9828t^3 + 61302t^4 + 248816t^5 \\ & + 688516t^6 + 1337738t^7 + 1857081t^8 + 1857081t^9 \\ & + 1337738t^{10} + 688516t^{11} + 248816t^{12} + 61302t^{13} \\ & + 9828t^{14} + 952t^{15} + 49t^{16} + t^{17}. \end{aligned} \quad (17.5)$$

Its coefficient sum is 8 408 566. An independent C++ enumeration of *all* these lattice points, based only on integer capacity allocations, reproduces (17.5); see Section 22.

For orientation only, a nonreal pair of p_0 is approximately $-1.8588440 \pm 0.1497675 i$. Its images yield two nonreal reciprocal pairs of h_{p_0} , approximately

$$-2.0461493 \pm 0.0565649 i, \quad -0.4883497 \pm 0.0135002 i.$$

The exact negative Laguerre value, rather than these approximations, is the certificate used in the proof. The accompanying exact computation does not need numerical roots.

17.4 What is old, what is transferred, and what is not settled

Polynomial (13.6) is the normalized descent polynomial in the older Stembridge example, and its interpretation as a bipartite matching-support polynomial is recorded in [8, Example 3.7]. We independently recompute it from the interval board so that the counterexample is specified without relying on a figure or a prior decimal root calculation.

For an explicit comparison, the older poset has chains $1 < 3 < \dots < 17$ and $2 < 4 < \dots < 16$, together with the relations

$$2 < 3, 4 < 5, 6 < 7, 8 < 9, 10 < 11, 12 < 13, 14 < 17, \quad 1 < 12, 3 < 14, 5 < 16.$$

Its cross-chain incomparable pairs have precisely the neighborhoods in (13.5), under $a_i = 2i - 1$, $b_j = 2j$. In the present counterexample these incomparable pairs are made into *new*

comparabilities, all oriented from A to B ; the old chain comparisons are discarded. The two posets are therefore not the same object.

This transfer supplies an independent negative answer to the assertion about the support polynomial of *every preorder*; the existence of such counterexamples was already claimed in the smaller example [11]. It does not contradict γ -positivity, palindromic unimodality, or simplicial-polytopal realizability. The 17-element construction is not smallest: the preceding family already gives an eight-element example. Adding isolated elements multiplies the support polynomial by $(1 + t)$, so counterexamples exist in every size $n \geq 17$; this operation alone says nothing about connected counterexamples of every size.

18 The canonical root polytope by orthant decomposition

Part I's construction (1.2) has a particularly concrete form in height two. Negate the coordinates in the lower shore A . This unimodular map sends the roots $\mathbf{e}_b - \mathbf{e}_a$ to $\mathbf{e}_b + \mathbf{e}_a$ and preserves the collection of signed coordinate vectors. Thus C_{P_G} is lattice equivalent to

$$D_G = \text{conv}(\{\pm \mathbf{e}_v : v \in V\} \cup \{\mathbf{e}_a + \mathbf{e}_b : ab \in E\}). \quad (18.1)$$

Only the positive sums are added; the polytope is not generally centrally symmetric. Also put

$$T_G = \text{conv}(\{0\} \cup \{\mathbf{e}_v : v \in V\} \cup \{\mathbf{e}_a + \mathbf{e}_b : ab \in E\}). \quad (18.2)$$

The full coordinate simplex is present, so both polytopes have dimension $n = |V|$, with the zero-dimensional empty case understood separately.

18.1 An integral gauge from a capacitated matching problem

For $y \in \mathbb{R}_{\geq 0}^V$, let $g_T(y)$ be the least $s \geq 0$ for which $y \in sT_G$. Write $I(G)$ for the independent vertex sets of G .

Lemma 18.1. *For every bipartite graph G and nonnegative y ,*

$$g_T(y) = \max_{I \in I(G)} y(I). \quad (18.3)$$

Consequently g_T is coordinatewise monotone and takes integer values on nonnegative integer vectors.

Proof. Represent y as a nonnegative combination of coordinate vectors and edge-sum vectors. If an edge receives weight w_{ab} , it supplies that amount at each endpoint, reducing the total coefficient mass by w_{ab} compared with using only coordinate vectors. The minimum total mass is therefore

$$|y| - \max \left\{ \sum_{ab \in E} w_{ab} : w_{ab} \geq 0, \sum_{e \ni v} w_e \leq y_v \right\}.$$

The maximum is the value of a flow from a source through A , then B , to a sink, with vertex capacities y_a, y_b and across-edge capacities larger than $|y|$. A minimum cut never cuts an across-edge arc. Such cuts correspond exactly to vertex covers C , with capacity $y(C)$: the cover consists of lower-shore vertices on the sink side and upper-shore vertices on the source side. Max-flow/min-cut gives the maximum as $\min_C y(C)$. Complements of vertex covers are independent sets, proving (18.3). This argument is valid for real capacities; compactness and the usual residual-path proof give max-flow/min-cut in this finite network. The two final conclusions follow directly from the displayed maximum over independent sets. $\square$

For $x \in \mathbb{R}^V$, let $x_v^+ = \max(x_v, 0)$ and $x_v^- = \max(-x_v, 0)$. Since $D_G = \text{conv}(T_G, \{-\mathbf{e}_v\})$, its gauge is

$$g_D(x) = g_T(x^+) + |x^-|. \quad (18.4)$$

Indeed, any representation $x = y - z$ with $y, z \geq 0$ has coefficient mass at least $g_T(y) + |z|$. Necessarily $y \geq x^+$ and $z \geq x^-$, so the lower bound in (18.4) follows from monotonicity. Taking $y = x^+$ and $z = x^-$ achieves it.

This also proves reflexivity without a triangulation. The gauge can be written as the finite maximum of integer linear forms

$$g_D(x) = \max_{I \in \mathcal{I}(G), R \subseteq V \setminus I} \left(\sum_{v \in I} x_v - \sum_{v \in R} x_v \right). \quad (18.5)$$

Every form is bounded by the right side of (18.4); equality is obtained by taking a maximizing independent subset of the positive coordinates and all negative coordinates in R . Thus D_G is cut out by integer inequalities with right side one. The origin is interior because D_G contains the crosspolytope. Its polar is the convex hull of the integer normals in (18.5), hence a lattice polytope.

18.2 Counting the lattice points in every orthant

The augmented edge polytope $B_{\widehat{G}}$ from Section 14 is lattice equivalent to T_G . Project away the two added coordinates. The inverse recovers them as

$$x_0 = 1 - \sum_{a \in A} x_a, \quad y_0 = 1 - \sum_{b \in B} y_b$$

on the affine span of $B_{\widehat{G}}$. These formulas preserve the full affine integer lattice. The projected vertices are exactly 0, the coordinate vectors, and the edge sums. Therefore the established identity (14.4) gives

$$h^*(T_G; t) = p_G(t). \quad (18.6)$$

For an integral gauge g of a bounded polytope containing the origin,

$$\text{Ehr}(t) = \frac{1}{1-t} \sum_{x \in \mathbb{Z}^n} t^{g(x)};$$

a lattice point of least dilation $g(x)$ occurs in every subsequent integer dilation. For T_G , the sum is over nonnegative lattice points. All series here are formal series with finitely many contributions to each coefficient.

Partition the lattice points x of $\mathbb{R}^V$ by the set R of strictly negative coordinates, and put $S = V \setminus R$. The negative coordinates contribute $(t/(1-t))^{|R|}$. The nonnegative coordinates contribute $(1-t) \text{Ehr}_{T_{G[S]}}(t)$, since independent sets restricted to S give precisely the gauge of the induced graph. Using (18.4),

$$\begin{aligned} \text{Ehr}_{D_G}(t) &= \frac{1}{1-t} \sum_{R \subseteq V} \left(\frac{t}{1-t} \right)^{|R|} (1-t) \text{Ehr}_{T_{G[V \setminus R]}}(t) \\ &= \frac{\sum_{R \subseteq V} t^{|R|} p_{G[V \setminus R]}(t)}{(1-t)^{n+1}}. \end{aligned} \quad (18.7)$$

In the numerator, a fixed matchable set W of size $2k$ occurs once for each $R \subseteq V \setminus W$. Its contribution is $t^k(1+t)^{n-2k}$. Hence:

Theorem 18.2 (Height-two root identity). *For every bipartite graph G , D_G is reflexive and*

$$h^*(C_{P_G}; t) = h^*(D_G; t) = \sum_{k \geq 0} |\mathcal{M}_k(G)| t^k (1+t)^{n-2k} = h_{P_G}(t). \quad (18.8)$$

This is a direct height-two proof of the corresponding identity in Part I (Corollary 4.2), independent of its general preorder fiber argument.

Remark 18.3 (A normalization trap). The symmetric edge polytope of type B in Ohsugi–Tsuchiya includes all signed edge sums. Its γ -polynomial is $p_G(4u)$ [8, 9]. Our D_G has only the positive edge sums, and its γ -polynomial is $p_G(u)$. These are different polytopes. The quick check is $h_1(P_G) = n + |E|$, not $n + 4|E|$.

19 Cactus graphs: matrices, stability, and interlacing

A cactus graph is one in which every edge belongs to at most one simple cycle. Distinct cycles may meet in a vertex, but not in an edge; isolated vertices and disconnected components are allowed. In the bipartite case every cycle has even length. Univariate real-rootedness of p_G for these graphs is already contained in Ohsugi–Tsuchiya [9, Theorem 7.2]. The purpose here is a concrete matrix proof that retains a multivariate and weighted structure.

19.1 A unit-modulus certificate for every matchable minor

Let $\zeta = e^{\pi i/3}$, so $\zeta^2 - \zeta + 1 = 0$ and $|1 - \zeta| = |1 - \bar{\zeta}| = 1$. Form an A -by- B matrix K supported on the edges of G . Initially every edge entry is one. For each cycle of length $2r$, choose one of its edges and replace that entry by $(-1)^r \zeta$. These choices do not conflict because cycles are edge-disjoint. All nonzero entries are sixth roots of unity.

Lemma 19.1 (The minor identity). *For $X \subseteq A$ and $Y \subseteq B$ with $|X| = |Y|$,*

$$|\det K_{X,Y}|^2 = \begin{cases} 1, & G[X \cup Y] \text{ has a perfect matching,} \\ 0, & \text{otherwise.} \end{cases} \quad (19.1)$$

Proof. If there is no perfect matching, every term in the determinant is zero. Otherwise fix a perfect matching M_0 of the induced subgraph. The symmetric difference of M_0 with another perfect matching is a disjoint union of M_0 -alternating cycles. In a cactus, all M_0 -alternating cycles are vertex-disjoint: if two shared a vertex, the unique matching edge there would have to belong to both cycles, contrary to edge-disjointness. Conversely, flipping any chosen subcollection of these alternating cycles gives a perfect matching. Thus all determinant terms are indexed by independent choices of cycle flips.

For an alternating cycle of length $2r$, the ratio of the products of edge entries in its two matching halves is $\Gamma = (-1)^r \zeta$ or $(-1)^r \bar{\zeta}$. There is exactly one modified entry on that cycle; modifications on other cycles are not on its edges. The permutation sign changes by $(-1)^{r-1}$ when the cycle is flipped. Factoring out the unit-modulus term for M_0 , the determinant therefore factors as

$$\text{a complex unit} \prod_{\text{alternating cycles}} (1 + (-1)^{r-1} \Gamma).$$

Each factor is $1 - \zeta$ or $1 - \bar{\zeta}$, hence has absolute value one. The determinant has absolute value one, proving the claim. $\square$

Theorem 19.2 (Determinant and spectral factorization). *Let ν be the maximum matching size of a bipartite cactus graph, and let $\lambda_1, \dots, \lambda_\nu$ be the positive eigenvalues of KK^* , with multiplicities. Then*

$$p_G(u) = \det(I + uKK^*) = \prod_{j=1}^{\nu} (1 + \lambda_j u), \quad (19.2)$$

$$h_{P_G}(t) = (1+t)^{n-2\nu} \prod_{j=1}^{\nu} ((1+t)^2 + \lambda_j t). \quad (19.3)$$

In particular, all roots of p_G and h_{P_G} are real and negative.

Proof. The principal-minor expansion and Cauchy–Binet give

$$\det(I + uKK^*) = \sum_{k \geq 0} u^k \sum_{\substack{X \subseteq A, Y \subseteq B \\ |X|=|Y|=k}} |\det K_{X,Y}|^2.$$

Lemma 19.1 turns its coefficients into $|\mathcal{M}_k(G)|$. The largest nonzero minor has size ν , so $\text{rank } K = \nu$. The Hermitian positive-semidefinite matrix KK^* has exactly ν positive eigenvalues, proving (19.2). Substitute in Theorem 13.2 to get (19.3). Each quadratic factor has discriminant $\lambda_j(\lambda_j + 4) > 0$ and two negative roots. $\square$

The distinction between a cactus and a general series-parallel graph is now sharp in a useful sense: the connected graphs Θ_m still have treewidth two, but for $m \geq 3$ their cycles overlap along paths and their support polynomials fail real-rootedness dramatically.

19.2 A stable multivariate refinement

Let

$$H_G = \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}.$$

This is an n -by- n Hermitian matrix indexed by all vertices. For a vertex set W , its principal minor is zero if the two shores have unequal sizes. When $|W \cap A| = |W \cap B| = k$, it equals $(-1)^k |\det K_{W \cap A, W \cap B}|^2$. Expanding a determinant along its diagonal entries therefore proves

$$\Phi_G(\mathbf{z}) := \det(\text{diag}(z_v) - H_G) = \sum_{W \text{ perfectly matchable}} (-1)^{|W|/2} \prod_{v \notin W} z_v. \quad (19.4)$$

In particular, the right side has real coefficients and does not depend on the particular chosen edges or phases in the construction.

Theorem 19.3 (Stability and deletion interlacing). *For a bipartite cactus graph, Φ_G is real stable: it is nonzero whenever $\text{Im } z_v > 0$ for every v . Its diagonal specialization*

$$q_G(x) := x^n p_G(-x^{-2}) = \det(xI - H_G) \quad (19.5)$$

is real-rooted, and q_{G-v} interlaces q_G for every vertex v .

Proof. If $(\text{diag}(\mathbf{z}) - H_G)w = 0$ for $w \neq 0$, taking the imaginary part of its scalar product with w would give

$$0 = \sum_v \text{Im}(z_v) |w_v|^2 > 0,$$

because w^*H_Gw is real. This proves stability. The diagonal identity follows from (19.4); it denotes a polynomial even at $x = 0$. Deleting v gives the principal submatrix of H_G . Its characteristic polynomial is q_{G-v} by the same minor identity, regardless of whether a previously modified edge survives as a bridge. The eigenvalues of a Hermitian principal submatrix interlace those of the whole matrix: the variational characterization of the j th eigenvalue, restricted to the coordinate hyperplane omitting v , gives the two inequalities between consecutive eigenvalues. This proves the stated polynomial interlacing. $\square$

The signed complementary-support polynomial in (19.4) is the stable object. We are not claiming that the unsigned polynomial $\sum_W \prod_{v \in W} z_v$ is stable; already $1 + z_a z_b$ for a single edge vanishes at $z_a = z_b = i$.

[Added 29 September 2026, batch 42: Part III shows that this construction is exhaustive. Only cacti carry a matrix satisfying (19.1), and on a cactus every such matrix is gauge-equivalent to one of $2^{\beta(G)}$ sixth-root matrices, the choice $(-1)^r \zeta$ being allowed on each cycle as well (Theorems 29.1 and 30.3). Real stability of the signed polynomial extends beyond cacti (Theorem 37.1).]

19.3 Positive weights and a probabilistic consequence

Given positive vertex weights w_v , replace K by $K_w = \text{diag}(\sqrt{w_a})K\text{diag}(\sqrt{w_b})$. The minor identity gives

$$\sum_{W \text{ perfectly matchable}} u^{|W|/2} \prod_{v \in W} w_v = \det(I + uK_w K_w^*). \quad (19.6)$$

Thus the weighted polynomial is real-rooted as well, and its γ -transform has the same negative-root factorization. This is a weighted graph refinement; no weighted preorder-polytope interpretation is presumed.

For the original unweighted preorder polytope, choose a lattice point uniformly and let S be its support size. Its probability generating function is $h_{P_G}(t)/h_{P_G}(1)$. Equation (19.3) identifies its distribution with a sum of $n - 2v$ independent Bernoulli variables of parameter $1/2$ and v independent variables $Y_j \in \{0, 1, 2\}$ with

$$\Pr(Y_j = 0) = \Pr(Y_j = 2) = \frac{1}{4 + \lambda_j}, \quad \Pr(Y_j = 1) = \frac{2 + \lambda_j}{4 + \lambda_j}.$$

Hence

$$\mathbb{E}S = \frac{n}{2}, \quad \text{Var}(S) = \frac{n - 2v}{4} + \sum_{j=1}^v \frac{2}{4 + \lambda_j}. \quad (19.7)$$

This is an equality in distribution, not a claimed bijection assigning independent coordinates to individual lattice points.

20 Crowns: explicit formulas with simple negative roots

Let the crown P_r have lower elements $a_1, \dots, a_r$ and upper elements $b_1, \dots, b_r$, with comparisons $a_i < b_i$ and $a_{i+1} < b_i$, where indices are cyclic. For $r \geq 2$, its comparability graph is C_{2r} . The graph polynomial formula below is known [7, Corollaries 3.4 and 3.14]; we derive it to identify precisely the polynomials in the geometric obstruction that follows.

A proper induced subgraph of a cycle is a forest, so each of its matchable vertex sets has a unique perfect matching. The whole even cycle has two perfect matchings, but contributes only

one vertex set. Therefore

$$p_{C_{2r}}(u) = M_{C_{2r}}(u) - u^r = \sum_{k=0}^r \frac{2r}{2r-k} \binom{2r-k}{k} u^k - u^r. \quad (20.1)$$

To check the matching coefficient, distinguish one of the k selected edges. There are $2r$ choices of its position and $\binom{2r-k-1}{k-1}$ choices of positive intervening gap lengths; divide by k to forget the distinction. The $k = 0$ case contributes one. The top matching coefficient is two and becomes one after the correction.

Put

$$L_{\pm}(t) = \frac{1 + t \pm \sqrt{1 + 6t + t^2}}{2}.$$

The cycle matching recurrence, or the two characteristic roots of the path recurrence, gives

$$h_{P_r}(t) = L_+(t)^{2r} + L_-(t)^{2r} - t^r. \quad (20.2)$$

The radical expression is a polynomial: its conjugate terms cancel all odd powers of the square root. In particular,

$$|Q(P_r) \cap \mathbb{Z}^{2r}| = (1 + \sqrt{2})^{2r} + (1 - \sqrt{2})^{2r} - 1. \quad (20.3)$$

For $r = 3$, this gives the six-element crown vector $(1, 12, 48, 75, 48, 12, 1)$ and total 197 of Part I's table in Section 8.

Proposition 20.1 (Explicit location of every crown root). *For $r \geq 2$, all r roots of $p_{C_{2r}}$ are simple and less than $-1/4$. They are*

$$u = -\frac{1}{4 \cos^2 \theta}, \quad 0 < \theta < \frac{\pi}{2}, \quad \cos(2r\theta) = \frac{(-1)^r}{2}. \quad (20.4)$$

All $2r$ roots of h_{P_r} are simple and belong to

$$(-3 - 2\sqrt{2}, -1) \cup (-1, -3 + 2\sqrt{2}).$$

Proof. The roots of $z^2 - z - u = 0$ are $z_{\pm} = (1 \pm \sqrt{1 + 4u})/2$, and $M_{C_{2r}}(u) = z_+^{2r} + z_-^{2r}$. Under the substitution in (20.4), these become $z_{\pm} = e^{\pm i\theta}/(2 \cos \theta)$. Hence

$$M_{C_{2r}}(u)/u^r = 2(-1)^r \cos(2r\theta).$$

The correction in (20.1) gives exactly the angle equation. As $2r\theta$ traverses $(0, r\pi)$, each interval $(j\pi, (j+1)\pi)$ contains one solution. Thus there are exactly r distinct roots, all strictly below $-1/4$. Each gives two negative reciprocal roots through

$$t = \frac{1 - 2u \pm \sqrt{1 - 4u}}{2u}.$$

This map has no double root for $u < 0$. Its endpoint behavior as $u \rightarrow -1/4$ and $u \rightarrow -\infty$ gives the two asserted open intervals. No roots from different u can coincide, because $u = t/(1+t)^2$ recovers the original value. $\square$

21 Induced crowns obstruct every lattice triangulation

A simplicial complex is *flag* if every set of pairwise adjacent vertices is a face. A *minimal nonface* is a set that is not a face although all its proper subsets are faces. Flagness is equivalent to all minimal nonfaces having size two. We show that the canonical root model has forced minimal nonfaces of unbounded size, even when its h^* -polynomial has only simple negative roots.

21.1 An exposed circuit face

Let a finite preorder P contain an induced crown on distinct elements a_i, b_i ($1 \leq i \leq r$), $r \geq 3$. Thus the only distinct-element comparisons among these $2r$ points are the crown comparisons. Consider C_P as in Theorem 13.6. Give a linear functional the coordinate values $-1/2$ at the a_i , $1/2$ at the b_i , and zero on all other elements. Every coordinate root has value at most $1/2$, and every difference root has value at most one. Equality occurs precisely at the crown roots

$$v_i = \mathbf{e}_{b_i} - \mathbf{e}_{a_i}, \quad w_i = \mathbf{e}_{b_i} - \mathbf{e}_{a_{i+1}}. \quad (21.1)$$

They therefore span an exposed face F of C_P .

The coordinate equations show that the unique affine dependence, up to scale, is

$$\sum_{i=1}^r v_i = \sum_{i=1}^r w_i. \quad (21.2)$$

Indeed, a dependence among the cycle edge vectors forces consecutive coefficients alternately equal and opposite. There is one free scalar, and the coefficients sum to zero. Hence F has dimension $2r - 2$, its $2r$ vertices form an affine circuit, and its Radon partition consists of $U = \{v_1, \dots, v_r\}$ and $W = \{w_1, \dots, w_r\}$.

After negating the a_i coordinates, the vertices of F are 0–1 edge-incidence vectors. A convex combination of 0–1 vectors can be integral only if all vectors with positive coefficient agree in every coordinate: an integer coordinate in $[0, 1]$ is an endpoint, forcing agreement. Thus

$$F \cap \mathbb{Z}^n = \{v_1, \dots, v_r, w_1, \dots, w_r\}. \quad (21.3)$$

There are no additional lattice points with which to subdivide this face.

21.2 The nonflag obstruction does not assume regularity

Lemma 21.1. *Every triangulation of F using its vertices has a minimal nonface of size r .*

Proof. Every proper subset T of U spans a simplex face of F . To see this, assign height zero to T , height $-r$ to $U \setminus T$, and height $-(r - |T|)$ to W . The sums of the assigned heights on the two halves are equal, so the unique affine relation (21.2) is respected. There is consequently an affine functional with exactly these values on the vertices. Its maximum face is precisely T . The vertices in T are affinely independent because a proper subset of an affine circuit is independent. The same argument works for proper subsets of W .

On the other hand, $\text{conv}(U)$ and $\text{conv}(W)$ meet in their relative interiors at the common average in (21.2). They cannot both be simplices of one triangulation: simplices with disjoint vertex sets may not intersect in a point which is not a common face. At least one of U, W is therefore missing. All of its proper subsets are forced simplex faces, so it is a minimal nonface of size r . $\square$

A lattice triangulation of C_P , or of its boundary, restricts to a triangulation of its exposed face F . By (21.3) the restriction has no extra vertices. Every proper subset used in the lemma is already a simplex face of C_P and cannot disappear under such a triangulation. A missing half cannot occur elsewhere as a simplex, since its geometric simplex lies in F . This proves the triangulation part of Theorem 13.6 for *all* lattice triangulations, not merely coherent, pulling, or regular ones. Since $r \geq 3$, the missing half is a clique in the one-skeleton, so the triangulation is not flag.

For a height-two graph, the hypothesis is exactly an induced cycle of length at least six in the bipartite comparability graph. Thus chordal bipartiteness (absence of such induced cycles) is a necessary condition for a flag lattice triangulation of D_G . Sufficiency for this particular root model is not proved here. Results about a fully signed symmetric edge polytope cannot be substituted without checking the different configuration.

21.3 An indispensable degree- r toric relation

Fix any field $\mathbb{k}$. Let $\mathcal{A} = C_P \cap \mathbb{Z}^n$, and consider the toric homomorphism

$$\mathbb{k}[X_a : a \in \mathcal{A}] \longrightarrow \mathbb{k}[s, z_1^{\pm 1}, \dots, z_n^{\pm 1}], \quad X_a \longmapsto sz^a.$$

Its kernel $I_{\mathcal{A}}$ is homogeneous in total degree. Define

$$M_U = \prod_{i=1}^r X_{v_i}, \quad M_W = \prod_{i=1}^r X_{w_i}.$$

Equation (21.2) gives $M_U - M_W \in I_{\mathcal{A}}$.

Proposition 21.2. *The binomial $M_U - M_W$ occurs, up to a nonzero scalar, in every binomial generating set of $I_{\mathcal{A}}$. For every monomial order, its leading monomial is a minimal generator of the corresponding initial ideal. Consequently neither a quadratic binomial generating set nor a quadratic Gröbner basis is possible when $r \geq 3$.*

Proof. Consider the monomial fiber with total degree r and vector sum $\sum_i v_i$. The exposing functional of F has value r on this sum, and every lattice point of C_P has value at most one. Every factor of any monomial in the fiber must therefore be a point of F . By (21.3), only its cycle vertices may occur. After the coordinate sign change, the required sum has a one at every cycle vertex. Its decompositions into edges are exactly the two perfect matchings of the cycle. Thus the fiber contains only M_U and M_W .

Moreover, every proper divisor of either monomial has a singleton fiber. The same face-functional argument restricts any competing factorization to cycle edges, while its endpoint set is a proper subset of the cycle. The induced subgraph is a forest and has at most one perfect matching. No proper divisor can therefore be a term of a nonzero toric binomial.

A binomial generating set connects each monomial fiber by multiplying its generators by monomials. In this two-element fiber, a connection cannot come from a generator of smaller degree, because that would require a proper divisor with another monomial in its fiber. The binomial itself is therefore indispensable. For an initial ideal, the leading monomial of this binomial has no proper divisor in the initial ideal by the same singleton-fiber argument. It is a minimal generator of degree r . $\square$

Corollary 21.3 (A spectral–triangulation separation). *For every $r \geq 3$ there is a height-two poset on $2r$ elements whose canonical root polytope has a real-rooted h^* -polynomial with simple zeros, but every lattice triangulation has a minimal nonface of size r and every toric initial ideal has a minimal generator of degree r .*

Proof. Take the crown P_r and combine Theorem 18.2, Proposition 20.1, and the preceding results. $\square$

The existential flag question is different. These obstructions apply to triangulations and the toric configuration of the canonical C_P . Athanasiadis–Chapoton Conjecture 5.1(d) asks whether *some* flag simplicial polytope has h -polynomial h_P . A different polytope might do so even when every lattice triangulation of C_P fails. The present result does not disprove that conjecture.

22 Exact verification and reproducibility

Part II ships three source programs and their recorded outputs, under the names given in Section 22.4; below they are called by their delivered names `verify.py`, `verify_theta.py` and `direct_counterexample.cpp`. (Part I's own verifier, `code/verify.py` in this directory, is a different program; see Section 9.) No numerical root finder is used by the certificate programs. Python computations use integers, rational numbers, or pairs of integers in $\mathbb{Z}[\zeta]$; the C++ program uses exact finite state sets and integer counts. The proofs establish the infinite-family results. The programs check finite instances and offer independent routes to the explicit counterexamples.

22.1 Independent representations, not repeated substitutions

The program `verify.py` represents a bipartite graph by the bitmask neighborhood of each upper vertex. It computes p_G by adding upper vertices one at a time and keeping a set of pairs of covered-shore masks. Different matchings with the same covered vertices are merged. For the 17-element graph it independently runs the ordered-board recurrence from Section 17; the entire 10-by-9 polynomial table is saved in `data/02-height-two-board_certificate.csv`.

For every one of the 512 labeled bipartite graphs with three vertices on each shore, the program directly enumerates lattice points from the ideal inequalities and compares their support polynomial with the matching-support transform. It independently compares the cardinality of $\mathcal{R}_G$ with $p_G(1)$. A further 24 graphs with shores of sizes four and three use the fixed seed 20260928.

The cactus tests do not infer determinants from matching counts. They compute determinants in $\mathbb{Z}[\zeta]$ by subset-memoized Laplace expansion, using $\zeta^2 = \zeta - 1$. If $z = a + b\zeta$, its exact squared modulus is $a^2 + ab + b^2$. The code compares that value with an independently enumerated matching-support indicator for every square minor. The tested graphs are cycles of lengths 4, 6, 8, 10, a square and a hexagon sharing a vertex, and that cactus with a two-edge bridge extension. There are 936 square minors in total, including empty minors.

22.2 The theta-family audit

The separate program `verify_theta.py` computes (13.4) by binomial expansion and compares it with the covered-shore enumeration for $1 \leq m \leq 9$. For $1 \leq m \leq 4$, it also enumerates the original lattice points directly from the ideal inequalities. In particular, the eight-element specialization has exactly 1281 lattice points with the coefficients displayed in Section 16.

A rational-arithmetic Sturm implementation independently checks the number of real roots of p_m for $1 \leq m \leq 20$. Its polynomial remainders use exact fractions, normalized only by positive factors. The variations at $-\infty$ and $+\infty$ are stored alongside every polynomial. These checks agree with the theorem, but the theorem does not extrapolate from them: (16.1) and the two sign-change arguments prove the count for all m . The small example's exact Laguerre defect is also recomputed.

22.3 All 8,408,566 points of the larger example

The C++17 program `direct_counterexample.cpp` is a third route, not a translation of the polynomial transform. For each lower-zero set Z , each upper coordinate has its own optional private unit, and there is one optional unit for each element of Z , assignable only to an adjacent upper coordinate. The set of resulting upper-load vectors is deduplicated.

This is exact because the upper-demand inequalities

$$b(T) \leq |T| + |N(T) \cap Z|$$

are precisely Hall’s conditions for assigning demand clones to the upper private resources and the lower resources. Integral demand vectors are therefore realizable by these allocations if and only if they satisfy the original ideal inequalities. Counting the positive coordinates and the fixed lower support gives h_P directly.

Each of the eight upper coordinates is encoded in three bits. Its maximum load is at most its degree plus one, at most seven in this graph, so the encoding is injective and never overflows a digit. A generation array deduplicates states without hash-dependent behavior. The resulting coefficient vector agrees exactly with (17.5), and its sum is 8 408 566. The source is included; a precompiled executable is not.

Certificate	Recorded verification
All labeled 3-by-3 graphs	512 direct support-polynomial and demand-count comparisons.
Additional 4-by-3 graphs	24 direct comparisons with a recorded seed.
Crown graphs	Formula checks for $2 \leq r \leq 8$; direct lattice checks for $r \leq 4$.
Cactus matrix certificate	936 exact square-minor norm comparisons.
Theta family	Support enumeration through $m = 9$, direct lattice enumeration through $m = 4$, exact Sturm counts through $m = 20$.
17-element example	Two graph-polynomial algorithms and a separate direct enumeration of all 8 408 566 lattice points.

All the listed checks passed in the supplied run. None is a Lean proof or a substitute for independent mathematical review.

22.4 Build commands

The manuscript’s own build commands, run from its package directory, were `python3` on `code/verify.py` and then on `code/verify_theta.py`, the compiled C++ program, and `latexmk` for the PDF. In this report every delivered file name carries the prefix `02-height-two-`:

- programs: `code/02-height-two-verify.py`, `code/02-height-two-verify_theta.py`, `code/02-height-two-direct_counterexample.cpp` and `code/02-height-two-Makefile`;
- recorded outputs: `data/02-height-two-verification.json`, `data/02-height-two-theta_verification.json`, `data/02-height-two-direct_counterexample.json` and `data/02-height-two-board_certificate.csv`.

The programs still use the delivered names internally: `verify_theta.py` imports a module `verify`, which in this directory would be Part I’s verifier, and the `Makefile`’s `verify` target would run Part I’s verifier and overwrite Part I’s data. Therefore rerun them only in a scratch copy with the delivered names restored, from this report’s directory:

```
W=/path/to/scratch; mkdir -p "$W/02/code" "$W/02/data"
for f in code/02-height-two-*; do cp "$f" "$W/02/code/${f#code/02-height-two-}";
done
for f in data/02-height-two-*; do cp "$f" "$W/02/data/${f#data/02-height-two-}";
done
mv "$W/02/code/Makefile" "$W/02/Makefile"; cd "$W/02"
```

```
python3 code/verify.py
python3 code/verify_theta.py
c++ -O3 -std=c++17 -Wall -Wextra -pedantic code/direct_counterexample.cpp \
    -o direct_counterexample
./direct_counterexample > data/direct_counterexample.json
```

The Python programs require Python 3.10 or later and only its standard library. The C++ program needs a C++17 compiler. This article is rebuilt as described in the report’s README; its preamble adds TikZ to Part I’s packages. No external image or font files are required.

23 Proof dependencies and a formalization plan

The main route has a short dependency structure:

$$\begin{aligned} &\text{two published root-polytope inputs} \implies \text{counting lemma} \\ &\implies \text{height-two formula} \implies \text{theta root distribution.} \end{aligned}$$

The canonical root identity is proved separately by orthants. The cactus minor proof and the crown circuit proof are independent of one another. The crown toric result depends only on an explicit exposed lattice face, not on the general root-polytope theorem of Part I.

The following checks are particularly important in an independent audit. First, all matching counts concern *covered vertex sets*; confusing these with individual matchings destroys both the theta and crown formulas. Second, a height-two poset here has singleton elements, not nontrivial preorder-equivalence blocks. Third, h_P counts support sizes of lattice points; it is not initially the Ehrhart numerator of $Q(P)$. The identity with an Ehrhart numerator concerns the different polytope C_P . Fourth, a nonflag triangulation obstruction for C_P does not preclude an unrelated flag polytope having the same h -vector.

A staged formalization could begin with finite graphs on two finite types, covered-shore pairs, and the elementary theta-support classification. The polynomial identity (13.4), the positive-interval rearrangement, and the Descartes/interval root count then form a compact algebraic project once the required real-polynomial lemmas are available. This would verify the entire graph-family theorem without any polytope formalization.

Transferring that theorem to preorder support polynomials requires either formalizing Lemma 14.1 and its classical inputs, or finding a direct combinatorial bijection. A finite-only route to the eight-element counterexample is different: certify its bounded ideal-inequality enumeration, then verify its polynomial and negative Laguerre value. That route need not formalize the general graph-to-polytope bridge, but it also does not establish the all-height-two theorem.

For the cactus result, exact sixth-root arithmetic can be implemented as a quadratic extension with relation $z^2 - z + 1 = 0$. The combinatorial symmetric-difference lemma is a natural intermediate target. The analytic stability proof can then be isolated as a general Hermitian determinant lemma. For the crown obstruction, the finite affine dependence and the two-monomial fiber give a clean algebraic certificate before tackling full triangulation definitions. No uncompiled Lean source is presented as if it had been checked.

24 Further research questions and topics

The results replace an undifferentiated real-rootedness question with several sharper problems. The following questions are proposed directions, not assertions that they have been settled by

this article. Except for explicitly attributed conjectures, they are directions arising from these proofs, not claims that no answer exists anywhere in the literature.

Research question 1 (Extend the matching-support interpretation). Does the γ -vector of a larger class of posets admit a similarly concrete nonnegative interpretation? Height three and preorders with nontrivial equivalence classes are natural next cases. Determine exactly which step of the lower-zero-set argument survives weighted blocks.

A successful extension must account for the fact that a coordinate in a nontrivial equivalence block need not be individually bounded by one. Simply replacing the preorder by its quotient poset loses information.

Research question 2 (A purely combinatorial bridge). Construct an explicit bijection between $\mathcal{R}_G$ and perfectly matchable vertex subsets, or an equally elementary proof of their equal cardinality. Can the choice be made compatible with induced subgraphs, vertex deletion, or positive weights?

Such a construction would remove the two root-polytope inputs from the height-two theorem and substantially reduce its formalization burden. The hypertree-vector bijection in Section 14 does not itself solve this problem.

Research question 3 (Which graph structures preserve real-rootedness?). Characterize useful hereditary classes of bipartite graphs for which p_G is real-rooted. Cacti are a positive class, while the treewidth-two family Θ_m shows that series-parallel structure alone is inadequate. What happens under vertex sums, edge subdivisions, or controlled cycle intersections?

The precise boundary between one shared vertex and a shared path is worth studying. The present examples do not imply that every noncactus graph fails real-rootedness. [Added 29 September 2026, batch 42: indeed not every noncactus fails. Part III shows that $K_{2,3}$ and every graph obtained from it by attaching trees have a real stable Φ_G , hence a real-rooted p_G (Corollary 37.4, Section 39). The question itself remains open; see Section 26.3.]

Research question 4 (Beyond the sixth-root cactus matrices). Which bipartite graphs admit a complex matrix supported on their edges whose square-minor norms are exactly the matchability indicators? Is this condition equivalent to a useful multivariate stability property, or strictly stronger?

A positive answer for a class immediately gives positive-weight real-rootedness through Cauchy–Binet. The theta counterexamples show that such representations cannot exist for all bipartite graphs. [Added 29 September 2026, batch 42: answered in Part III. Exactly the cacti admit such a matrix (Theorem 29.1), and the condition is strictly stronger than real stability of Φ_G : $K_{2,3}$ and all its tree attachments are real stable without it (Theorem 37.1, Corollary 37.4). See Section 26.3.]

Research question 5 (Asymptotic zero geometry for Θ_m). The real-root count is exact, but how are the nonreal roots distributed as $m \rightarrow \infty$? Determine their limiting curves and scaling regimes, including the roots near zero and the large negative real root. What is the corresponding distribution after $u = t/(1+t)^2$?

Formula (13.4) gives a concrete exponential-versus-polynomial equation from which such asymptotics may be developed; the present paper does not assert a limiting measure.

Research question 6 (Flag triangulations of the canonical model). For height-two posets, does chordal bipartiteness suffice for D_G to admit a flag unimodular triangulation? More generally, characterize which induced configurations obstruct flagness beyond the crown circuits.

The necessity of chordal bipartiteness follows here. Sufficiency is a separate problem for D_G , even if related statements hold for other signed root configurations.

Research question 7 (Existential flag realizations). Can the crown polynomial h_{P_r} be realized as the h -polynomial of a flag simplicial polytope by a construction unrelated to lattice triangulations of C_{P_r} ? Can one do this for all height-two posets?

This is the distinction needed to make progress on AC Conjecture 5.1(d), rather than only testing one canonical construction.

Research question 8 (Exact toric degree bounds). How much of the largest indispensable degree of $I_{C_P \cap \mathbb{Z}^n}$ is controlled by induced crowns? Is there an upper bound in terms of the largest such crown, or do other faces produce independent unbounded obstructions?

The theorem supplies a lower bound, not a characterization. Additional coordinate roots may change global toric relations while leaving the exposed-face obstruction intact.

Research question 9 (Minimum counterexamples with checkable completeness). Give a fully reproducible exhaustive certificate for the smallest counterexample to preorder support real-rootedness, including nontrivial preorder blocks. The eight-element example is explicit, while Athanasiadis–Chapoton report successful tests through size seven [1]; a complete independent certificate for that lower bound is not supplied here.

A smaller search space restricted to height-two posets may be practical for identifying every minimal forbidden example and testing structural conjectures about their bipartite graphs.

Research question 10 (Efficient exact enumeration and formal certificates). Develop a tree-decomposition algorithm for matchable *vertex-set* counts that avoids counting matching multiplicities. Can it output a small certificate that a proof assistant can check independently of the search procedure?

The theta family gives a useful stress test: treewidth is small, yet root behavior is not controlled by real-rootedness. Enumeration and spectral regularity should therefore be treated as distinct algorithmic issues.

25 Conclusion

The height-two identity identifies the γ -coefficients of preorder support polynomials with a classical graph invariant. Its value is not merely that it reproduces an isolated counterexample: it exposes a complete root-distribution family, a determinantal positive class, and a sharp distinction between root behavior and canonical triangulations.

The negative examples do not destroy γ -positivity in height two; that property is guaranteed by the matching-support formula. Likewise, real-rooted crown polynomials do not rescue flag lattice triangulations of the canonical root polytope. The remaining questions therefore have different mathematical mechanisms and require different constructions. The accompanying proofs and exact certificates are intended to make those mechanisms, and the limits of the novelty claims, available for direct independent review.

Part III. Cactus rigidity: exact square-minor models and stability

Part III was added on 29 September 2026 in batch 42 of ProveIt's incoming-report intake, from one manuscript written against Parts I and II as printed above. It answers Part II's Research question 4, "Beyond the sixth-root cactus matrices": the bipartite graphs that admit a complex matrix on their edges whose square-minor norms are exactly the matchability indicators are precisely the cacti, and this property is strictly stronger than real stability of the signed matching-support polynomial. Parts I and II are unchanged except for dated pointers to this addition. Section 26 records the provenance, fixes the notation and states what is answered.

26 Provenance, conventions and what Part III answers

26.1 One manuscript, one addition

Part III is batch-42 manuscript 05, *Cactus Rigidity for Matching-Support Determinants: Exact classification, a sharp approximation gap, and stability beyond vertex determinants*, an AI-assisted research manuscript prepared for Vladimir Reshetnikov and dated 29 September 2026 (24 US-Letter pages as delivered). It arrived as the archive `ProveIt_Cactus_Rigidity.zip` (inner directory `ProveIt_Cactus_Rigidity/`) in ProveIt commit 8315d24e3 and was placed in 3609d0473. It is pinned to ProveIt commit 9754e8360 (the full hash is printed in Section 27) and records the Git blob 8739a224fa8956adab5d8df9f2d5012a6a220959 of this report's `article.tex`. That blob is Parts I and II exactly as printed above: the file was last changed by the batch-39 write 6a354f52f and is unchanged at the pin and at the placement commit. The manuscript therefore cites Part II in its printed numbering; its "Section 19", "Section 24" and "Theorem 13.2" are Section 19, Section 24 and Theorem 13.2, and are replaced here by cross-references. The manuscript's program, recorded outputs, build file and notes are shipped beside Part I's files with the prefix 03-cactus- (Part I's unprefixed files count as 01, Part II's are 02-); Section 40.3 lists them. Its manuscript, delivery README and PDF are not shipped and survive in the arrival commit; its checksum list `SHA256SUMS.txt` was verified (10/10) at placement and retired.

The manuscript is printed here as Sections 27 to 45: its Section k is Section $k + 26$ here, its Appendices A and B are Sections 44 and 45, and its theorem numbers shift accordingly (its Theorem 3.1 is Theorem 29.1); its equations, numbered (1)–(26) consecutively there, are numbered by section here, as in the rest of this report. Its nine research questions continue Part II's numbering as Research questions 11–19. Every result, proof, remark, limitation, question and non-claim of the manuscript is printed; three displays that repeat Part II's formulas are printed once, in Part II (Section 26.4). Its title-page abstract and status note are reproduced as delivered.

Abstract of manuscript 05 (as delivered). For a finite bipartite graph, can one assign unit complex numbers to its edges so that every square minor has squared absolute value equal to the indicator that its vertex support is perfectly matchable? The ProveIt preorder-polytope report constructs such assignments for cactus graphs and explicitly asks for the exact boundary of the construction. We prove that the boundary is precisely the cactus class. Equivalently, the matching-support polynomial has a Hermitian vertex-diagonal determinant representation exactly for cacti; equivalently, the graph is the support pattern of a matrix over $\mathbb{F}_3$ with no vanishing structurally nonzero minor. A cactus of cycle rank β has exactly 2^β phase-gauge classes of complex certificates.

The obstruction is quantitative. Every unit-phase matrix on every noncactus has a matchable minor whose squared absolute value differs from one by at least

$$\varepsilon_* = \frac{\sqrt{17} - 3}{2},$$

and this constant is attained on $K_{2,3}$. We obtain the classification and the uniform gap by reducing induced theta graphs to three explicit matrices. In contrast, the five-variable matching-support polynomial of $K_{2,3}$ is real stable, as certified by three exact sums-of-squares identities. Attaching arbitrary trees produces infinitely many connected stable examples without a Hermitian vertex determinant. We also prove that bounded-order minor tests cannot recognize exact representability, even on subcubic graphs of treewidth two.

Status (manuscript 05, as delivered). Complete ordinary mathematical proofs are supplied for the stated results. This AI-assisted manuscript is unrefereed and has not been verified in Lean or another proof assistant. The cactus construction and its established real-rootedness consequences are prior work, not claimed here. The classification answers a specified repository question; worldwide priority for its formulations and refinements is not asserted.

Priority and prior work (from the manuscript’s status and source notes). The cactus sixth-root construction, its determinant factorization and cactus stability are Part II’s (Section 19); univariate cactus real-rootedness is Ohsugi–Tsuchiya’s [9, Theorem 7.2]; the height-two matching-support transform is Part II’s Theorem 13.2. The binary (forest) observation and the general phenomenon of stability without a determinant representation are not claimed as new; the latter is established elsewhere, for example for Vámos-type polynomials [15]. A related earlier characterization of binary fundamental transversal matroids [16] was located (publisher abstract only; its full text was not available); possible connections to older fundamental-transversal-matroid characterizations deserve further literature review. The source audit was targeted, not exhaustive, and does not establish the absence of an older equivalent theorem; no worldwide publication priority is claimed for the classification, the finite-field reformulations, the gauge count, the quantitative refinements or the stability formulations. The only specialized external theorem used is Brändén’s multiaffine Rayleigh criterion [12, Theorem 5.6], for the stability of $K_{2,3}$.

26.2 Notation of Part III

Part III uses Part II’s graph conventions: $G = (A \sqcup B, E)$ is a finite simple bipartite graph with $p = |A|$, $q = |B|$, $n = p + q$; a cactus is a graph in which every edge lies on at most one simple cycle; p_G and $\mathcal{M}_k(G)$ count perfectly matchable *vertex sets*; $\zeta = e^{\pi i/3}$; ν is the matching number and λ_j are the positive eigenvalues of KK^* . Table 2 lists the symbols Part III adds or uses differently. One symbol of the manuscript was renamed: its theta graph $\Theta(a, b, c)$ with three paths of lengths a, b, c is written $\vartheta(a, b, c)$ here, because Part II’s Θ_m is a different family with m paths. No normalization was changed.

Table 2: Symbols of Part III. The last column prints the tempting false reading, usually a Part I or Part II symbol with the same letter.

Symbol	Meaning in Part III	Not to be confused with
$\vartheta(a, b, c)$	a theta: three internally disjoint paths of lengths a, b, c between two common endpoints; one length may be 1 (renamed from the manuscript's $\Theta(a, b, c)$)	Part II's Θ_m , which has m paths, all of length three. Only $\Theta_3 = \vartheta(3, 3, 3) = \mathsf{T}$ is both; Θ_m for $m \neq 3$ is no $\vartheta(a, b, c)$. Part II's “theta family” means Θ_m .
E, D, T	the three cores $\vartheta(2, 2, 2) = K_{2,3}$, $\vartheta(1, 3, 3)$, $\vartheta(3, 3, 3)$	the edge set E ; Part II's polytope D_G and its positive part T_G .
$\Phi_G(\mathbf{z})$	the signed complementary-support polynomial (28.2), defined for <i>every</i> bipartite G	Part II defines $\Phi_G := \det(\text{diag}(\mathbf{z}) - H_G)$ for cacti; the two agree there (19.4). For a noncactus, Φ_G is <i>no</i> such determinant (Theorem 29.1).
$\Phi_G(s, \dots, s)$	$s^n p_G(-s^{-2})$, for every G	Part II's $q_G(s)$ of (19.5), stated there for cacti only.
K , faithful, H , H_K	a phase matrix; faithful if (28.3) holds; a Hermitian vertex-indexed matrix; the bipartite Hermitian matrix of K	Part II's K is <i>one</i> faithful certificate (choice $(-1)^r \zeta$ on every cycle) and its H_G is H_K for it; Part II's bipartite graph $H = (X \sqcup Y, E)$; Part I's triangulation K . Part II's D_G ; Part I's neighbourhoods D_i .
D_A, D_B , gauge	diagonal unitary row and column scalings	
T	a spanning forest	Part II's T_G ; Part I's τ .
$\beta(G), c(G)$	cycle rank $ E - A - B + c(G)$; number of components	—
$\delta_G(K), d(G), \varepsilon_*, d_0$	minor defect of K , its minimum over phases, the constant $(\sqrt{17} - 3)/2$, the constant $(\sqrt{5} - 1)/2$	the vertex d of Θ_m ; Part I's diagonal vertices d_i .
p, q, r in Lemma 36.6; q in Proposition 35.3	negated real inner products of unit vectors; a field size ($\mathbb{F}_q$)	the shore sizes $p = A $, $q = B $.
I, J	a row set and a column set of a square minor	Part I's order ideals $I \in \mathcal{J}(\tau)$; Part II's independent sets.
U, V, W in (37.2)	the monomials xy, xz, yz	W as a perfectly matchable set elsewhere.
L in Theorem 38.1	an odd path length	Part I's shore index i_L .

26.3 What Part III answers in Part II

Status after Part III, as of 29 September 2026:

- **Research question 4 (“Beyond the sixth-root cactus matrices”) is answered.** A bipartite graph has a complex matrix on its edges whose square-minor norms are exactly the matchability indicators if and only if it is a cactus; equivalently, Φ_G is a Hermitian vertex-diagonal determinant, or G carries a support-exact matrix over $\mathbb{F}_3$ (Theorem 29.1). Part II’s sixth-root matrices are therefore all there is: every such matrix is gauge-equivalent to one of $2^{\beta(G)}$ sixth-root matrices (Theorem 30.3). The condition is **strictly stronger** than real stability of Φ_G , the multivariate property Part II proves for cacti (Theorem 19.3): $\Phi_{K_{2,3}}$ and Φ_G for every tree attachment to $K_{2,3}$ are real stable with no such matrix (Theorem 37.1, Corollary 37.4). The obstruction is quantitative, with the sharp constant ε_* (Theorem 36.3). Graphs with real stable Φ_G are not classified (Research question 11 below).
- **Part II’s remark after that question** (“the theta counterexamples show that such representations cannot exist for all bipartite graphs”) is consistent with Part III and gives a second, non-quantitative route for one core. *Merge note, not in manuscript 05:* a faithful matrix gives $p_G(u) = \det(I + uKK^*)$ (Section 39), which is real-rooted, whereas $h_{p_{\Theta_3}}$, and hence p_{Θ_3} (Proposition 15.1), is not real-rooted (Theorem 13.3); so $\mathsf{T} = \Theta_3$ has no faithful matrix. Every Θ_m with $m \geq 3$ contains Θ_3 as an induced subgraph. Part III’s Lemma 36.6 gives the quantitative form, and Lemma 31.1 explains why some core occurs in every noncactus.
- **Research question 3 (which structures preserve real-rootedness) is re-scoped, not answered.** Its remark notes that the examples of Part II do not imply that every noncactus fails real-rootedness. Part III supplies such noncacti: $K_{2,3}$, with $p_{K_{2,3}}(u) = 1 + 6u + 3u^2$, and every connected graph obtained from it by attaching trees has a real stable Φ_G and hence a real-rooted p_G (Section 39). The question of useful hereditary classes stays open.
- Research question 1 (a γ -interpretation beyond height two) remains open; Research question 18 below proposes stable signed-support methods as one route.

Part III refutes no statement of Parts I and II, and nothing in them is retracted.

26.4 Where the merge had to choose

- *Placement.* Part III follows Part II and precedes Part I’s appendices, whose letters and numbers are unchanged. The manuscript’s own Appendices A and B are printed as the ordinary Sections 44 and 45 of Part III.
- *Self-citation.* The manuscript cites this report as an external repository report (its reference “Repo”, with the pinned commit, Sections 19 and 24 and Theorem 13.2). Here those citations are replaced by references to Part II, and “the repository(s)” by “Part II(s)” where it means this report in running text; two headings, in Sections 27 and 39, keep the manuscript’s wording. The pin itself is kept in Section 27.
- *Duplicates.* The manuscript’s cactus factorization (Lemma 30.1) is Part II’s Lemma 19.1, by the same route; it is printed as the manuscript states it because it displays both admissible phases on each cycle, which Theorem 30.3 needs. The manuscript’s height-two formula, Cauchy–Binet identity and two spectral factorizations (its displays (23)–(26), in its Section 13) are Theorem 13.2 and Theorem 19.2; they are printed once, in Part II, and cited in Section 39. The manuscript’s recall of the cactus stability argument (Section 37) is the proof of Theorem 19.3 and is kept as a short recall.

- *Notation.* $\Theta(a, b, c)$ is renamed $\vartheta(a, b, c)$ (Table 2); nothing else is renamed.
- *File names.* The delivered names (`code/verify.py`, `data/verification.json`, the `Makefile`) are replaced by the shipped names in Section 40, because `code/verify.py` in this directory is Part I’s verifier; the build commands are replaced by commands that run on a copy.
- *Bibliography.* The manuscript’s entries for Ohsugi–Tsuchiya’s matching-generating-polynomial paper and for Davis–Kohl are Part II’s references [9] and [7]; its five other external entries are appended.
- *Labels and front matter.* Labels carry the prefix `cac:`. The conclusion and the dependency ledger received labels, and Part II’s Research question 4 received the label `cac:q:beyond` so that Part III can cite it. The manuscript’s title page, abstract and status note are printed in this section; its table of contents is dropped.

27 The question, its provenance, and the answer

27.1 The repository question being resolved

The starting point is Part II of this report, *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, whose height-two extension is entitled *Height-Two Preorder Polynomials: Matching supports, nonreal-root families, and crown obstructions*. Section 19 of Part II constructs a sixth-root-of-unity matrix for each bipartite cactus. Its Section 24 asks, under the heading “Beyond the sixth-root cactus matrices” (Research question 4), which bipartite graphs admit a matrix whose square-minor norms reproduce the matchability indicators, and whether this requirement is equivalent to multivariate stability.

This article answers both parts, with the precise meaning of the representation fixed below:

$$\text{Exact minor representation} \iff \text{cactus graph} \implies \text{real stability},$$

where the last implication is strict. The necessity direction is the main extension of Part II’s result. In particular, adding arbitrary complex phases does not extend the exact model to a larger graph class than the existing sixth-root construction.

The source is pinned to commit

9754e83603e223811b7515eb892f22c847144593.

The relevant `article.tex` has Git blob

8739a224fa8956adab5d8df9f2d5012a6a220959.

The main branch may subsequently change. All references to the source’s sections concern this pinned version, not an unspecified current checkout. [Merge note: that blob is this report’s `article.tex` before Part III was added, so the cited sections are those of Part II above; see Section 26.1.]

27.2 Why exact support matters

For an ordinary matrix, a determinant is a signed sum over perfect matchings. The invariant studied here counts a vertex set once whenever at least one perfect matching exists. It must not count the number of such matchings. These two tasks agree on forests but diverge on cycles.

Phases can correct the overcount on isolated cycles; the question is whether they can do so consistently when cycles overlap.

This is a concrete obstruction problem with implications for determinant proofs of stability, weighted real-rootedness, and exact certificate design. It does not settle general γ -positivity or flag realization questions for preorders. Nor does it characterize all graphs with real-rooted univariate support polynomials.

27.3 A map of the contributions

The results are organized around four statements. Theorem 29.1 gives the exact graph classification, including its Hermitian and ternary versions. Theorem 30.3 classifies all complex certificates up to diagonal phase changes. Theorem 36.3 gives the sharp universal separation from noncacti. Theorem 37.1 and Corollary 37.4 separate real stability from the exact vertex determinant property. Theorem 38.1 shows why testing a fixed collection of minor orders cannot substitute for the structural theorem.

The proofs of the classification and approximation gap use only finite graph arguments, determinants, and elementary geometry of unit complex numbers. The stability example uses Brändén's established multiaffine Rayleigh criterion [12, Theorem 5.6]. This is the only specialized external theorem needed for the new stability conclusion.

28 Definitions and distinctions

Throughout, $G = (A \sqcup B, E)$ is a finite simple bipartite graph with a specified bipartition. Write $p = |A|$, $q = |B|$, $n = p + q$, and let $c(G)$ count connected components, including isolated vertices. Its cycle rank is

$$\beta(G) = |E| - |A| - |B| + c(G).$$

Empty shores and isolated vertices cause no difficulty. A cactus is a graph in which every edge belongs to at most one simple cycle. Equivalently, every nontrivial block is an edge or a cycle. We allow disconnected cacti.

Definition 28.1 (Matching supports). A vertex set $W \subseteq A \sqcup B$ is *perfectly matchable* when the induced graph $G[W]$ has a perfect matching. The empty set is perfectly matchable. Let $\mathcal{M}_k(G)$ be the set of such W with $|W| = 2k$. Define

$$p_G(u) = \sum_{k \geq 0} |\mathcal{M}_k(G)| u^k, \quad (28.1)$$

$$\Phi_G(\mathbf{z}) = \sum_{k \geq 0} \sum_{W \in \mathcal{M}_k(G)} (-1)^k \prod_{v \notin W} z_v. \quad (28.2)$$

The polynomial p_G is a classical perfectly matchable set polynomial; relevant prior work includes Ohsugi–Tsuchiya and Davis–Kohl [9, 7]. Our symbol Φ_G agrees with the signed multivariate refinement used in Section 19 of Part II, equation (19.4), on the cacti treated there; here it is defined for every bipartite graph. Every monomial in (28.2) has coefficient 1 or -1 , regardless of the number of matchings on its omitted vertex set.

Definition 28.2 (Faithful phase certificate). A matrix $K \in \mathbb{C}^{A \times B}$ is a *phase matrix* on G if

$$K_{ab} = 0 \quad (ab \notin E), \quad |K_{ab}| = 1 \quad (ab \in E).$$

It is *faithful* if, for every $I \subseteq A$, $J \subseteq B$ with $|I| = |J|$,

$$|\det K_{I,J}|^2 = \mathbf{1}\{G[I \sqcup J] \text{ has a perfect matching}\}. \quad (28.3)$$

The empty determinant is one.

There is no extra restriction in calling the edge entries phases: the one-by-one instances of (28.3) force precisely these moduli and this zero pattern. A nonmatchable square submatrix has determinant zero for every choice of entries on its pattern, since every permutation product contains a zero.

Definition 28.3 (Support-exact matrices over a field). For a field F , a matrix $L \in F^{A \times B}$ is *support-exact* for G if its nonzero entries occur precisely on E and every square submatrix whose support has a perfect matching is nonsingular.

This is the pattern-sensitive notion often called superregularity: structurally zero minors are permitted, but no other minor may vanish [13]. We use “support-exact” to avoid confusion with the different convention that a superregular matrix has *every* square minor nonzero.

Remark 28.4 (Two weaker conditions are not the same problem). Requiring merely that every nonzero complex minor have modulus one permits matchable minors to vanish. That weaker requirement belongs to the theory of complex unimodular and sixth-root-of-unity matrices [14]; it is not (28.3). Likewise, an univariate identity $p_G(u) = \det(I + uM)$ alone forgets the individual vertex supports and does not imply faithfulness.

29 The classification theorem

Theorem 29.1 (Exact boundary of the determinant model). *For a finite simple bipartite graph G , the following are equivalent:*

- (i) G is a cactus.
- (ii) G has a faithful complex phase certificate.
- (iii) G has a faithful certificate all of whose nonzero entries are sixth roots of unity.
- (iv) There is an $n \times n$ Hermitian matrix H , indexed by $V(G)$, for which

$$\Phi_G(\mathbf{z}) = \det(\text{diag}(z_v : v \in V(G)) - H). \quad (29.1)$$

- (v) G has a support-exact matrix over $\mathbb{F}_3$.

No zero pattern for H is assumed in (iv); the polynomial identity forces the required bipartite pattern.

The implication (i) $\Rightarrow$ (iii) is the construction already present in Part II (Theorem 13.5 and Lemma 19.1). We reproduce its proof in Section 30 to keep the argument self-contained and to prepare the classification of all certificates. Sections 31–33 prove the converse. Section 34 proves the precise Hermitian equivalence, and Section 35 proves the finite-field characterization.

Remark 29.2 (The strength and the limit of the answer). The theorem rules out every Hermitian matrix of the form (29.1), not only the matrix chosen by Part II’s construction. It does *not* rule out arbitrary larger determinantal pencils, substitutions involving auxiliary variables, sums of determinants, or representations of powers of a polynomial. Those are different questions.

30 Cactus certificates and their complete phase freedom

Set $\zeta = e^{\pi i/3}$. Thus $\zeta^2 - \zeta + 1 = 0$ and

$$|1 - \zeta| = |1 - \bar{\zeta}| = 1.$$

Choose a spanning forest T of a cactus. Each cycle has exactly one edge outside T , and no edge belongs to two cycles. Put $K_{ab} = 1$ on the forest edges. On the remaining edge of each cycle of length $2r$, put

$$K_{ab} = (-1)^r \zeta \quad \text{or} \quad (-1)^r \bar{\zeta}, \quad (30.1)$$

independently from cycle to cycle. All nonedges have entry zero.

Lemma 30.1 (The cactus factorization; prior construction). *Every matrix constructed by (30.1) is faithful.*

Proof. Fix a square induced support admitting a perfect matching M . The symmetric difference of M with any other perfect matching consists of vertex-disjoint M -alternating cycles. In a cactus, any two M -alternating cycles are themselves vertex-disjoint: if they met at a vertex, its unique edge of M would have to belong to both cycles, contradicting the cactus condition.

Consequently all perfect matchings on the support are obtained by independently flipping a subset of its M -alternating cycles. A flip on a cycle of length $2r$ changes the permutation sign by $(-1)^{r-1}$. The ratio of the products of edge entries in the two alternating halves is $(-1)^r \zeta$ or $(-1)^r \bar{\zeta}$, depending on the orientation of the comparison. Therefore, after factoring out the unit contribution of M , the determinant is a product of factors $1 - \zeta$ and $1 - \bar{\zeta}$. Its modulus is one. If the support has no perfect matching, every determinant term is zero. $\square$

This is the content of Part II's cactus minor lemma (Lemma 19.1, which uses the choice $(-1)^r \zeta$ on every cycle), now with both allowed choices on each cycle displayed. The proof does not invoke any general matrix representability theorem.

Definition 30.2 (Phase gauge). Two phase matrices on the same labeled bipartite graph are gauge-equivalent if

$$K' = D_A K D_B$$

for diagonal matrices D_A, D_B whose diagonal entries have modulus one. No row permutation, column permutation, pivot, or complex conjugation is included in this equivalence.

Such changes preserve all square-minor moduli. They are also precisely the relevant diagonal unitary changes of the corresponding bipartite Hermitian matrix, after conjugating the column-side factor as appropriate.

Theorem 30.3 (Rigidity and moduli of faithful certificates). *For a cactus G , every faithful certificate is gauge-equivalent to exactly one of the matrices (30.1), relative to a fixed spanning forest. There are exactly $2^{\beta(G)}$ gauge classes. In particular, arbitrary unit complex phases produce no additional classes beyond sixth-root phases.*

Proof. On each tree component, select a root. Choose row and column phases successively away from the root so that every tree edge becomes one. A residual gauge preserving all forest entries uses a common phase on the row vertices of a component and its reciprocal on the column vertices; it changes no edge entry anywhere in that component. Thus normalization gives a unique representative of each gauge class.

A simple cycle in a cactus is induced. In its normalized square submatrix, all entries are one except for the single non-tree entry t , and there are exactly two perfect matchings. Up to a unit sign, its determinant is

$$1 + (-1)^{r-1}t.$$

Faithfulness and $|t| = 1$ imply

$$\operatorname{Re}((-1)^{r-1}t) = -\frac{1}{2}.$$

The only possibilities are $t = (-1)^r \zeta$ and $t = (-1)^r \bar{\zeta}$. This proves necessity of every listed cycle choice. Lemma 30.1 shows that the choices are independent and sufficient. A cactus has $\beta(G)$ cycles, proving the count. $\square$

Corollary 30.4 (When real phases suffice). *A bipartite graph has a faithful real matrix if and only if it is a forest. For a forest, every assignment of signs to the edges is faithful and all assignments are gauge-equivalent.*

Proof. A forest has at most one perfect matching on each vertex support, because the symmetric difference of two would contain a cycle. Hence every nonzero determinant is a signed product of edge entries. Conversely, a faithful real matrix would be a faithful complex matrix and hence have cactus support. On any cycle the required normalized entry (30.1) is nonreal. One can also argue directly on a shortest induced cycle: its two determinant terms are ± 1 , so the determinant is 0 or ± 2 , never of modulus one. $\square$

31 Why every noncactus contains the right obstruction

A *theta* consists of three internally vertex-disjoint paths with common distinct endpoints. Our convention permits one of the paths to have length one; at most one can do so in a simple graph. An *induced theta* has no edges besides those in the three paths. This convention matters: excluding the length-one case would miss two cycles sharing an edge.

Lemma 31.1 (Induced-theta lemma for bipartite graphs). *Every finite noncactus bipartite graph contains an induced theta. The three path lengths have the same parity.*

Proof. A noncactus has a two-connected block B that is not a cycle or an edge. To see the relevant block fact, two cycles sharing an edge lie in a common two-connected subgraph, and a maximal such block is not a single cycle. Choose a shortest cycle C of B . It is induced, and B has vertices outside C ; otherwise an extra edge would be a chord contradicting shortestness.

First suppose that a vertex $v \notin C$ has at least two neighbors on C . If it has exactly two, $C \cup \{v\}$ is an induced theta. If it has at least three, take three consecutive neighbors x, y, z in their cyclic order. Retain v and the arc of C from x through y to z having no further neighbors of v in its interior. The three paths from v to y are the edge vy and the two paths through x and z . The arc has no chord, and its endpoints x, z are in the same bipartition class, so there is no extra edge xz . This is again an induced theta.

We may therefore assume that each vertex outside C has at most one neighbor on C . Some component of $B - V(C)$ has at least two distinct attachment vertices on C : if it had only one, that vertex would disconnect B . Choose a shortest path with distinct endpoints on C and all internal vertices outside C . It has no chord. It also has no additional edge from an internal vertex to C . Such an edge would produce a shorter path of the same kind to one of the two endpoints; at the first or last internal vertex, the assumed uniqueness of its neighbor on C excludes the only exceptional adjacency. The union of this path with C is therefore an induced theta.

In a bipartite graph, every path between two fixed vertices has parity determined by the shores of its endpoints. Hence all three lengths have the same parity. $\square$

The lemma is tailored to induced subgraphs because deleting arbitrary edges from a phase matrix need not preserve any minor identity. Our proof never makes that invalid inference. The only graph reductions used below are restriction to an induced subgraph and a precisely controlled path replacement.

32 A path-compression identity that preserves all supports

Consider a path

$$a - v - u - b, \quad a, u \in A, \quad v, b \in B,$$

whose internal vertices v, u have degree two. Assume that a and b are not adjacent. Delete u, v and replace the three-edge path by the edge ab . For a matrix supported on the original graph, define the new entry by

$$K'_{ab} = -\frac{K_{av}K_{ub}}{K_{uv}}, \quad (32.1)$$

and leave the other surviving entries unchanged. This operation shortens one path by two while preserving bipartiteness.

Lemma 32.1 (Compression preserves support and determinant magnitude). *Suppose $K_{uv} \neq 0$. For every equally sized pair I, J of surviving row and column sets,*

$$\det K_{I \cup \{u\}, J \cup \{v\}} = \pm K_{uv} \det K'_{I, J}. \quad (32.2)$$

Moreover, the two induced supports in (32.2) are perfectly matchable simultaneously. If all edge entries of K have modulus one, the same is true for K' , and the squared determinant magnitudes in (32.2) are equal.

Proof. Put row u and column v last. The block determinant identity gives the factor K_{uv} times the determinant of the Schur complement. The column at v has only one other possible nonzero entry, in row a , and the row at u has only one other possible nonzero entry, in column b . Thus the only altered surviving entry is (32.1). Restoring the original order contributes only a sign.

For matchings there is an explicit bijection. A matching on the compressed support not using ab lifts by adding uv . A matching using ab lifts by replacing it with av and ub . Conversely, a matching of the augmented original support either uses uv , or must match v to a and u to b . These alternatives also cover the cases in which I or J omits one of the endpoints. Finally, (32.1) is a quotient of unit complex numbers with a minus sign, hence has modulus one. $\square$

Corollary 32.2 (Preservation of exactness). *Faithful complex certificates and support-exact matrices over any field descend through this compression. Both properties also descend to induced subgraphs.*

Repeated compression of an induced theta produces one of the following three graphs:

Parity and short path	Final path lengths	Name used here
Even	(2, 2, 2)	$E = K_{2,3}$
Odd, a length-one path present	(1, 3, 3)	D
Odd, no length-one path	(3, 3, 3)	T

In the even case shorten only paths of length at least four. In the odd case shorten only paths of length at least five. Consequently no parallel edge is created. At each step the internal vertices have degree two, and the replacement edge does not already exist. These details are why the Schur-complement identity applies throughout.

33 Three finite phase obstructions

We now exclude a faithful matrix for each core. Row and column phase changes normalize a spanning tree to ones. Every remaining parameter below has modulus one.

33.1 The even core $E = K_{2,3}$

A normalized matrix is

$$K_E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & z & w \end{pmatrix}. \quad (33.1)$$

Its three two-by-two supports are all matchable, so faithfulness requires

$$|1 - z| = |1 - w| = |z - w| = 1.$$

The first two equalities imply $z, w \in \{\zeta, \bar{\zeta}\}$. Their distance is then either zero or $\sqrt{3}$, contradicting the last equality. Geometrically, three points on the unit circle cannot have all three chord lengths equal to one.

33.2 Two squares sharing an edge: D

A normalized matrix is

$$K_D = \begin{pmatrix} 1 & 1 & 1 \\ x & 1 & 0 \\ y & 0 & 1 \end{pmatrix}. \quad (33.2)$$

The two cycle minors have determinants $1 - x$ and $1 - y$. They force $x, y \in \{\zeta, \bar{\zeta}\}$. The full determinant is $1 - x - y$, with squared modulus zero if the phases are conjugate and three if they are equal. The full support is matchable, so both possibilities are forbidden.

33.3 Three odd paths: T

A normalized matrix is

$$K_T = \begin{pmatrix} 0 & 1 & 1 & 1 \\ x_1 & 1 & 0 & 0 \\ x_2 & 0 & 1 & 0 \\ x_3 & 0 & 0 & 1 \end{pmatrix}. \quad (33.3)$$

Every pair of arms gives a three-by-three cycle minor with determinant $-(x_i + x_j)$. Faithfulness would imply

$$\operatorname{Re}(x_i \bar{x}_j) = -\frac{1}{2} \quad (i \neq j).$$

But the full determinant is $-(x_1 + x_2 + x_3)$, and these equalities give

$$|x_1 + x_2 + x_3|^2 = 3 + 2 \sum_{i < j} \operatorname{Re}(x_i \bar{x}_j) = 0.$$

Its support has a perfect matching, a contradiction.

Proof of (ii) $\Rightarrow$ (i) in Theorem 29.1. If G were a noncactus with a faithful matrix, Lemma 31.1 would give an induced theta. Restrict the matrix to it, and apply Lemma 32.1 until one of the three cores remains. The resulting faithful matrix contradicts the corresponding calculation above. $\square$

34 The Hermitian representation has no hidden extra freedom

For a phase matrix K , put

$$H_K = \begin{pmatrix} 0 & K \\ K^* & 0 \end{pmatrix}.$$

On a balanced vertex set $W = I \sqcup J$, with $|I| = |J| = k$, its principal determinant is

$$\det(H_K)_W = (-1)^k |\det K_{I,J}|^2. \quad (34.1)$$

On an unbalanced vertex set the principal determinant vanishes: one of the two off-diagonal rectangular blocks has more rows than columns and hence produces a kernel vector.

Proposition 34.1 (Coefficient recovery). *Identity (29.1) holds for a Hermitian H if and only if $H = H_K$ for a faithful phase matrix K , with vertices ordered by the two shores.*

Proof. The diagonal expansion is

$$\det(\text{diag } \mathbf{z} - H) = \sum_{W \subseteq V} (-1)^{|W|} \det H_W \prod_{v \notin W} z_v. \quad (34.2)$$

If K is faithful, (34.1) and the vanishing of unbalanced minors make this exactly (28.2).

Conversely, compare coefficients for omitted singleton sets. They give $H_{vv} = 0$. For an omitted pair $\{v, w\}$, the coefficient is

$$\det H_{\{v,w\}} = -|H_{vw}|^2.$$

The coefficient in Φ_G is -1 for an edge and zero otherwise. Therefore $H_{vw} = 0$ on nonedges and $|H_{vw}| = 1$ on edges. In particular its same-shore blocks vanish, so $H = H_K$. Comparing all remaining balanced coefficients using (34.1) yields faithfulness. $\square$

This proves (ii) $\Leftrightarrow$ (iv). It also explains exactly which information disappears under diagonal specialization. For example, $K_{2,3}$ has

$$p_{K_{2,3}}(u) = 1 + 6u + 3u^2 = (1 + (3 + \sqrt{6})u)(1 + (3 - \sqrt{6})u).$$

Thus its univariate polynomial has a positive-semidefinite two-by-two determinant representation using a diagonal matrix. Nevertheless, there is no representation (29.1) for its five-variable polynomial. A one-variable factorization is not a counterexample to the classification.

35 The same boundary over the smallest fields

Theorem 35.1 (Binary and ternary support patterns). *A finite bipartite graph admits a support-exact matrix over $\mathbb{F}_2$ exactly when it is a forest, and over $\mathbb{F}_3$ exactly when it is a cactus. A cactus has a unique support-exact $\mathbb{F}_3$ gauge class under independent nonzero row and column scalings.*

Proof. Over $\mathbb{F}_2$, all edge entries are one. A shortest cycle is induced, and its two perfect-matching terms sum to zero. Conversely, a forest has at most one perfect matching on any support, so every structurally nonzero determinant is one.

For $\mathbb{F}_3$, first consider a cactus. Normalize a spanning forest to ones. On the remaining edge of a cycle of length $2r$, choose

$$t = (-1)^{r+1} \in \mathbb{F}_3.$$

The flip ratio including its determinant sign is $(-1)^{r-1}t = 1$. The same independent-cycle factorization as in Lemma 30.1 expresses every structurally nonzero determinant as a nonzero matching term times a product of factors $1 + 1 = 2$. It cannot vanish in $\mathbb{F}_3$.

For necessity, induced restriction and path compression are valid over every field. Normalize each of the three cores as above, now with all parameters in $\mathbb{F}_3^* = \{1, -1\}$. In (33.1), three column ratios cannot all be distinct, so some two-by-two minor vanishes. In (33.2), nonvanishing of $1 - x$ and $1 - y$ forces $x = y = -1$, but $1 - x - y = 3 = 0$. In (33.3), nonvanishing of every $x_i + x_j$ forces the three parameters to be equal, but their sum is again zero. No core is support-exact, so the graph must be a cactus.

Finally, on a normalized cactus cycle, $1 + (-1)^{r-1}t$ is nonzero for precisely the choice $t = (-1)^{r+1}$. The spanning-forest gauge is unique as in Theorem 30.3, proving uniqueness. $\square$

Together with the preceding sections, this completes every equivalence in Theorem 29.1.

Remark 35.2 (Reduction of the complex certificate). The ring homomorphism $\mathbb{Z}[\zeta] \rightarrow \mathbb{F}_3$ sending ζ to -1 is well-defined because $(-1)^2 - (-1) + 1 = 3$. It sends both complex cycle choices to the unique ternary choice. This provides another explanation of why the 2^β complex classes collapse to one ternary class. The determinant factorization, rather than just entrywise reduction, ensures that no required minor is lost.

Proposition 35.3 (A first larger-field contrast). *For $m \geq 1$ and a finite field $\mathbb{F}_q$, the complete bipartite graph $K_{2,m}$ admits a support-exact matrix over $\mathbb{F}_q$ if and only if $m \leq q - 1$.*

Proof. Scale the columns so that the first row is all ones. The second row consists of nonzero elements $t_1, \dots, t_m$. The two-by-two determinants are $t_j - t_i$, so support-exactness is equivalent to the t_i being pairwise distinct. There are exactly $q - 1$ available elements. $\square$

This elementary observation shows that the cactus classification is genuinely tied to the ternary field in its finite-field formulation; it is not a field-independent prohibition on overlapping cycles. Broader finite-field classifications are not asserted.

36 A sharp universal approximation gap

Exact equality can be fragile in numerical experiments. Here the obstruction is not merely that some equations cannot hold simultaneously: they cannot all be approximately correct beyond an explicit dimension-independent threshold.

Definition 36.1 (Minor defect). For a phase matrix K on G , let

$$\delta_G(K) = \max_{\substack{I \subseteq A, J \subseteq B, |I|=|J| \\ G[I \sqcup J] \text{ perfectly matchable}}} \left| |\det K_{I,J}|^2 - 1 \right|, \quad d(G) = \min_K \delta_G(K). \quad (36.1)$$

The minimum is over unit phases on all edges. It exists because the phase torus is compact and the maximum is over finitely many continuous functions. The empty minor guarantees a nonempty indexing set, even for an edgeless graph.

Nonmatchable minors always have their exact target value zero and need not be included. The normalization $|K_{ab}| = 1$ is essential: allowing unrestricted edge magnitudes would define a different optimization problem.

Lemma 36.2 (Defect is monotone under the reductions). *Induced restriction and the path compression of Lemma 32.1 cannot increase the defect of a phase matrix.*

Proof. Restriction only removes minors from the maximum. Every minor of a compressed matrix is, by (32.2), equal in squared modulus to an augmented minor of the original matrix, and the two supports are matchable simultaneously. Thus every quantity tested after compression already occurs before it. $\square$

Theorem 36.3 (Sharp universal noncactus gap). *Put*

$$\varepsilon_* = \frac{\sqrt{17} - 3}{2} = 0.5615528128 \dots$$

For every finite bipartite noncactus G and every phase matrix K on it,

$$\delta_G(K) \geq \varepsilon_*. \quad (36.2)$$

The constant is optimal over all noncactus graphs: $d(K_{2,3}) = \varepsilon_$. In particular,*

$$G \text{ cactus} \iff d(G) = 0, \quad \inf_{G \text{ noncactus}} d(G) = \varepsilon_*.$$

The proof occupies the next two subsections. It yields more information than the exact classification: a phase fit with all matchable minor squared moduli in $(1 - \varepsilon_*, 1 + \varepsilon_*)$ certifies that the graph is a cactus. This statement concerns all minors, not a sampled subset.

36.1 The exact optimization on $K_{2,3}$

Proposition 36.4 (Three-chord minimax problem). *For the even core, $d(E) = \varepsilon_*$. A minimizing normalized matrix is*

$$K_* = \begin{pmatrix} 1 & 1 & 1 \\ e^{-i\varphi} & 1 & e^{i\varphi} \end{pmatrix}, \quad \cos \varphi = \frac{\sqrt{17} - 1}{4}. \quad (36.3)$$

Its three nontrivial minor squared moduli are $1 - \varepsilon_$, $1 - \varepsilon_*$, and $1 + \varepsilon_*$.*

Proof. After column normalization, the second-row entries are three unit complex numbers. Every nontrivial determinant is their pairwise difference, so the problem is to approximate one by all three squared chord lengths.

Suppose first that the defect δ is less than one; otherwise the desired lower bound is immediate. All squared chord lengths are then less than two, so all pairwise real inner products are positive. The three points lie in an arc of angular length $\theta < \pi/2$. To justify this, rotate one point to angle zero and represent the other two angles in $(-\pi/2, \pi/2)$. Their difference is in $(-\pi, \pi)$, and positivity of their mutual inner product forces its absolute value below $\pi/2$. Thus the interval between the extreme angles has the asserted length.

Order the points on that arc. At least one of the two successive gaps is at most $\theta/2$. Write $c = \cos(\theta/2)$, so $\sqrt{2}/2 < c \leq 1$. The smallest squared chord length is at most $2 - 2c$, whereas the largest is exactly $2 - 2\cos \theta = 4(1 - c^2)$. It follows that

$$\delta \geq \max\{2c - 1, 3 - 4c^2\}. \quad (36.4)$$

The first expression increases with c and the second decreases. Their unique intersection in the indicated interval occurs at

$$c_* = (\sqrt{17} - 1)/4, \quad 2c_* - 1 = 3 - 4c_*^2 = \varepsilon_*.$$

Therefore (36.4) gives $\delta \geq \varepsilon_*$.

For (36.3), the consecutive gaps are equal to φ . The corresponding squared chord lengths are $2 - 2c_* = 1 - \varepsilon_*$ and $4(1 - c_*^2) = 1 + \varepsilon_*$. Thus the lower bound is attained. The argument also shows that any minimizing triple has equal consecutive gaps, up to rotation, reflection, and permutation of its points. $\square$

36.2 The other two cores are separated by at least $3/5$

Lemma 36.5 (A stronger-than-needed bound for D). *Every phase matrix on D has defect at least $3/5$.*

Proof. Use (33.2) and suppose the defect is less than $3/5$. The two cycle minors imply

$$a = \operatorname{Re} x, \quad b = \operatorname{Re} y \in (1/5, 4/5).$$

Depending on whether the imaginary parts have the same or opposite signs, the squared modulus of the full determinant is

$$g_{\pm}(a, b) = 3 - 2a - 2b + 2ab \pm 2\sqrt{1 - a^2}\sqrt{1 - b^2}.$$

On the closed rectangle $[1/5, 4/5]^2$, g_- is convex in each variable separately and g_+ is concave in each variable separately. Indeed the second derivative with respect to a is respectively

$$\frac{2\sqrt{1 - b^2}}{(1 - a^2)^{3/2}} \quad \text{or its negative,}$$

and the same holds with the variables exchanged. Hence the maximum of g_- and the minimum of g_+ occur at corners. The corner values give

$$g_- \leq \frac{9}{25}, \quad g_+ \geq \frac{9}{5}.$$

For the off-diagonal corners the respective values are $(33 - 12\sqrt{6})/25$ and $(33 + 12\sqrt{6})/25$, so these bounds follow by direct comparison. Both alternatives are outside the interval $(2/5, 8/5)$ required for a full minor of defect less than $3/5$. This is a contradiction. $\square$

Lemma 36.6 (A planar Gram obstruction for T). *Every phase matrix on T has defect at least $3/5$.*

Proof. Use (33.3). Suppose its defect is less than $3/5$ and put

$$p = -\operatorname{Re}(x_1\bar{x}_2), \quad q = -\operatorname{Re}(x_1\bar{x}_3), \quad r = -\operatorname{Re}(x_2\bar{x}_3).$$

The three pair-cycle minors imply $p, q, r \in (1/5, 4/5)$. As three unit vectors in the real plane, the x_i have a singular real Gram matrix. Expanding its determinant yields

$$p^2 + q^2 + r^2 + 2pqr = 1. \tag{36.5}$$

Write $s = p + q + r$. The full minor has squared modulus $3 - 2s$, so its lower bound $3 - 2s > 2/5$ implies $s < 13/10$.

For each $t \in [1/5, 4/5]$, the inequality $(t - 1/5)(t - 4/5) \leq 0$ gives $t^2 \leq t - 4/25$. Also $pqr \leq (s/3)^3$ by the arithmetic–geometric mean inequality. Therefore

$$1 = p^2 + q^2 + r^2 + 2pqr \leq s - \frac{12}{25} + 2(s/3)^3 < \frac{13}{10} - \frac{12}{25} + 2(13/30)^3 = \frac{13267}{13500} < 1,$$

a contradiction. $\square$

Proof of Theorem 36.3. Take an induced theta in G and compress it to a core. The defect cannot increase by Lemma 36.2. On E it is at least ε_* by Proposition 36.4; on the other cores it is at least $3/5 > \varepsilon_*$ by Lemmas 36.5 and 36.6. Formula (36.3) attains ε_* on the noncactus $K_{2,3}$, proving optimality. Cacti have defect zero by Lemma 30.1. $\square$

Remark 36.7 (Why a weaker four-minor bound is not sharp). On T, let $a_{ij} = |x_i + x_j|^2 - 1$ and $b = |x_1 + x_2 + x_3|^2 - 1$. Then $b = a_{12} + a_{13} + a_{23} - 1$, so the triangle inequality alone gives $\delta \geq 1/4$. It loses the fact that the three complex phases live in a real plane. The Gram identity (36.5) restores that constraint and makes the sharp universal theorem possible.

37 Real stability is strictly more general

A nonzero polynomial with real coefficients is *real stable* if it does not vanish when every variable has strictly positive imaginary part. For a cactus certificate, (29.1) immediately gives stability: if $(\text{diag } \mathbf{z} - H)u = 0$ for a nonzero vector u , then

$$0 = \text{Im}(u^*(\text{diag } \mathbf{z} - H)u) = \sum_v \text{Im}(z_v)|u_v|^2 > 0,$$

a contradiction. This determinant argument is already used in Part II, as the proof of Theorem 19.3.

The converse fails on only five vertices. Denote the variables on the two shores of $K_{2,3}$ by a, b and x, y, z . Direct support enumeration gives

$$f = \Phi_{K_{2,3}} = abxyz - (a+b)(xy+xz+yz) + (x+y+z). \quad (37.1)$$

The size-four supports are the three choices of two vertices on the larger shore together with both vertices on the smaller shore, explaining the last three terms.

Theorem 37.1 (A stable polynomial without a Hermitian vertex determinant). *Polynomial (37.1) is real stable, but there is no Hermitian 5×5 matrix H with $f = \det(\text{diag}(a, b, x, y, z) - H)$.*

Proof. Nonrepresentability follows from Theorem 29.1, or directly from the three-chord obstruction (33.1). For stability, use Brändén's criterion: a nonzero real multiaffine polynomial f is real stable if and only if

$$\Delta_{ij}f = (\partial_i f)(\partial_j f) - f \partial_i \partial_j f \geq 0$$

for all real choices of all variables and all distinct pairs [12, Theorem 5.6]. Symmetry leaves only three types of pairs.

For the pair on the smaller shore, put $U = xy$, $V = xz$, $W = yz$. Direct differentiation gives

$$\begin{aligned} \Delta_{a,b}f &= U^2 + V^2 + W^2 + UV + UW + VW \\ &= \frac{1}{2}((U+V+W)^2 + U^2 + V^2 + W^2) \geq 0. \end{aligned} \quad (37.2)$$

For two variables on the larger shore,

$$\begin{aligned} \Delta_{x,y}f &= (a^2 + ab + b^2)z^2 - (a+b)z + 1 \\ &= \left(1 - \frac{a+b}{2}z\right)^2 + \frac{3a^2 + 2ab + 3b^2}{4}z^2 \geq 0. \end{aligned} \quad (37.3)$$

For a cross-shore pair,

$$\begin{aligned} \Delta_{a,x}f &= (byz)^2 - byz(y+z) + y^2 + yz + z^2 \\ &= \left(byz - \frac{y+z}{2}\right)^2 + \frac{3y^2 + 2yz + 3z^2}{4} \geq 0. \end{aligned} \quad (37.4)$$

The remaining quadratic forms are nonnegative because $3u^2 + 2uv + 3v^2 = 2(u^2 + v^2) + (u+v)^2$. The three identities cover all ten pairs by shore symmetries. The criterion proves stability. $\square$

This is not a claim that stability without determinant representations is new as a general phenomenon. Such separations are established in other settings, including work on Vámos-type polynomials [15]. The result here answers Part II's specific question (Research question 4) for its particular matching-support refinement, with a small explicit example and elementary certificates.

37.1 Leaf extension and an infinite connected family

Lemma 37.2 (Exact leaf recurrence). *Let $G + \ell$ be obtained by adding a new leaf ℓ adjacent only to a vertex v . Then*

$$\Phi_{G+\ell} = z_\ell \Phi_G - \Phi_{G-v} = z_\ell \Phi_G - \partial_{z_v} \Phi_G. \quad (37.5)$$

Proof. A matchable support omitting ℓ gives the first term. A support containing ℓ must also contain v , and its perfect matching must use ℓv . Removing these two vertices gives a unique perfectly matchable support in $G - v$, with the opposite sign in (28.2). Finally, differentiation with respect to z_v retains exactly the supports omitting v and removes that variable from their monomials. $\square$

Lemma 37.3 (Stability of the leaf operation). *If a real multiaffine polynomial f is stable, then $wf - \partial_{z_v} f$, where w is a new variable, is stable.*

Proof. Fix all variables other than z_v in the upper half-plane. Write $f = Az_v + B$ at these fixed values. It is not the zero polynomial, since stability would otherwise fail. If $A = 0$, then $f = B \neq 0$ and the claimed expression is $wB \neq 0$. If $A \neq 0$, its root $\rho = -B/A$ has imaginary part at most zero. For z_v in the upper half-plane,

$$\frac{\partial_{z_v} f}{f} = \frac{1}{z_v - \rho}$$

has strictly negative imaginary part. It cannot equal w in the upper half-plane. Multiplying by the nonzero f proves the claim. $\square$

Corollary 37.4 (Infinitely many connected examples). *Every graph obtained from $K_{2,3}$ by attaching finite trees at its vertices has a real stable polynomial Φ_G and has no Hermitian vertex determinant representation (29.1). These examples can be arbitrarily large and connected.*

Proof. Construct the attached trees by successive leaf additions and apply Theorem 37.1 and Lemmas 37.2–37.3. The original $K_{2,3}$ remains an induced subgraph, so the result is still a noncactus. The classification excludes the representation. $\square$

38 No fixed minor order can test exact representability

The classification has a finite structural obstruction mechanism, but not a bounded matrix-size obstruction. Compression may require adjoining many eliminated rows and columns before a core minor becomes a minor of the original matrix.

Theorem 38.1 (Unbounded order of minor witnesses). *For every positive integer r , there is a connected bipartite noncactus G of maximum degree three and treewidth at most two such that every phase matrix on G satisfies (28.3) for every square minor of order at most r . Nevertheless no phase matrix on G satisfies all orders.*

Proof. Choose an odd integer $L > r$, $L \geq 3$, and take three internally disjoint paths of length L between two endpoints. This theta is bipartite, subcubic, connected, and not a cactus. Every simple cycle consists of two of the three paths, so its girth is $2L > 2r$.

A square minor of order at most r uses at most $2r$ vertices. Its induced graph therefore has no cycle. Such a forest has at most one perfect matching; its determinant is zero when there is none and a unit product when there is one, for every phase assignment. Thus all the small-minor tests pass. Theorem 29.1 rules out a faithful matrix globally.

For completeness, a width-two tree decomposition is explicit. Let the common endpoints be s, t and write the internal vertices of path i as $v_{i,1}, \dots, v_{i,L-1}$. Use a central bag $\{s, t\}$. Along path i use the chain of bags

$$\{s, v_{i,1}, v_{i,2}\}, \dots, \{s, v_{i,L-2}, v_{i,L-1}\}, \{s, v_{i,L-1}, t\},$$

attaching its last bag to the central bag. Every edge is covered and the bags containing any given vertex are connected. Each bag has at most three vertices. $\square$

Corollary 38.2 (Failure of uniform low-order numerical testing). *For every r , unit-phase matrices can pass all minor tests through order r exactly while some higher-order matchable minor has defect at least ε_* .*

Proof. Combine Theorem 38.1 with Theorem 36.3. $\square$

Thus the approximation gap does not justify replacing the complete minor maximum by a fixed-order sample. Even zero observed error on all two-by-two, three-by-three, or any other bounded range of minors can be misleading.

39 Consequences for the repository's preorder program

For a height-two poset P_G with comparability graph G on n vertices, Part II establishes the height-two formula $h_{P_G}(t) = (1+t)^n p_G(t/(1+t)^2)$ with a precise lattice-support interpretation (Theorem 13.2). We use this as a prior bridge rather than reprove it or claim its γ -positivity conclusion.

For a faithful matrix, the principal-minor expansion and Cauchy–Binet give $p_G(u) = \det(I + uKK^*)$. Indeed, the coefficient of u^k is the sum of $|\det K_{I,J}|^2$ over all k -element row and column sets, hence counts $\mathcal{M}_k(G)$. If $\lambda_1, \dots, \lambda_\nu$ are the positive eigenvalues of KK^* , where ν is the matching number, the factorizations (19.2) and (19.3) of Part II's Theorem 19.2 follow; that proof uses only faithfulness, so it applies to every certificate of Theorem 30.3. [Merge note: manuscript 05 printed these four formulas again as displays; they are printed once, in Part II.] These cactus formulas and their real-rootedness consequences are old relative to this article; univariate cactus real-rootedness is also contained in Ohsugi–Tsuchiya [9, Theorem 7.2]. Our contribution identifies the exact boundary of this *support-by-support* determinant route.

For the stable noncactus $K_{2,3}$, the diagonal specialization of (37.1) is

$$\Phi_{K_{2,3}}(s, s, s, s, s) = s^5 - 6s^3 + 3s = s(s^2 - (3 + \sqrt{6}))(s^2 - (3 - \sqrt{6})),$$

and Theorem 13.2 gives

$$h_{P_{K_{2,3}}}(t) = 1 + 11t + 31t^2 + 31t^3 + 11t^4 + t^5.$$

The graph is not excluded from real-rootedness; only its exact vertex determinant is excluded. More generally, stable Φ_G implies real-rooted p_G : for $s \neq 0$,

$$\Phi_G(s, \dots, s) = s^n p_G(-s^{-2}).$$

The left-hand side has only real zeros. Any nonzero zero of p_G would otherwise give a nonreal zero on the left after solving $u = -s^{-2}$. Since p_G has positive coefficients and constant term one, its zeros must consequently be negative. This applies to all the tree-attached examples of Corollary 37.4.

There are three logically different classes: faithful support determinants, stable signed support polynomials, and real-rooted univariate support polynomials. We prove the first is properly contained in the second. The second is contained in the third by specialization. We do not assert a classification of the latter two classes or strictness of that second containment.

40 Exact verification and reproducibility

The included verifier, shipped as `code/03-cactus-verify.py` (delivered as `code/verify.py`), uses only the Python standard library and exact arithmetic. Its checks were executed with Python 3.13.5. The full output is saved in `data/03-cactus-verification.json` (delivered as `data/verification.json`). The tests supplement the proofs; in particular, checking sixth-root assignments does not by itself exclude arbitrary complex phases.

40.1 Independent combinatorial and algebraic representations

Graphs with fixed labeled shores are encoded by binary edge masks. Cactus recognition enumerates simple cycles and checks that no edge belongs to two of them. This recognition routine does not use determinants. A separate spanning-forest procedure identifies the free normalized edge entries.

For each support pattern, the verifier enumerates every normalized assignment over $\mathbb{F}_3^*$, or over the six complex sixth roots, as appropriate. Structurally nonzero determinant terms are obtained by permutation expansion. Minors with one matching term automatically have unit magnitude; ambiguous minors are evaluated exactly. Complex values are pairs of integers $a + b\zeta$, with

$$(a + b\zeta)(c + d\zeta) = (ac - bd) + (ad + bc + bd)\zeta, \quad |a + b\zeta|^2 = a^2 + ab + b^2.$$

The canonical cactus certificates are checked again by evaluating *all* square minors, including empty minors. Their expected zero-or-one values come from a separate covered-shore dynamic program that merges different matchings with the same covered vertices. Thus the determinant checker is compared with a support-set enumeration, not with its own permutation count.

Check	Recorded count
All labeled 3-by-3 bipartite graphs	512
Cacti among those graphs	460
Normalized sixth-root assignments tested	5,872
Ambiguous minors evaluated in that search	7,792
All labeled 3-by-4 bipartite graphs	4,096
Cacti among those graphs	3,122
Normalized ternary assignments tested	10,576
Ambiguous minors evaluated in that search	14,717
Canonical cactus minor checks, both graph families	118,470
Compression: augmented-minor norm/support comparisons	100
Rayleigh identity classes checked symbolically	3
Independent leaf-attachment support enumerations	5

All checks passed. In a separate check, each of the three small cores has zero valid assignments among its 36 normalized sixth-root assignments and four normalized ternary assignments. The 3-by-3 cacti consist of 328 forests and 132 one-cycle graphs. The 3-by-4 cacti consist of 1,856 forests, 1,248 one-cycle graphs, and 18 two-cycle graphs. The exhaustive normalized counts agree with 2^β complex classes and one ternary class on every tested cactus, and zero on every tested noncactus.

40.2 Symbolic and compression checks

A small rational multivariate polynomial implementation expands and compares the three Rayleigh identities (37.2)–(37.4); by symmetry they cover all ten pairs. The polynomial (37.1)

is independently reconstructed from matchable supports. Each of the five possible leaf attachments is enumerated directly and compared with (37.5).

The compression checker uses theta path lengths $(2, 2, 4)$, $(1, 3, 5)$, and $(3, 3, 5)$ with deterministic, nonconstant sixth-root phases. For all compressed square minors it compares the augmented original determinant norm and matching-support indicator. Exact arithmetic in $\mathbb{Q}(\sqrt{17})$ also verifies the two algebraic equalities defining the extremal constant. None of these finite checks proves the compactness, geometric minimax, or all-graph statements by extrapolation; those are proved in the text.

40.3 Build and rerun

The manuscript's own commands, run from its package root, were `python3 code/verify.py` followed by two `pdflatex` runs on its `article.tex`; its Makefile offered the same. In this report every delivered file name carries the prefix `03-cactus-`:

- program and build file: `code/03-cactus-verify.py` and `code/03-cactus-Makefile`;
- recorded outputs: `data/03-cactus-verification.json` (the recorded run),
`data/03-cactus-verification_log.txt` (its console output, byte-identical to the JSON)
and `data/03-cactus-build_status.json` (the manuscript's record of its own PDF build);
- notes: `03-cactus-STATUS.md` and `03-cactus-SOURCES.md`.

The program still uses the delivered names. Run in this directory, it would write a new unprefixed `data/verification.json` beside Part I's files; the Makefile's `verify` target would run Part I's `code/verify.py`, which without an output option rewrites Part I's shipped data; and its default target would rebuild this whole article in place. Rerun the program only in a scratch copy with the delivered names restored, from this report's directory:

```
W=/path/to/scratch; mkdir -p "$W/03/code" "$W/03/data"
for f in code/03-cactus-*; do cp "$f" "$W/03/code/${f#code/03-cactus-}"; done
for f in data/03-cactus-*; do cp "$f" "$W/03/data/${f#data/03-cactus-}"; done
mv "$W/03/code/Makefile" "$W/03/Makefile"; cd "$W/03"
python3 code/verify.py # rewrites data/verification.json of the copy
```

The program needs Python 3.10 or later and only its standard library; no network access or repository checkout is needed, and its elapsed-time field differs from run to run. This article is rebuilt as described in the report's README; Part III needs no package beyond those of Parts I and II.

41 Proof audit, priority, and formalization boundary

41.1 The parts most worth independent scrutiny

The decisive combinatorial step is the induced-theta lemma, especially the length-one path case. The decisive algebraic step is (32.2), together with its exact matching bijection. Deleting arbitrary edges or suppressing arbitrary degree-two vertices would not be a valid substitute. In the Hermitian equivalence, the singleton and pair coefficients must be compared before concluding that the matrix has the graph's support. In the quantitative theorem, unit moduli of the pivot entries are essential for preserving the defect.

The sharp constant is not inferred from optimization software. Proposition 36.4 proves the lower bound and constructs an extremizer. The other cores are bounded away by $3/5$, a

strictly larger rational number. The final step uses induced restriction and compression, not a supposition that every noncactus literally contains an induced copy of one of the three *small* cores.

For stability, all ten Rayleigh inequalities are covered by symmetry, and the input theorem applies to nonhomogeneous multiaffine polynomials as well as homogeneous ones. The leaf-preservation proof is independent of a general classification of stability-preserving operators.

41.2 What is and is not claimed to be new

The following separation of status is part of the mathematical result, not a bibliographic afterthought.

Statement	Status in this report
Cactus sixth-root construction and its determinant factorization	Prior result of Part II; reproduced and credited.
Cactus univariate real-rootedness	Prior literature and Part II result; not claimed.
Height-two preorder matching-support transform	Prior result of Part II; used only for consequences.
Cactus necessity, full gauge classification, and ternary support characterization	Proved here as answers/extensions to Part II's Research question 4; absolute priority not established.
Sharp universal defect ε_* and the extremal $K_{2,3}$ phases	Proved here; no global publication-priority claim.
Explicit stable $K_{2,3}$ separation and its tree extensions	Proved here in Part II's signed-support setting; general stability/determinant separation is prior.
Failure of all fixed minor-order tests on subcubic width-two graphs	Proved here as a consequence of high-girth theta supports.

The targeted source audit inspected the pinned repository discussion (Part II), the perfectly matchable set literature [9, 7], Brändén's stability criterion [12], pattern-sensitive superregularity [13], and sixth-root-of-unity matrix terminology [14]. Related fundamental-transversal-matroid literature was also located, including an earlier binary characterization [16]; the present report does not claim novelty for the elementary binary corollary. This audit is not a complete survey of representable transversal matroids or every equivalent formulation of the support-pattern problem. A literature-priority determination would require that broader comparison. The proofs and the resolution of the identified repository question do not depend on a claim that no prior equivalent theorem exists.

41.3 A feasible formalization decomposition

A formal development should begin with finite bipartite graphs on two finite types and supports encoded by pairs of finite subsets. One layer proves that a forest support has at most one perfect matching and that symmetric differences decompose into alternating cycles. A second layer proves the exact compression bijection. A third proves the block determinant identity and the three core obstructions.

The graph-theoretic necessity can then be isolated from analysis. The ternary version needs only finite graph arguments and algebra over $\mathbb{F}_3$. The complex classification additionally needs elementary norm identities and the two allowed unit-circle solutions. The sharp gap adds real inequalities, the compact phase torus, and the planar Gram identity; the existence of the minimizer can be omitted if one first states the theorem for every phase assignment and gives the explicit extremizer.

Finally, the stability example can be represented as exact polynomial identities plus a separately imported or formalized Rayleigh criterion. This separation prevents the external analytic theorem from entering the trust boundary of the cactus classification. No uncompiled Lean files are supplied or described as checked. The Python verifier is executable evidence for finite cases, not a substitute for these formal layers.

42 Further research questions and topics

The original determinant-classification question is answered under the precise hypotheses stated here. The following directions remain outside the proved results. Except where attributed, they are questions generated by this investigation, not assertions that no answer is known anywhere in the literature.

Research question 11 (Classify real-stable signed support polynomials). Which bipartite graphs have real stable Φ_G ? The class contains all cacti and all tree attachments to $K_{2,3}$, but the determinant classification shows that no purely faithful-phase characterization can cover it. Determine a useful structural description, at least for graphs of treewidth two.

The three Rayleigh identities suggest searching for graph decompositions that preserve nonnegative Rayleigh differences without preserving a vertex determinant. A first target is a stable gluing theorem more general than leaf attachment.

Research question 12 (Determine the remaining core optimization exactly). Compute $d(T)$ exactly and classify its minimizing phase configurations. The present bound $d(T) \geq 3/5$ suffices for the sharp universal theorem but is not claimed optimal. Determine how the value changes when individual minors receive prescribed positive penalties.

The constraints reduce to three unit vectors and their rank-two real Gram matrix. This gives a small semialgebraic problem, but a numerical stationary point alone is not an exact certificate.

Research question 13 (Describe graphwise sharp defects). Which noncactus graphs satisfy $d(G) = \varepsilon_*$? Having an induced or compressed $K_{2,3}$ gives a lower bound, not an extension of its optimal phases to every other edge and minor. Characterize when the three-point extremizer can be extended without introducing a larger defect.

This asks for more than the universal infimum, which Theorem 36.3 already determines. Additional cycles can impose new norm constraints not visible in one chosen theta.

Research question 14 (Larger finite fields and prescribed support). For $q \geq 4$, characterize useful classes of bipartite support patterns admitting a support-exact $\mathbb{F}_q$ matrix. Proposition 35.3 gives a first field-size obstruction. Can the general question be expressed by a manageable family of induced-and-compressed obstructions?

Any proposed characterization must distinguish a fixed zero pattern from unrestricted representability of an associated abstract matroid. Changing the presentation can change the support question.

Research question 15 (Larger determinantal models). For the stable polynomial $\Phi_{K_{2,3}}$, determine the smallest useful Hermitian pencil representation allowing auxiliary dimensions or structured substitutions. How does that size grow under the tree attachments of Corollary 37.4?

Theorem 29.1 addresses only the vertex-indexed diagonal pencil. A no-go result for that form should not be used as an obstruction to every possible determinant construction.

Research question 16 (Quantitative obstructions with variable edge magnitudes). Replace the unit-phase constraint by $1 - \eta \leq |K_{ab}|^2 \leq 1 + \eta$ on edges. Find an explicit lower bound on the largest support-minor defect, in terms of η , for noncacti. Does a useful dimension-independent gap persist for all sufficiently small η ?

Path compression now multiplies and divides magnitudes, so the proof of Lemma 36.2 no longer gives an exact contraction. Long subdivisions are the natural stress test.

Research question 17 (Short checkable obstruction certificates). Design an algorithm that, given a noncactus graph and edge phases in an exact number field, outputs a matchable minor with defect at least ε_* , together with a compact certificate. Optimize the number of rows and columns of the output witness.

The constructive proof finds a theta and traces a failing core minor back through augmented minors. Theorem 38.1 proves that no constant bound on witness order is possible; it does not determine the best bound in terms of girth, cycle overlap, or input size.

Research question 18 (Transfer beyond height-two posets). Can stable signed-support methods, rather than faithful matrices, extend Part II's γ -interpretation (Theorem 13.2; compare Research question 1) to height-three posets or nontrivial preorder-equivalence blocks? Identify an explicit intermediate class where the combinatorial support model remains valid.

Neither the cactus theorem nor the $K_{2,3}$ example answers general γ -positivity or existential flag-polytope realization. Those questions require an additional bridge beyond Theorem 13.2.

Research question 19 (Kernel-checked certificate theory). Formalize the ternary support-pattern classification first, then extend it to the complex phase and sharp-gap theorems. Can the induced-theta reduction and the augmented-minor map be packaged as a reusable verified obstruction framework?

A successful implementation would convert a structural nonexistence proof into independently checkable finite witnesses while leaving the harder stability theory in a separate module.

43 Conclusion

The phase construction for cactus matching supports is not merely one successful family among many exact vertex determinant models: it exhausts them. Cycle overlaps produce induced theta obstructions, and a support-preserving Schur compression reduces every such obstruction to one of three small matrices. The same reduction determines the ternary support boundary and yields an optimal, dimension-independent approximation threshold.

At the same time, real stability survives beyond that boundary. The explicit $K_{2,3}$ sums of squares and their leaf extensions show why searching only for a Hermitian vertex determinant would miss valid stability phenomena. The next stage of this report's program should therefore distinguish structural determinant certificates from more general stability certificates, and should distinguish both from one-variable real-rootedness tests.

44 A sharper calculation for the shared-edge core

(Appendix A of manuscript 05.) Although the bound $3/5$ is enough for the universal theorem, the other small core D also has an exact elementary minimax value.

Proposition 44.1 (The exact defect of two squares sharing an edge). *For $D = \mathfrak{D}(1, 3, 3)$,*

$$d(D) = d_0 := \frac{\sqrt{5} - 1}{2}.$$

Proof. In (33.2), every ambiguous proper square minor is one of the two cycle minors $1 - x$, $1 - y$; the only remaining ambiguous minor is the full determinant $1 - x - y$. Thus these three norms determine the defect.

Assume a defect $\delta < d_0$. The real parts a, b lie strictly inside the rectangle $[l, h]^2$, where

$$l = (1 - d_0)/2, \quad h = (1 + d_0)/2, \quad l + h = 1.$$

Use the separately convex/concave functions g_- , g_+ from Lemma 36.5. The two equal-coordinate corners of g_- have value $d_0^2 = 1 - d_0$. At an off-diagonal corner,

$$g_-(l, h) = (\sqrt{1 - l^2} - \sqrt{1 - h^2})^2 = \frac{(h - l)^2}{(\sqrt{1 - l^2} + \sqrt{1 - h^2})^2} \leq d_0^2.$$

The denominator is at least one: $l + h = 1$ gives $(1 - l^2) + (1 - h^2) = 1 + 2lh \geq 1$. Therefore opposite imaginary signs force $|1 - x - y|^2 \leq 1 - d_0 < 1 - \delta$, impossible.

For the same-sign case, the equal-coordinate corners of g_+ have values $3 - 2d_0$ and $3 + 2d_0$. Both exceed $1 + d_0$, since $d_0 < 2/3$. For the off-diagonal corner, $lh = d_0/4$ and

$$g_+(l, h) = 1 + \frac{d_0}{2} + 2\sqrt{\frac{d_0}{2} + \frac{d_0^2}{16}} > 1 + d_0.$$

The last strict inequality follows from $\sqrt{d_0/2 + d_0^2/16} > d_0/4$. Separate concavity therefore forces $|1 - x - y|^2 > 1 + d_0 > 1 + \delta$, another contradiction. Hence $d(D) \geq d_0$.

For attainment, take $x = e^{i\alpha}$ and $y = e^{-i\alpha}$ with $\cos \alpha = (1 + d_0)/2$. The two cycle squared norms are $1 - d_0$, while the full squared norm is $d_0^2 = 1 - d_0$. Each has defect d_0 , and all other nonzero minors have norm one. This proves equality. $\square$

This additional calculation is not needed for the main classification or for the sharp universal constant. It leaves the three-odd-path core as the remaining small exact optimization target.

45 A compact dependency ledger

(Appendix B of manuscript 05.)

Result	Inputs
Cactus sufficiency	Symmetric differences of matchings; independent cycle flips; sixth-root norm identity.
Cactus necessity	Induced-theta lemma; support-preserving Schur compression; three core calculations.
Hermitian equivalence	Diagonal coefficient expansion; singleton and pair recovery; bipartite block determinant.
Ternary classification	Same graph reduction; three computations in $\mathbb{F}_3$; independent cycle factors.
Gauge count	Spanning-forest normalization; two possible normalized phases per cycle.
Sharp universal gap	Defect monotonicity; exact three-chord minimax; rational bounds on D, T .
Stable noncactus family	Three Rayleigh identities; Brändén's criterion; direct leaf-preservation argument.
No bounded minor-order test	High-girth odd theta; uniqueness of a forest perfect matching; classification.

A A hypertree proof of the known root identity

This appendix reconstructs Proposition 2.2 without using the support-enumerator theorem of [2]. It is included for conceptual clarity and as an independent route to the same normalization. Its one substantial imported result is the Kálmán–Postnikov theorem [3].

A.1 Hypertrees and internal inactivity

For a connected bipartite graph with left shore L , a left hypertree is a vector $a \in \mathbb{Z}_{\geq 0}^L$ realized by a spanning tree T through $a_i = \deg_T(i_L) - 1$. Distinct trees realizing the same vector contribute only one hypertree. Order the left shore. A coordinate i is internally inactive when there is a smaller coordinate j such that replacing a by $a - \mathbf{e}_i + \mathbf{e}_j$ gives another hypertree. The interior polynomial is the sum of $t^{\text{number of internally inactive coordinates}}$ over hypertrees.

The Kálmán–Postnikov theorem identifies this interior polynomial with the Ehrhart numerator of the bipartite root polytope. Their paper uses a reversed convention for the triangulation h -polynomial in its Theorem 1.1; the Ehrhart-basis formulation immediately following that theorem, equation (1.2), identifies the coefficients with the usual h^* -numerator used here. Thus there is no additional reversal of t in the statement we apply.

Lemma A.1. *The left hypertrees of $\widehat{G}_\tau$ are exactly*

$$a = (n - x(E), x_1, \dots, x_n), \quad x \in P_\tau. \quad (\text{A.1})$$

Proof. A spanning tree of $\widehat{G}_\tau$ has $2n + 1$ edges, so the sum of its left degree-minus-one vector is $(2n + 1) - (n + 1) = n$. For a nonempty $A \subseteq [n]$, the tree edges incident with A_L form a forest on $A_L \cup N(A)$. Their number is $a(A) + |A|$, and a forest on this nonempty vertex set has at most $|A| + |N(A)| - 1$ edges. Hence

$$a(A) \leq |N(A)| - 1 = |\downarrow A|.$$

Lemma 2.1 now shows that $x = (a_1, \dots, a_n)$ belongs to P_τ , and the sum condition determines a_0 .

Conversely, let $x \in P_\tau$. Replace the left element i by x_i distinct clones, each adjacent to the right set $\downarrow i \subseteq [n]$. A set of clones whose labels lie in A has at most $x(A) \leq |\downarrow A|$ members. Thus Hall's matching condition holds, and all clones can be matched to distinct right vertices. This is also a constructive statement: if an augmenting-path matching leaves a clone unmatched, the alternating paths reachable from it produce a set of clones with fewer neighbors, contradicting Hall's condition.

Build a tree by first taking all edges $i_L 0_R$ for $0 \leq i \leq n$. Each right vertex j_R , $j > 0$, is then attached to the left label of its matched clone, or to 0_L if it is unmatched. The first edges form a star; every subsequent edge attaches a new leaf. Hence the result is a spanning tree. Its left degrees are $x_i + 1$ for $i > 0$ and $n - x(E) + 1$ for $i = 0$, exactly as required. $\square$

Lemma A.2. *Order the left vertices with 0 first. For the hypertree in (A.1), internal inactivity is $|\text{supp}(x)|$.*

Proof. The first coordinate is never internally inactive. A zero coordinate cannot transfer a unit, so it is also not internally inactive. If $x_i > 0$, then $x - \mathbf{e}_i \in P_\tau$ because all the defining inequalities are upper bounds and the coordinates remain nonnegative. By Lemma A.1, the vector obtained by transferring one unit from coordinate i to coordinate 0 is another hypertree. Thus every positive nonzero-index coordinate is internally inactive, and there are no others. $\square$

Summing over (A.1) gives an interior polynomial equal to $h_\tau(t)$. The Kálmán–Postnikov theorem identifies it with $h^*(B_\tau, t)$, proving Proposition 2.2 by this second route. Notice that the argument counts hypertree *vectors*, not spanning trees; counting all realizing trees would introduce incorrect multiplicities.

B Proof dependencies and normalization audit

Claim or input	Where it comes from
$h^*(B_\tau) = h_\tau$	Known: [2, Theorem 1.2], specialized using Lemma 2.1. Alternatively Appendix A plus [3].
Ideal Hall presentation	Proposition 3.2; transitivity is used explicitly.
Integer fiber factorization	Theorem 4.1; all coordinates and inverse parameters are displayed.
$h^*(C_\tau) = h_\tau$	Corollary 4.2; a consequence of the fiber factorization and the known root identity.
Unimodularity	Leaf elimination in Lemma 5.1, followed by Proposition 5.3.
Polytopal pulling boundary	The established pulling lemma [4, Lemma 2.1].
Palindromicity, unimodality	Dehn–Sommerville and the g -theorem applied to the constructed simplicial polytope; an independent reciprocity proof of palindromicity is in Section 6.
Alternative polytopal route	The explicit special simplex and [4, Theorem 3.5].

There are several easy places to make an otherwise plausible argument wrong. The support polynomial of Q_τ is not asserted to be $h^*(Q_\tau)$. It is the numerator of the different polytopes B_τ and C_τ . The larger denominator has exponent $2n + 1$ and the smaller one exponent $n + 1$. Both auxiliary vertices, and the edge between them, are required. The edge orientation convention is $j \leq_\tau i \iff i_L j_R$ is present. The integer-fiber parameters consist of $n + 1$ nonnegative entries; the exponent n in (4.9) results only after dividing by the additional slack factor in the Ehrhart series of C_τ .

No argument substitutes a rational equivalence for an integral one. The fiber base is integral because $a_\tau(y)$ is an integer for integral y . The determinant statements refer to the full relevant lattices. Finally, regularity or shellability alone is not used to claim that a sphere is polytopal: the specific pulling-boundary theorem supplies that step.

C Reproducing the accompanying computations

The archive contains `article.tex`, `article.pdf`, a build script, `code/verify.py`, the full CSV enumeration, a JSON result report, and status notes. From the extracted directory, run

```
python3 code/verify.py --output data
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier requires Python 3.10 or later and no third-party Python packages. Do not use Python’s `-O` option, since assertions are part of the checks. The LaTeX document uses standard TeX Live packages, including `newpxtext`, `newpxmath`, `tcolorbox`, and `hyperref`.

The CSV uses zero-based bit positions: the row value 5 represents the set $\{0, 2\}$. Row i records the principal ideal of element i , not its filter. A polynomial is stored as coefficients from

degree zero upward. The JSON report also records explicit edge sets for matching certificates and complete Ehrhart counts for the independently tested instances.

[Added 28 September 2026, batch 39: this appendix describes Part I’s package; Part II’s programs and how to rerun them are in Section 22.4.]

The calculations in the supplied report completed without failed assertions. Their runtime is recorded separately as an observation from this environment, not as a performance guarantee. No claim of formal verification is made.

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